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(54) **INFORMATION PROCESSING APPARATUS,  
INFORMATION PROCESSING METHOD,  
AND STORAGE MEDIUM**

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**Foreign Application Priority Data**

Oct. 11, 2018 (WO) ..... PCT/JP2018/038000

(57)

**ABSTRACT**

An information processing apparatus includes: an information management unit that registers boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and a display management unit that causes a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and registered biometrics information, which is the biometrics information registered by the information management unit.

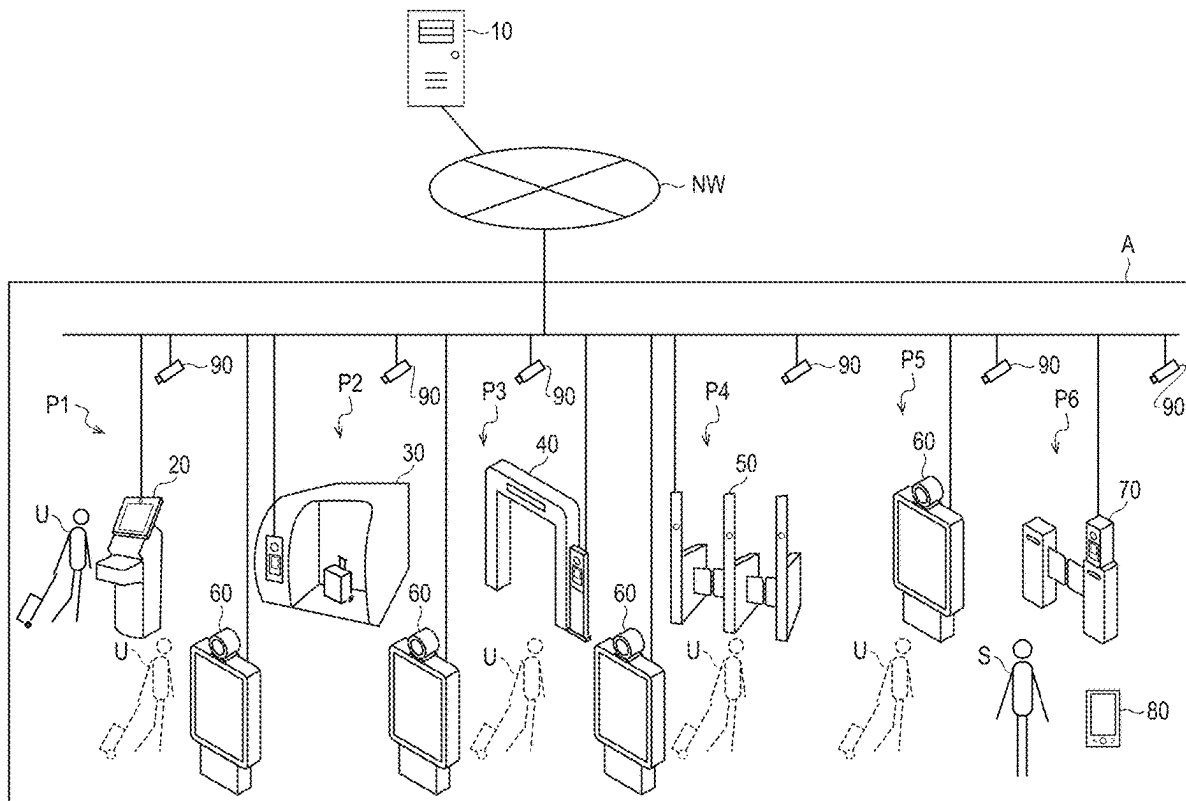


FIG. 1

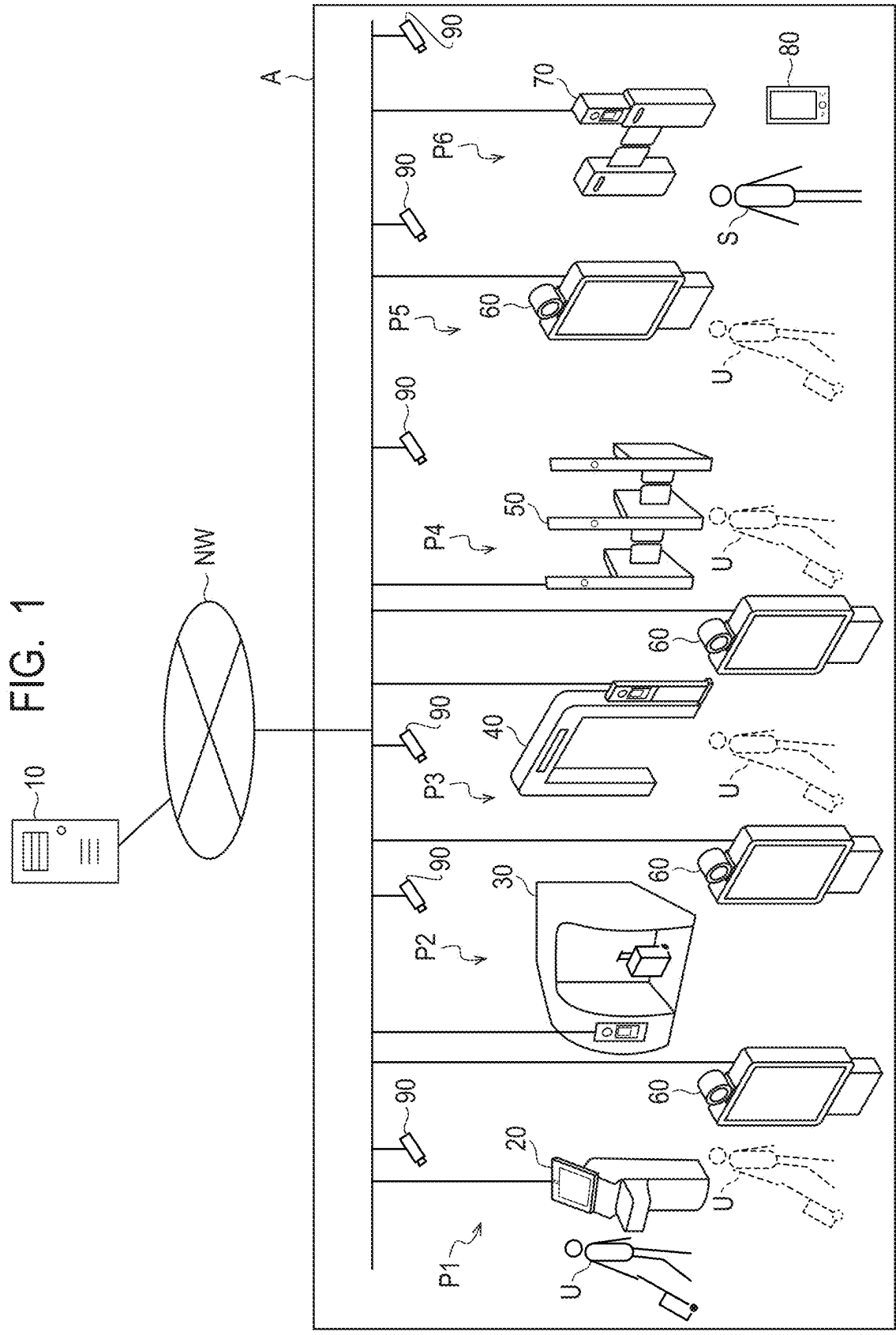


FIG. 2

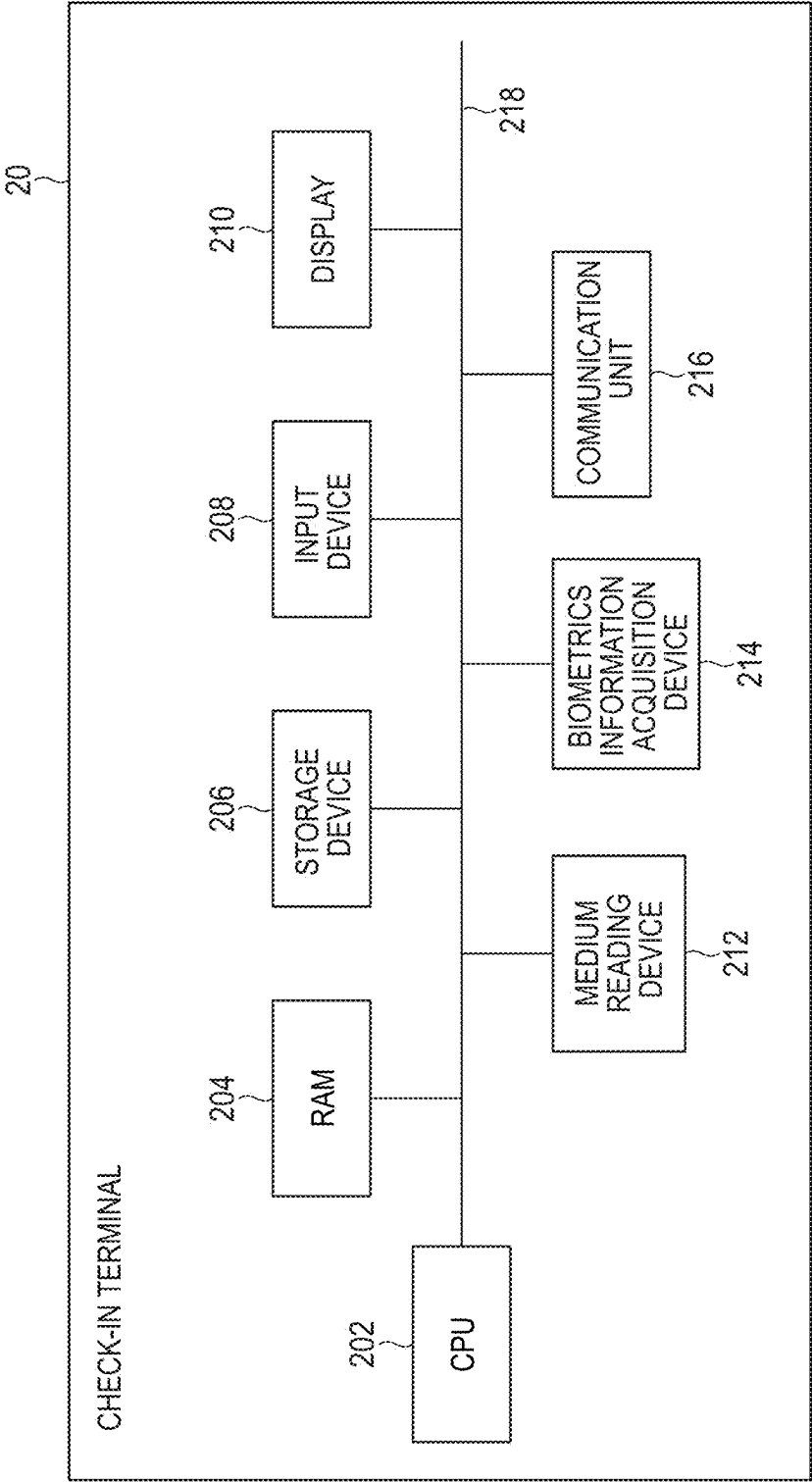


FIG. 3

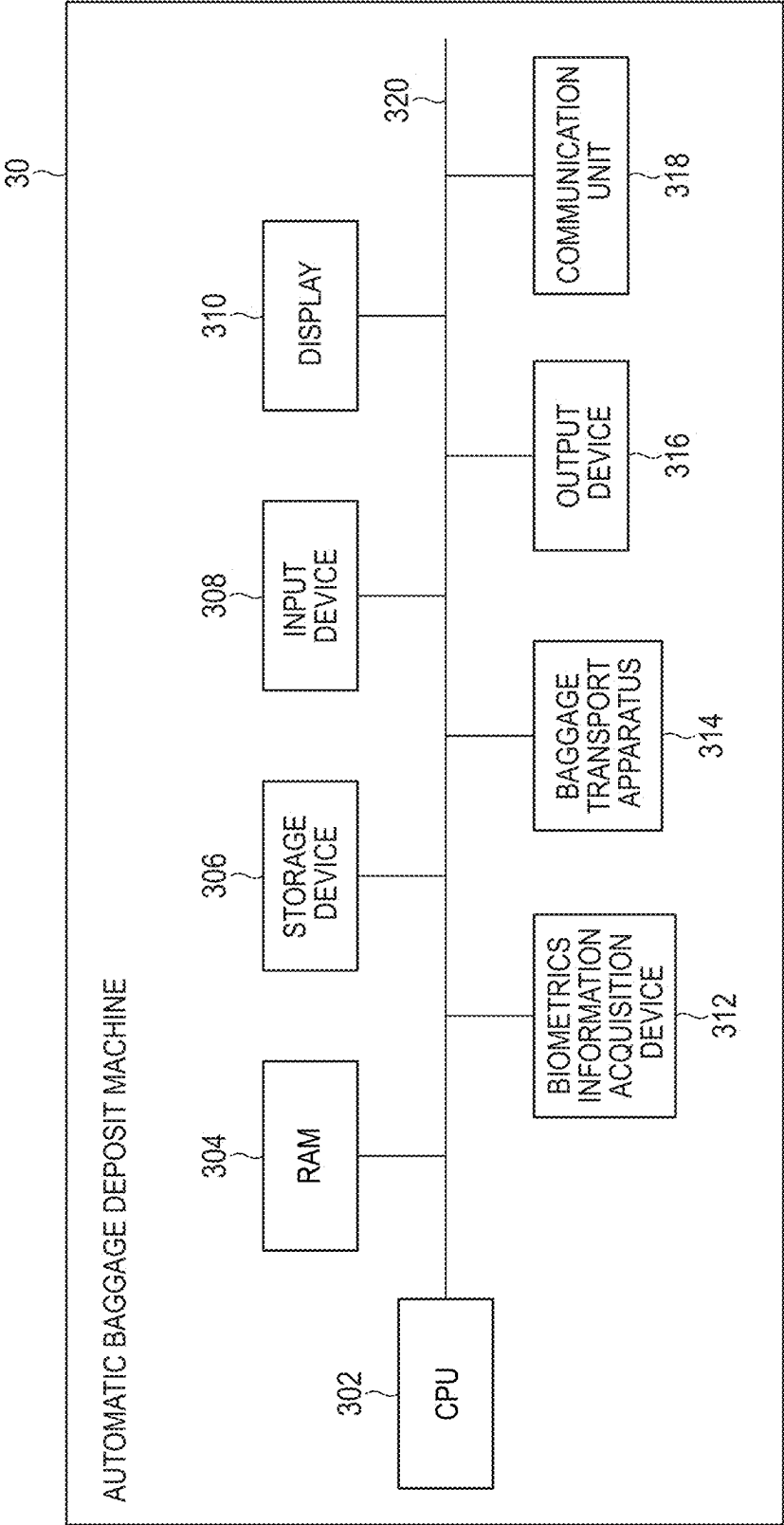


FIG. 4

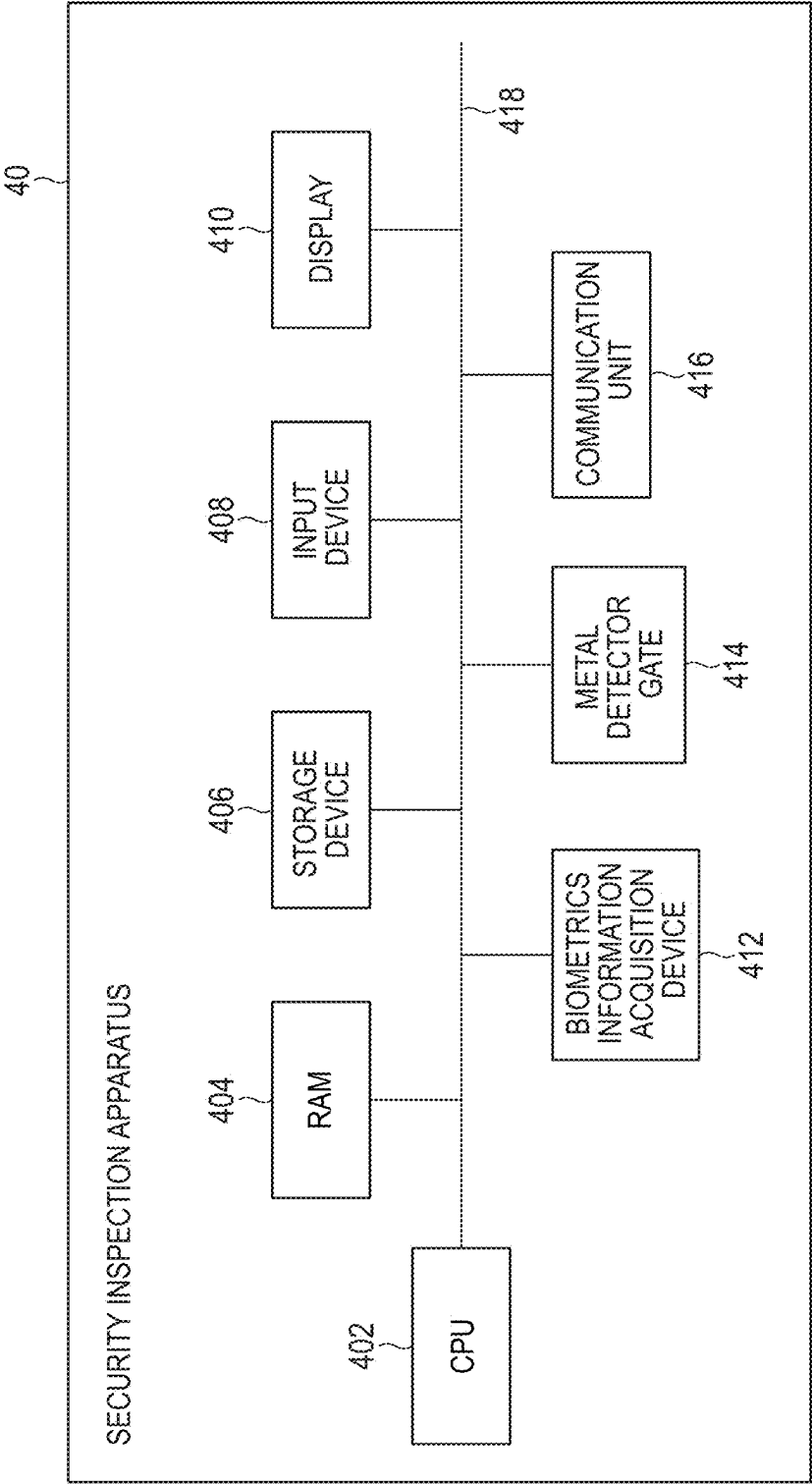


FIG. 5

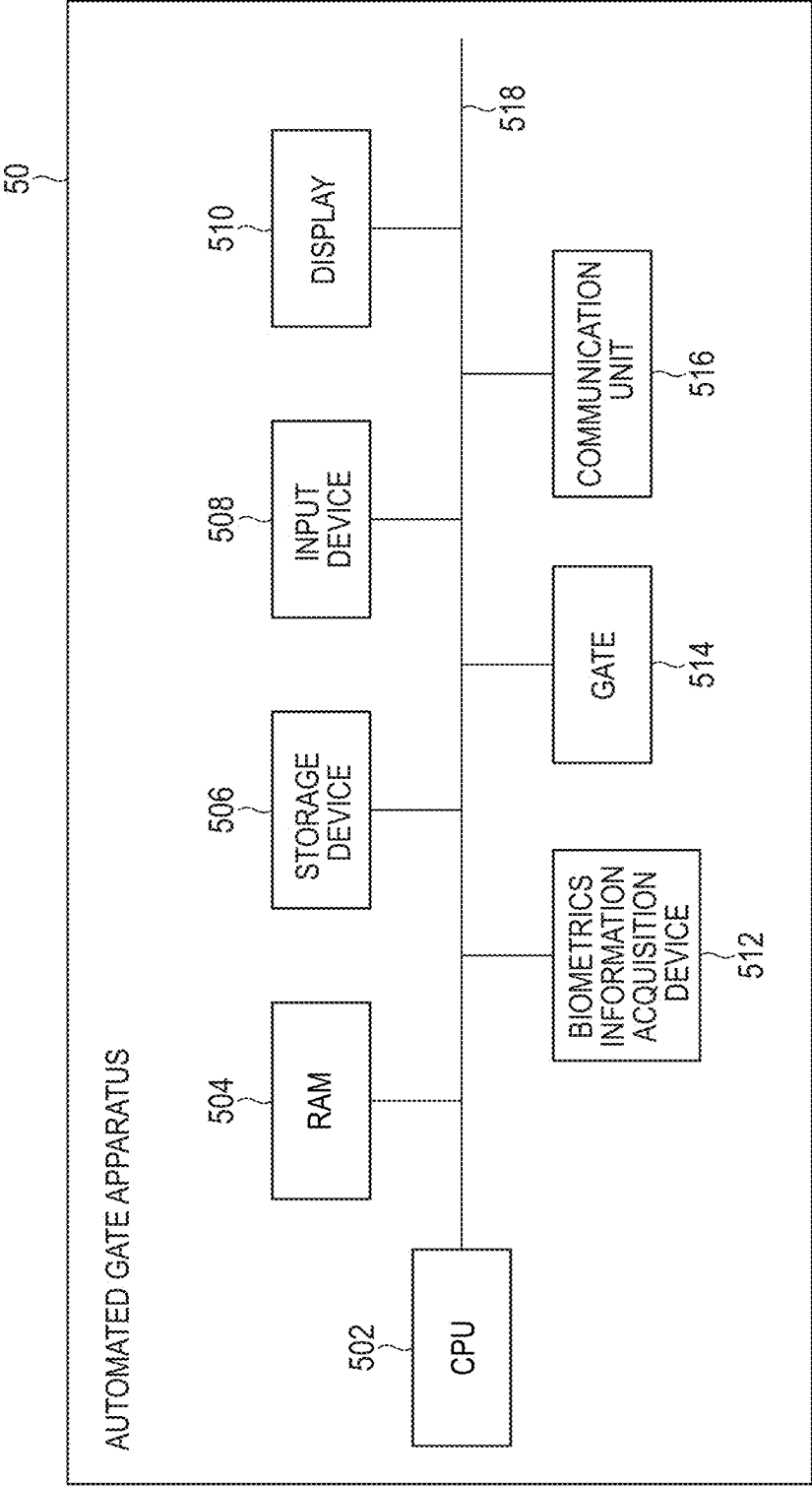


FIG. 6

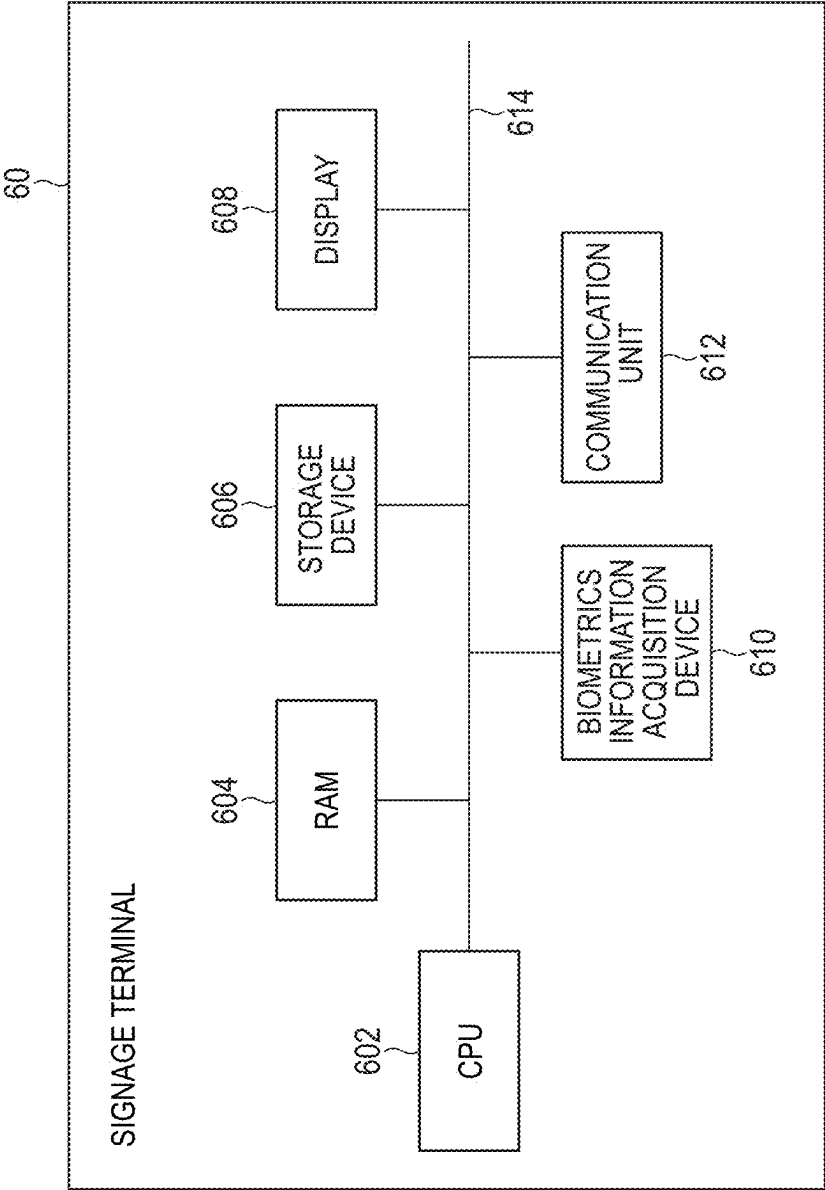


FIG. 7

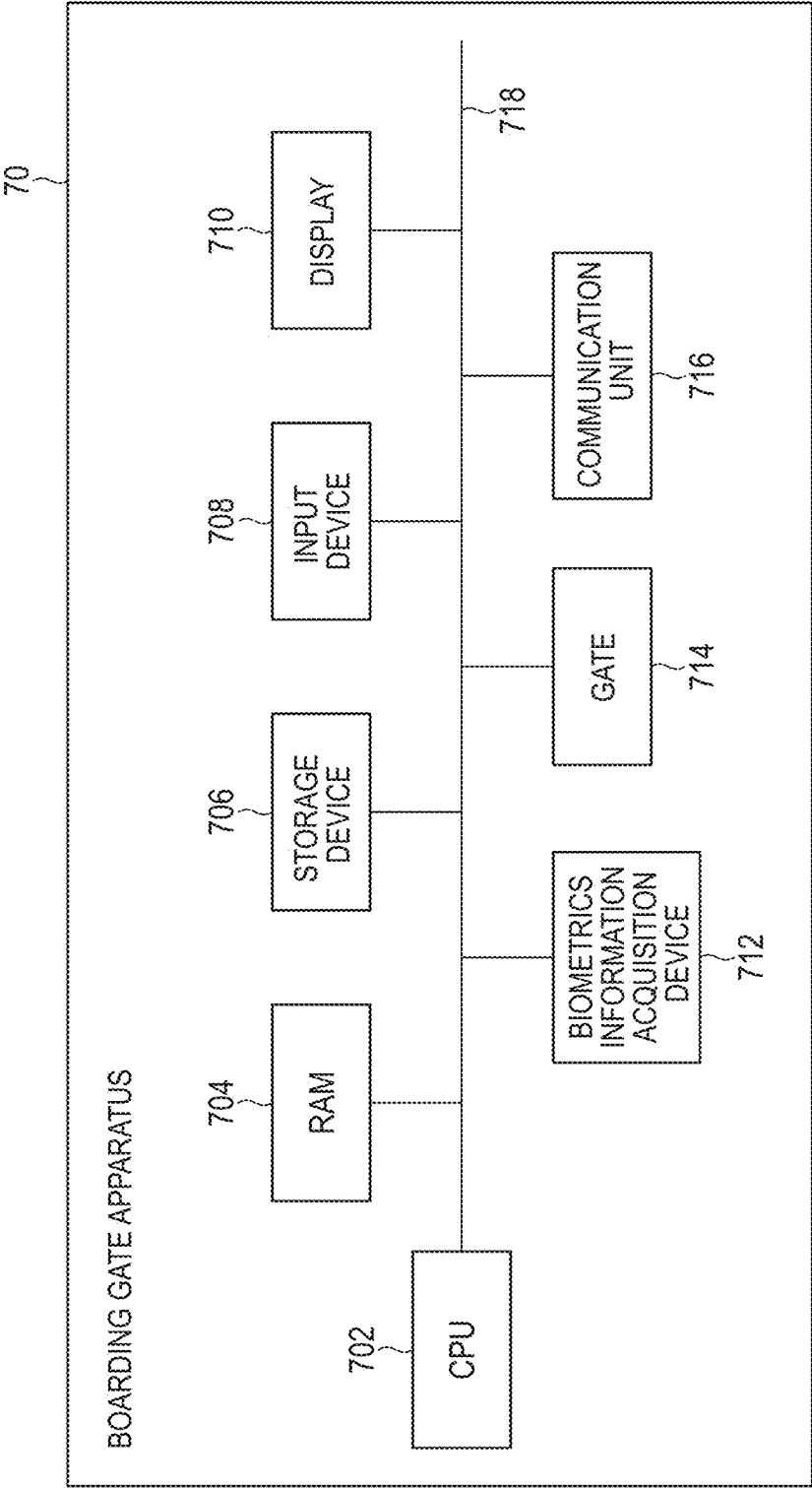




FIG. 8

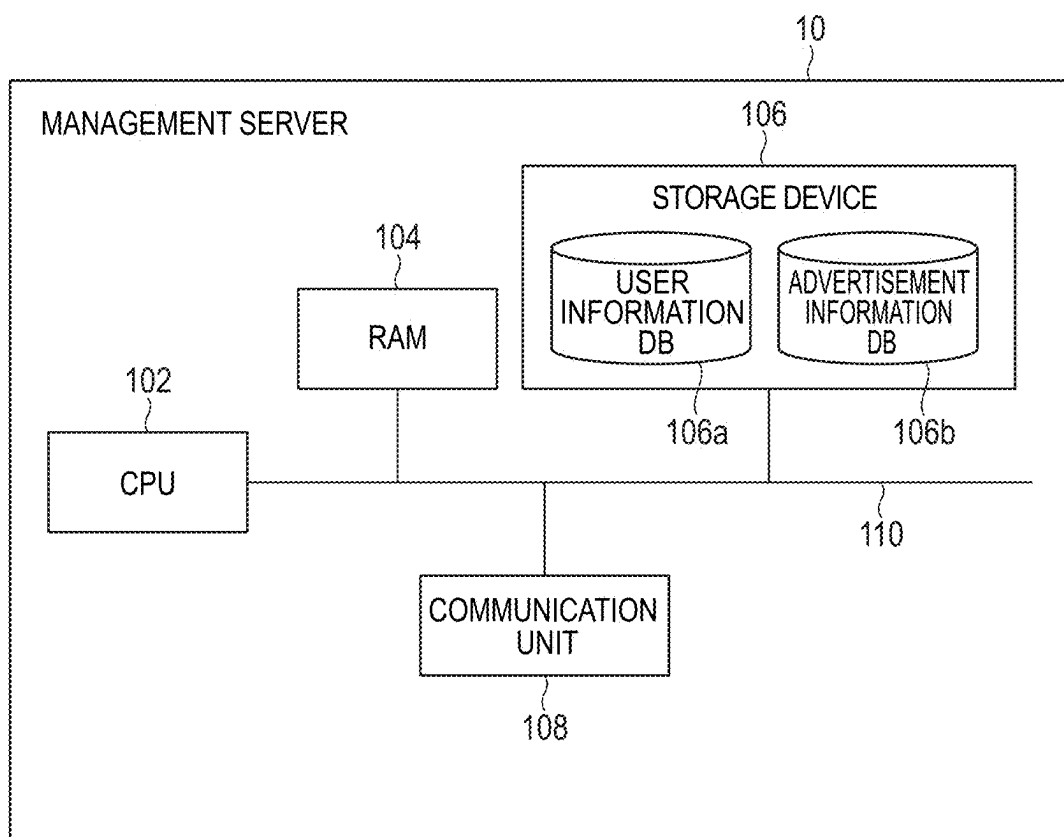


FIG. 9

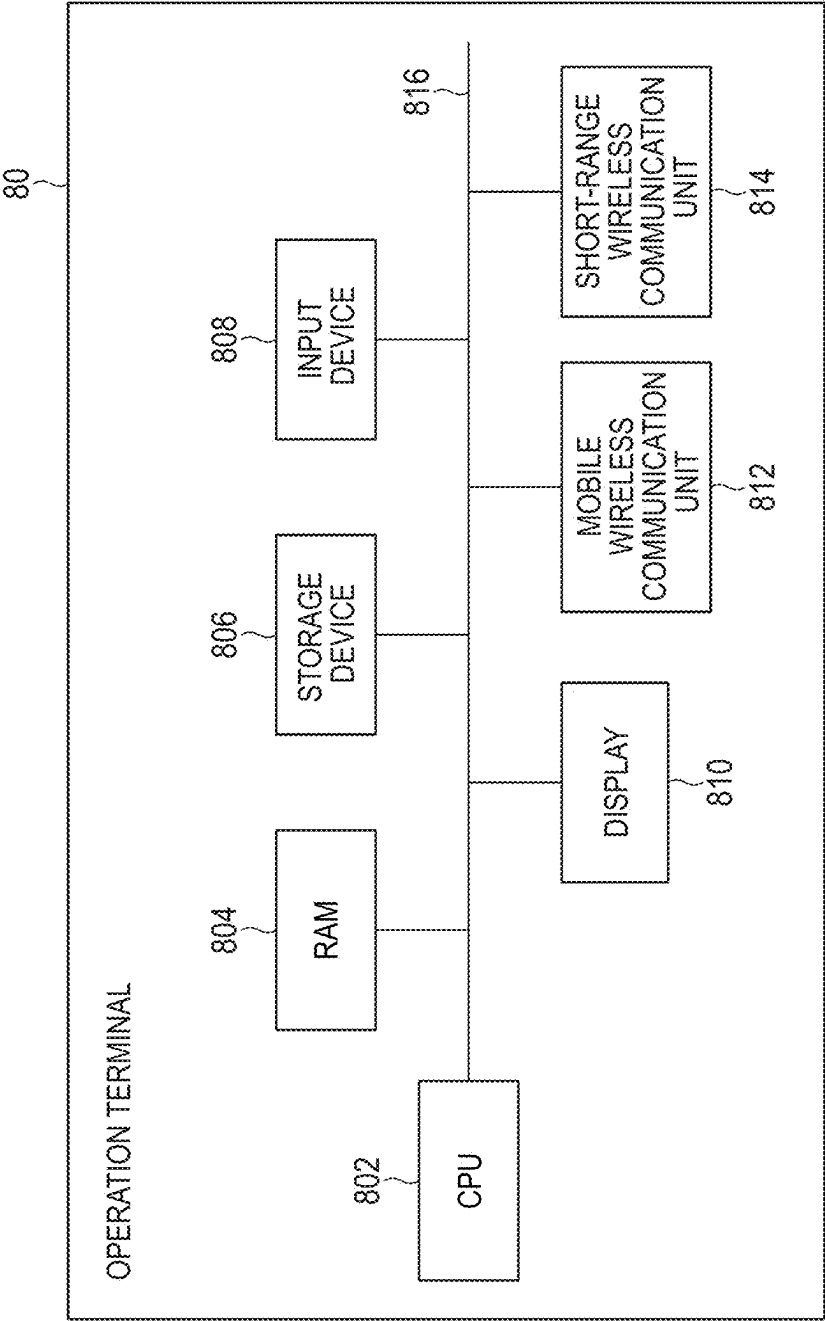


FIG. 10

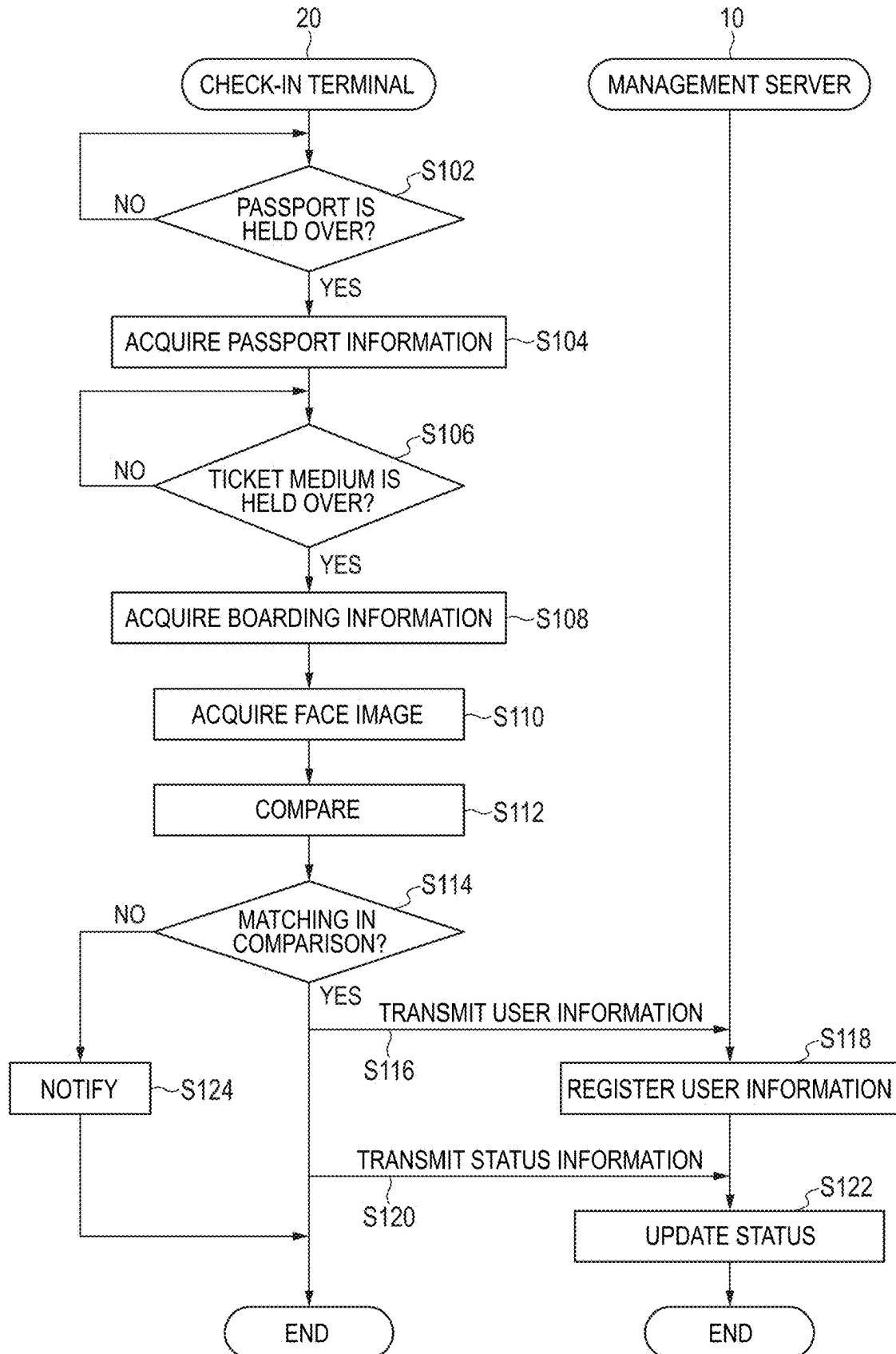


FIG. 11

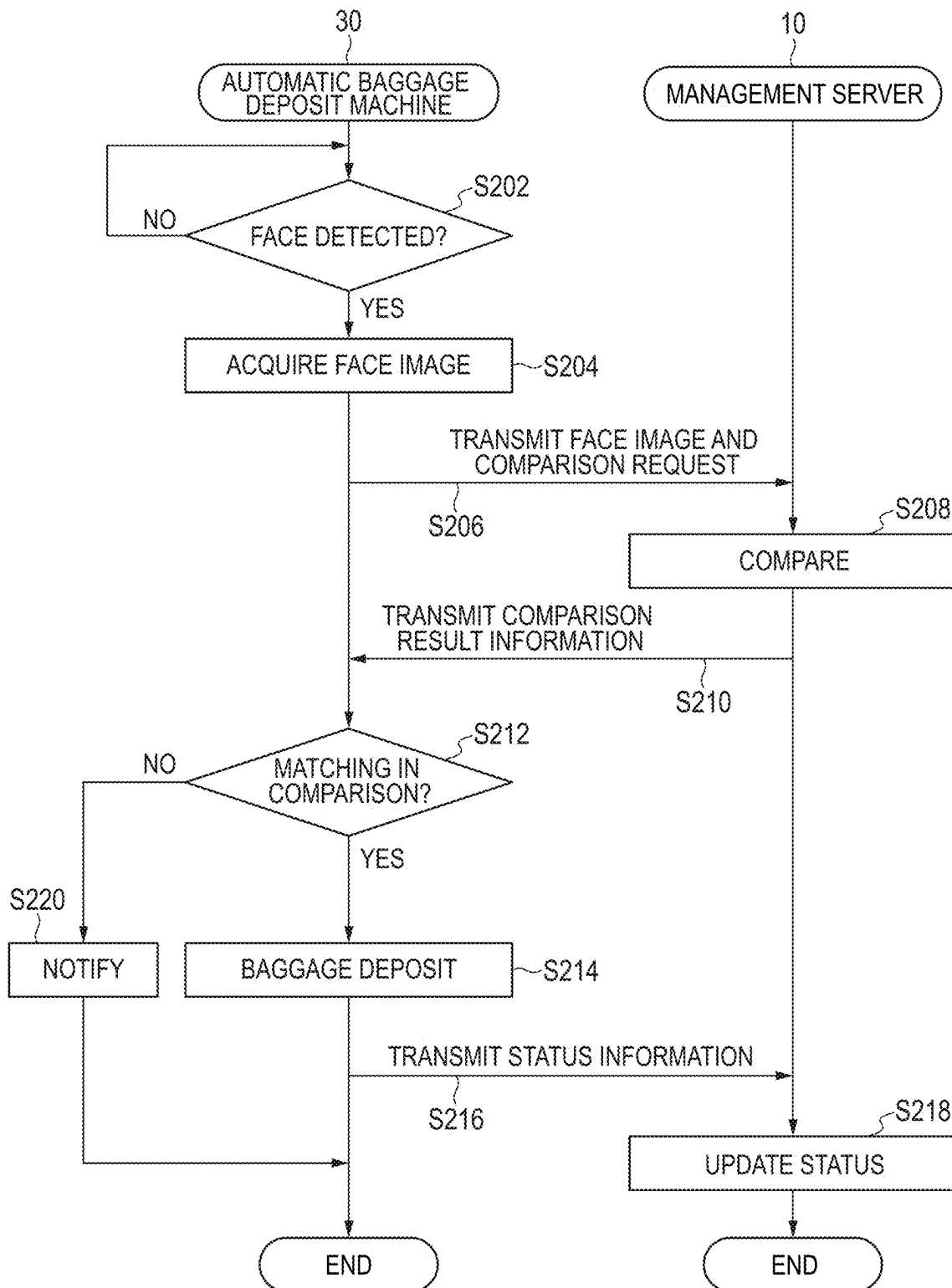


FIG. 12

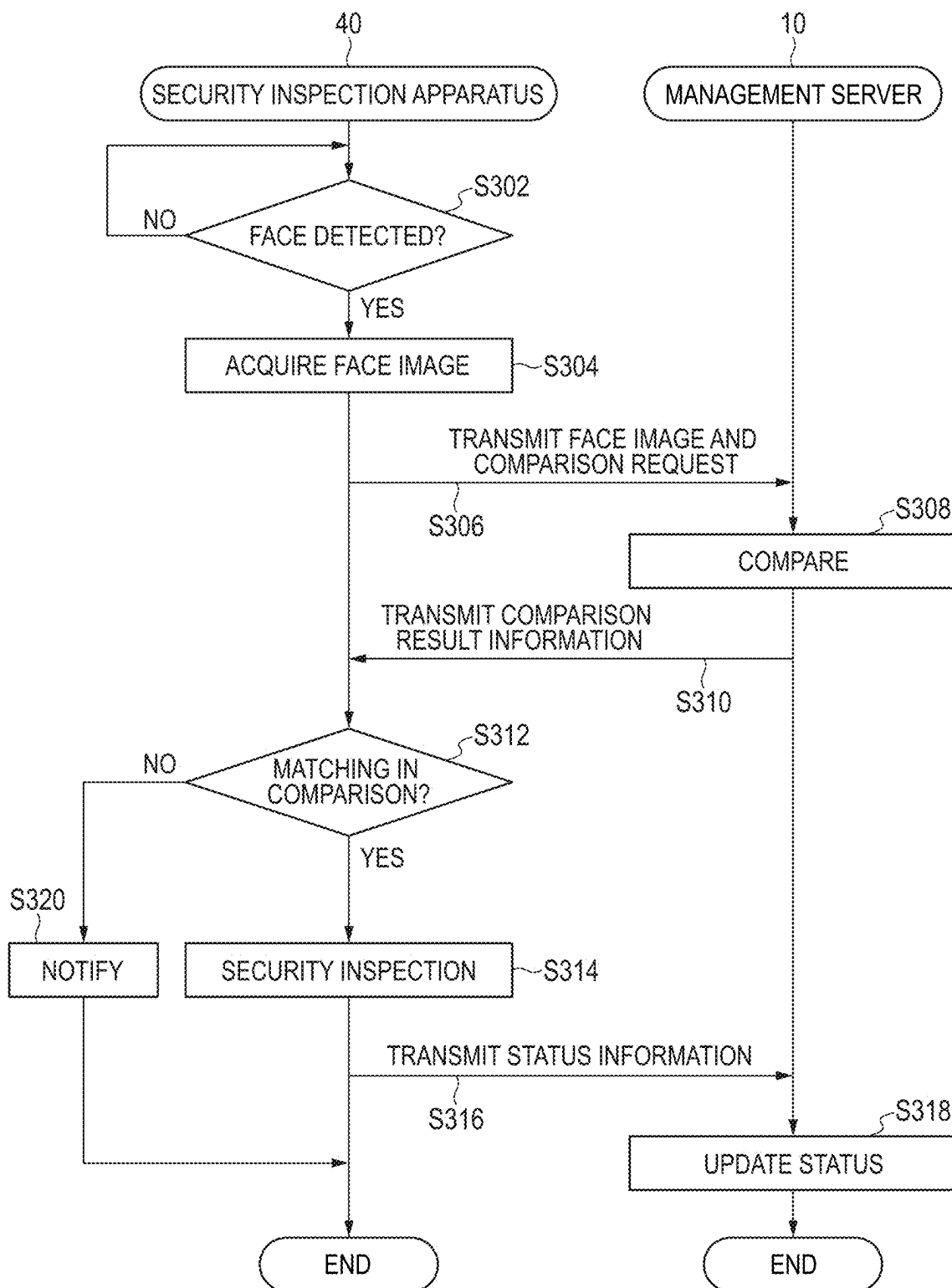


FIG. 13

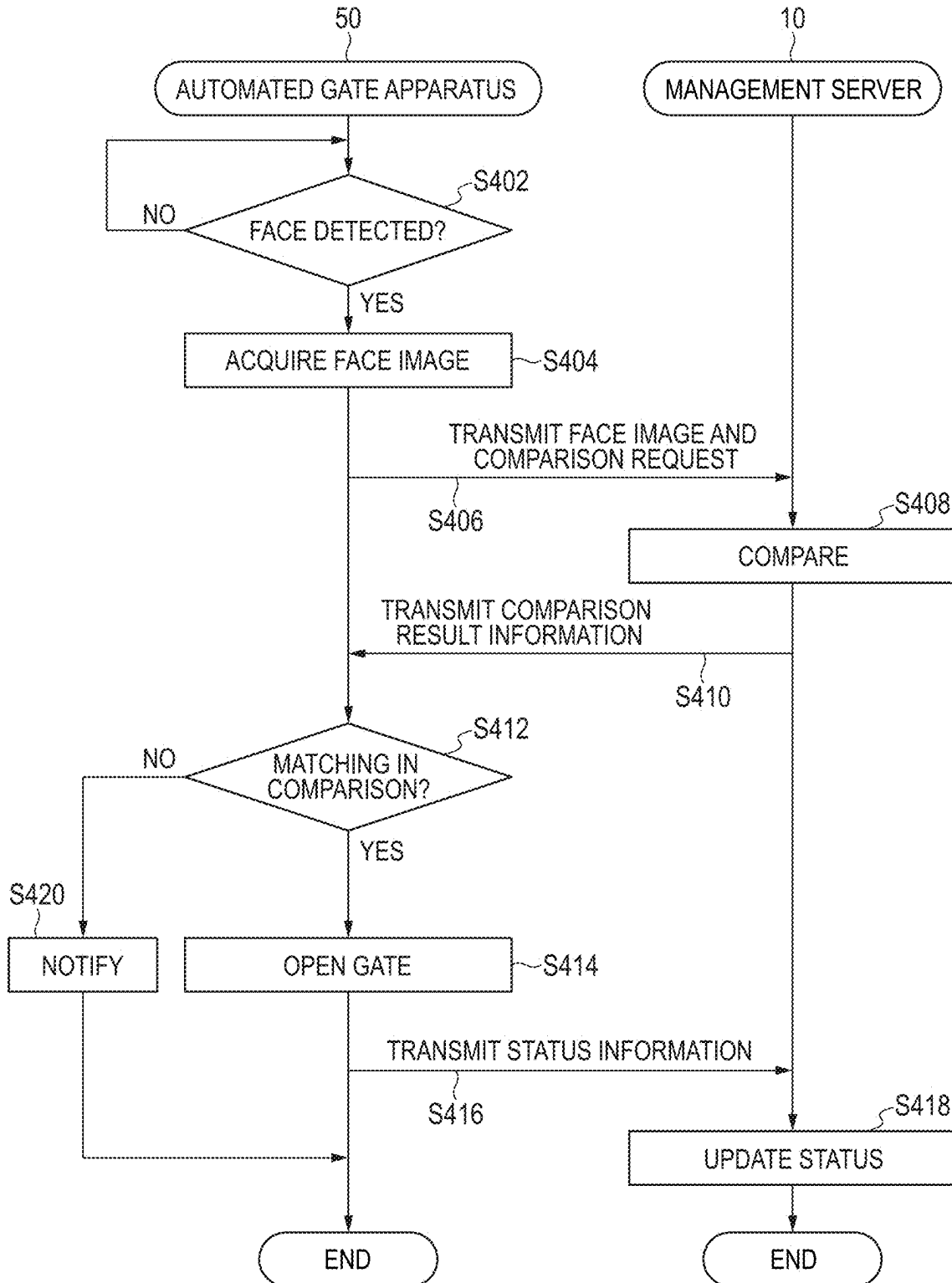


FIG. 14

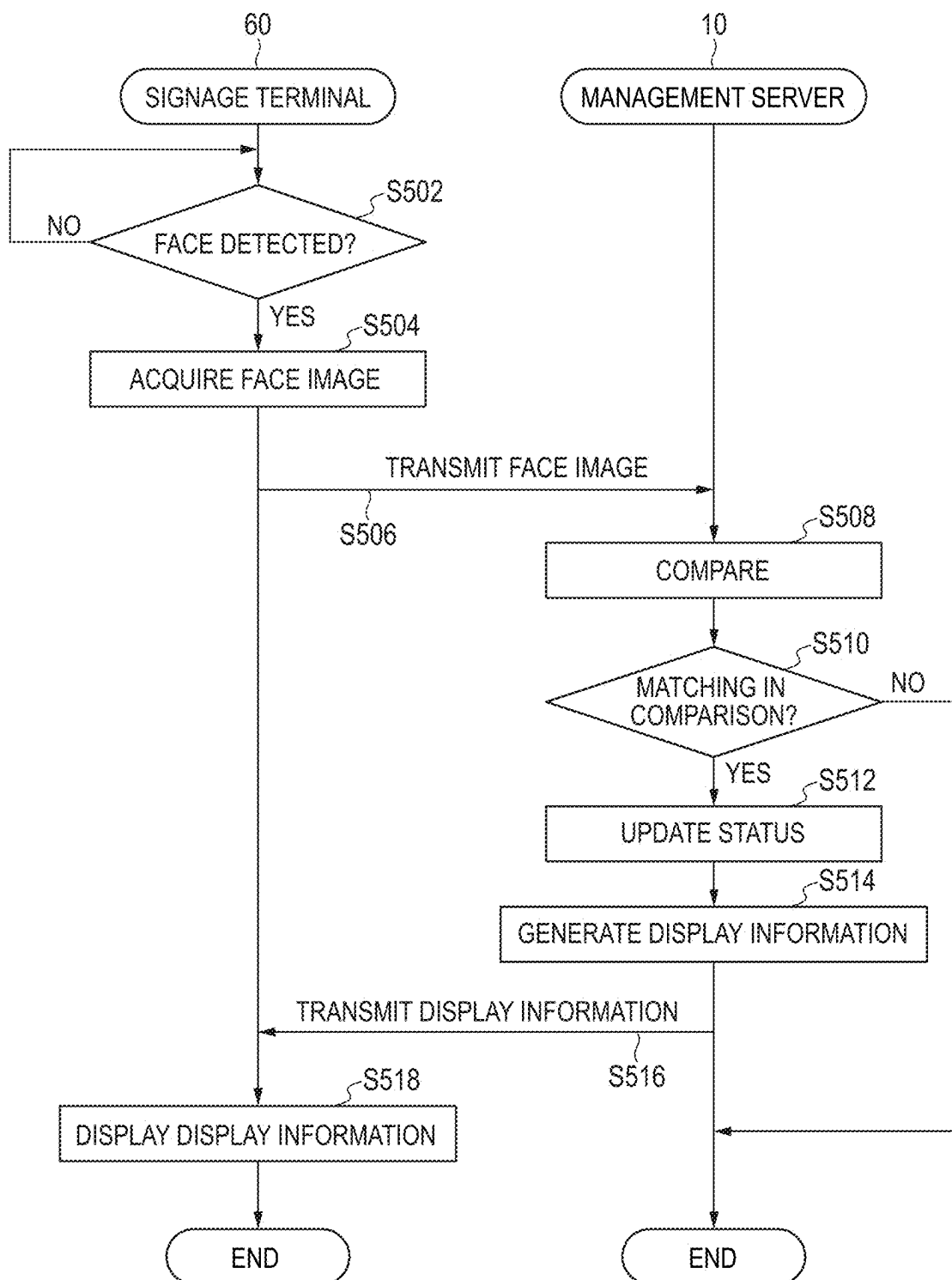


FIG. 15

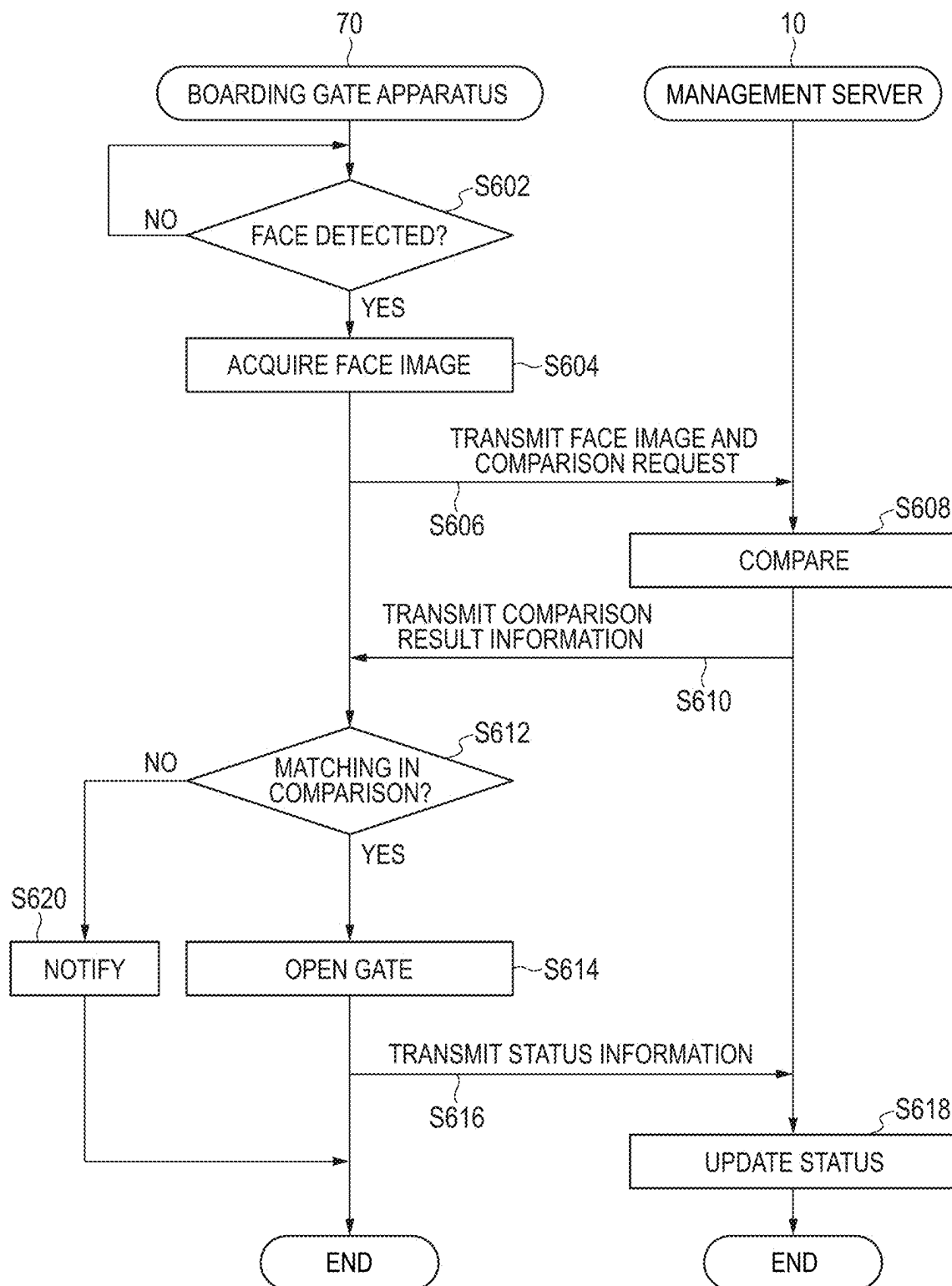






FIG. 17

| User ID | Boarding time | Deadline time             |                               |                       |                                                  |                      |
|---------|---------------|---------------------------|-------------------------------|-----------------------|--------------------------------------------------|----------------------|
|         |               | Baggage deposit procedure | Security inspection procedure | Immigration procedure | Presence or passage in front of signage terminal | Boarding on airplane |
| 10101   | 5:50          | 4:50                      | 5:20                          | 5:30                  | 5:35                                             | 5:40                 |
| 10102   | 6:00          | 5:00                      | 5:30                          | 5:40                  | 5:45                                             | 5:50                 |
| 10103   | 6:10          | 5:10                      | 5:40                          | 5:50                  | 5:55                                             | 6:00                 |
| 10104   | 6:20          | 5:20                      | 5:50                          | 6:00                  | 6:05                                             | 6:10                 |
| 10105   | 6:30          | 5:30                      | 6:00                          | 6:10                  | 6:15                                             | 6:20                 |
| 10106   | 6:40          | 5:40                      | 6:10                          | 6:20                  | 6:25                                             | 6:30                 |
| 10107   | 6:50          | 5:50                      | 6:20                          | 6:30                  | 6:35                                             | 6:40                 |
| 10108   | 7:00          | 6:00                      | 6:30                          | 6:40                  | 6:45                                             | 6:50                 |
| 10109   | 7:10          | 6:10                      | 6:40                          | 6:50                  | 6:55                                             | 7:00                 |
| 10110   | 7:20          | 6:20                      | 6:50                          | 7:00                  | 7:05                                             | 7:10                 |
| 10111   | 7:30          | 6:30                      | 7:00                          | 7:10                  | 7:15                                             | 7:20                 |
| 10112   | 7:40          | 6:40                      | 7:10                          | 7:20                  | 7:25                                             | 7:30                 |
| ⋮       |               |                           |                               |                       |                                                  |                      |

FIG. 18

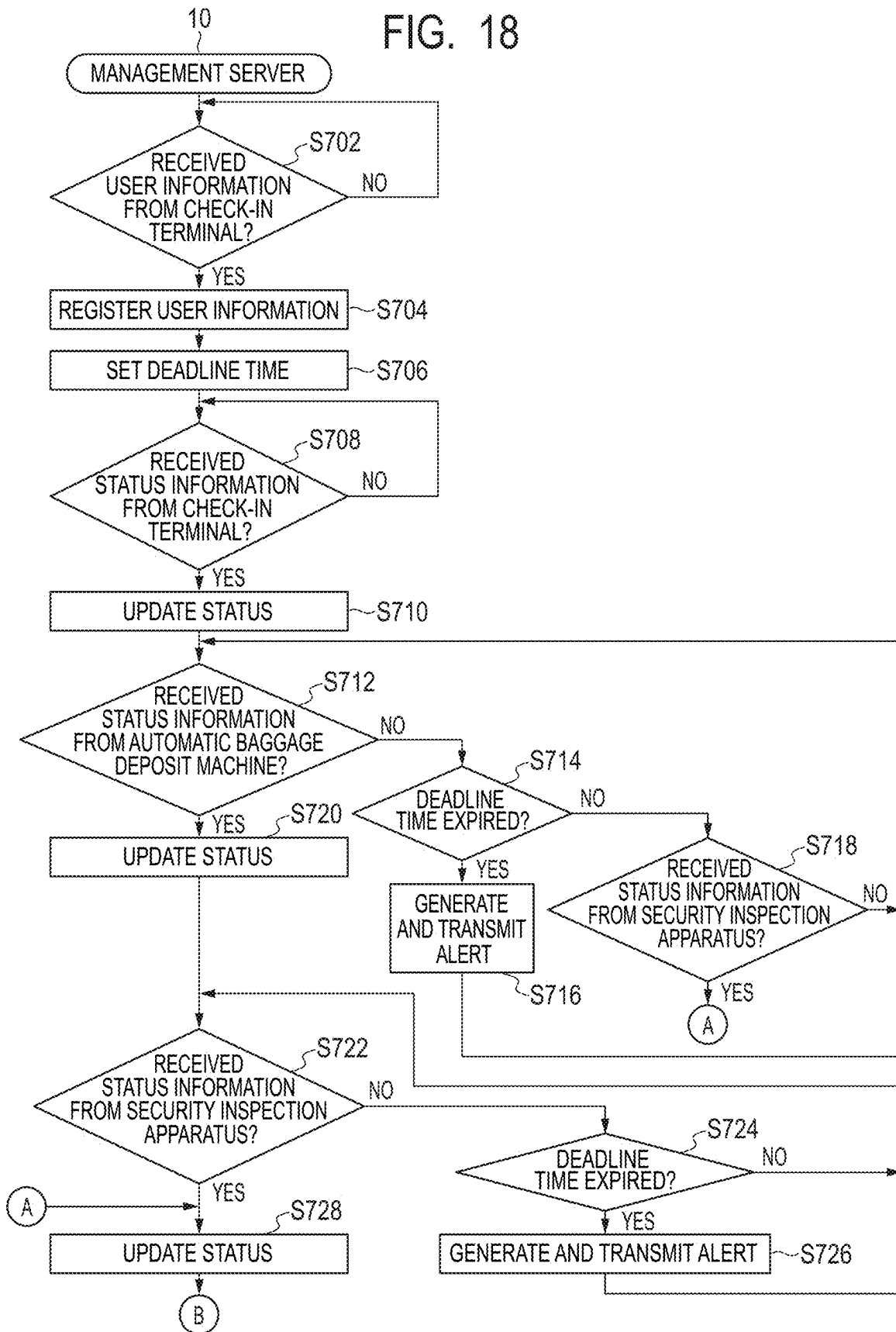


FIG. 19

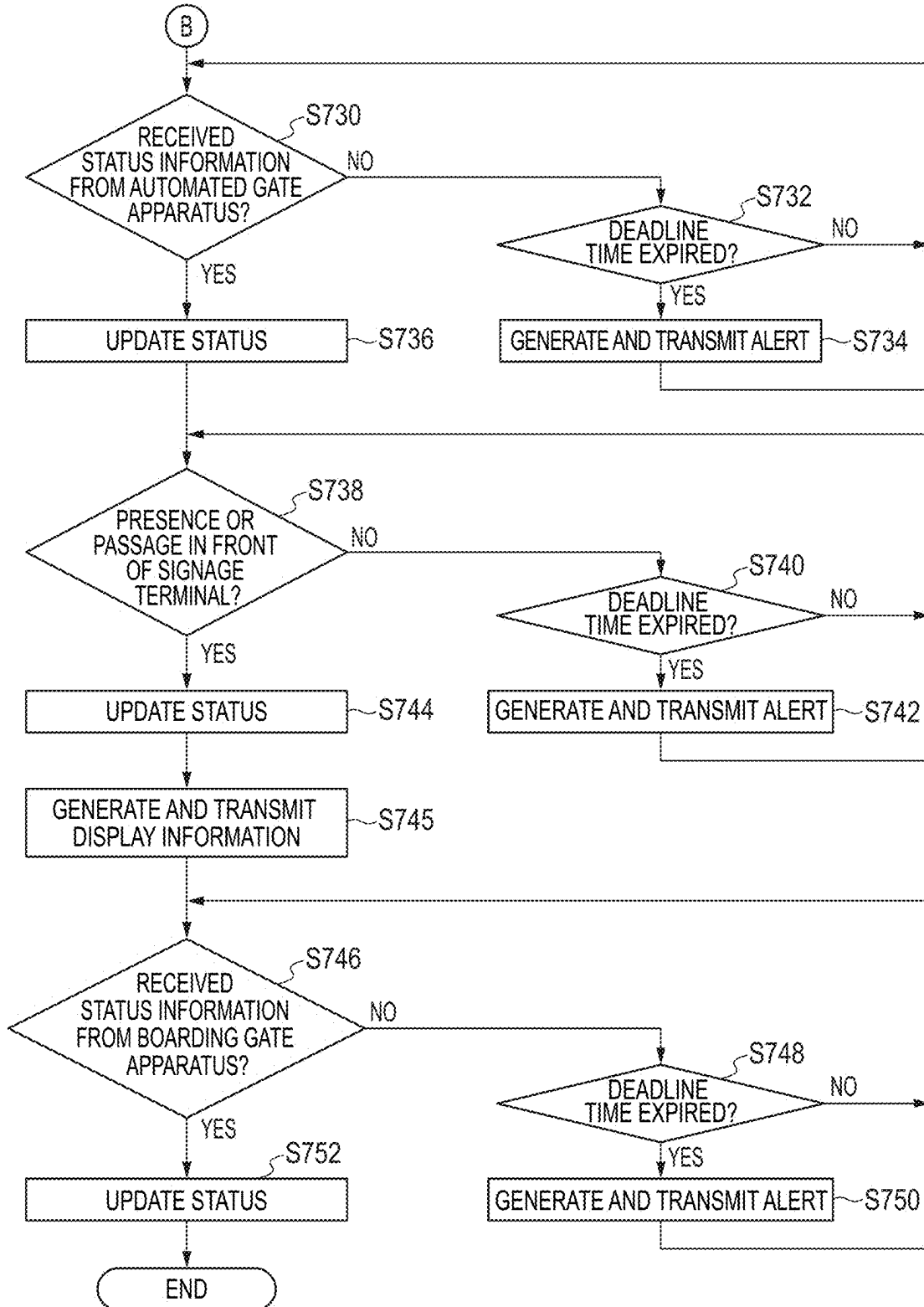


FIG. 20A

AL

User ID: 10105

Name: EEE eee

Nationality: JPN

Flight number: AB105

AL10

AL14

AL12

|                                                  |              |
|--------------------------------------------------|--------------|
| Check-in procedure                               | : Completed  |
| Baggage deposit procedure                        | : Completed  |
| Security inspection procedure                    | : Unfinished |
| Immigration procedure                            | : Unfinished |
| Presence or passage in front of signage terminal | : Unfinished |
| Boarding on airplane                             | : Unfinished |

Deadline time has expired

FIG. 20B

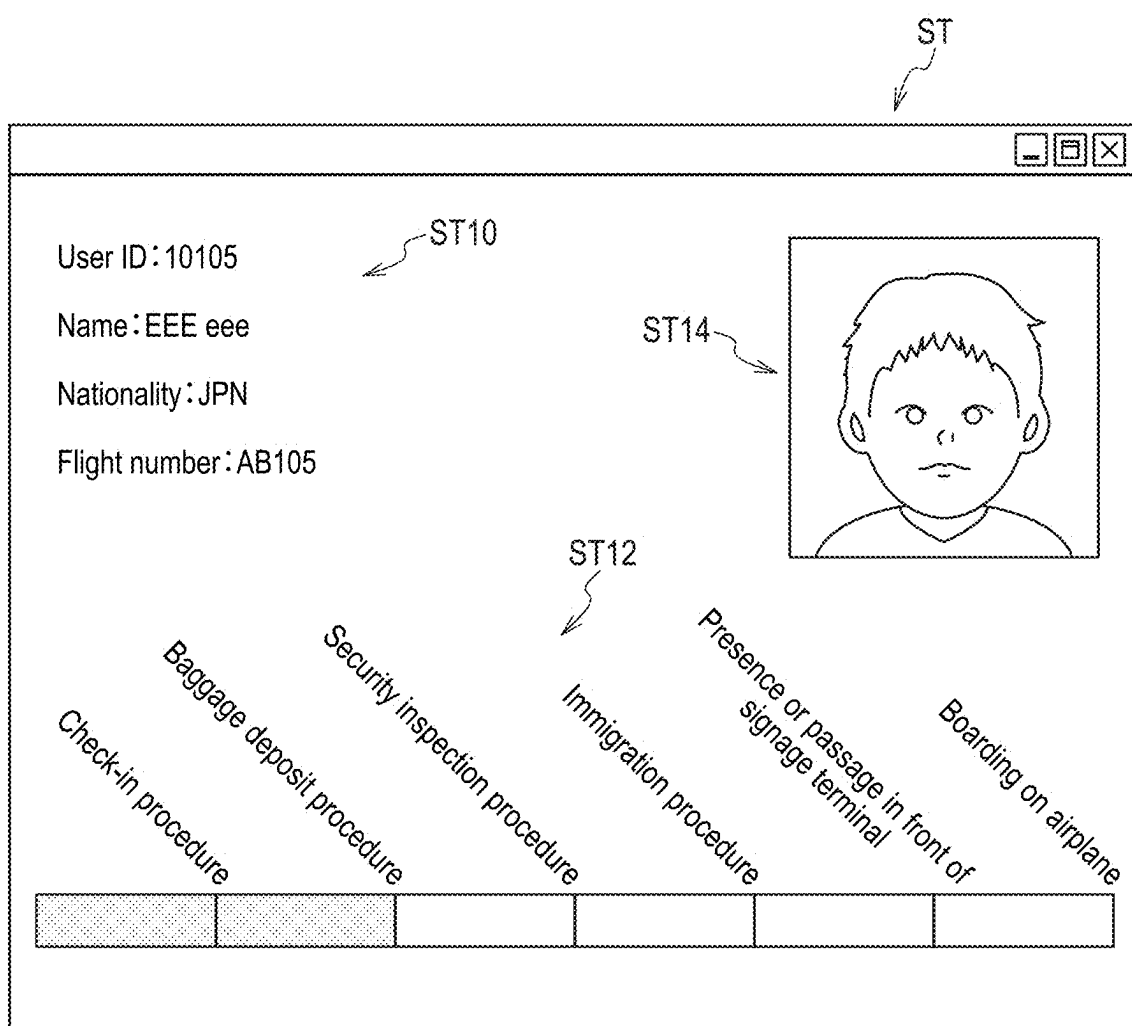


FIG. 20C

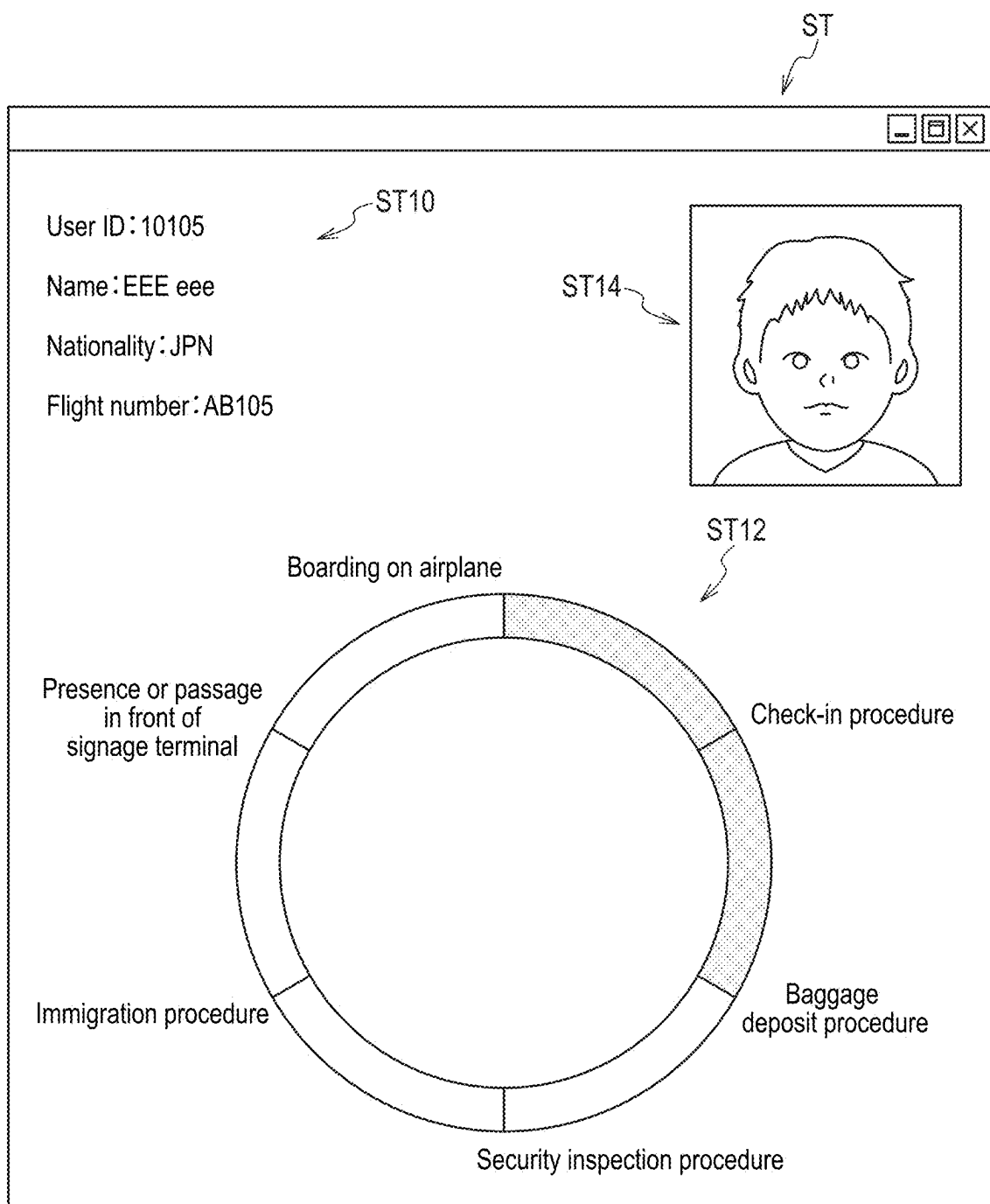


FIG. 21

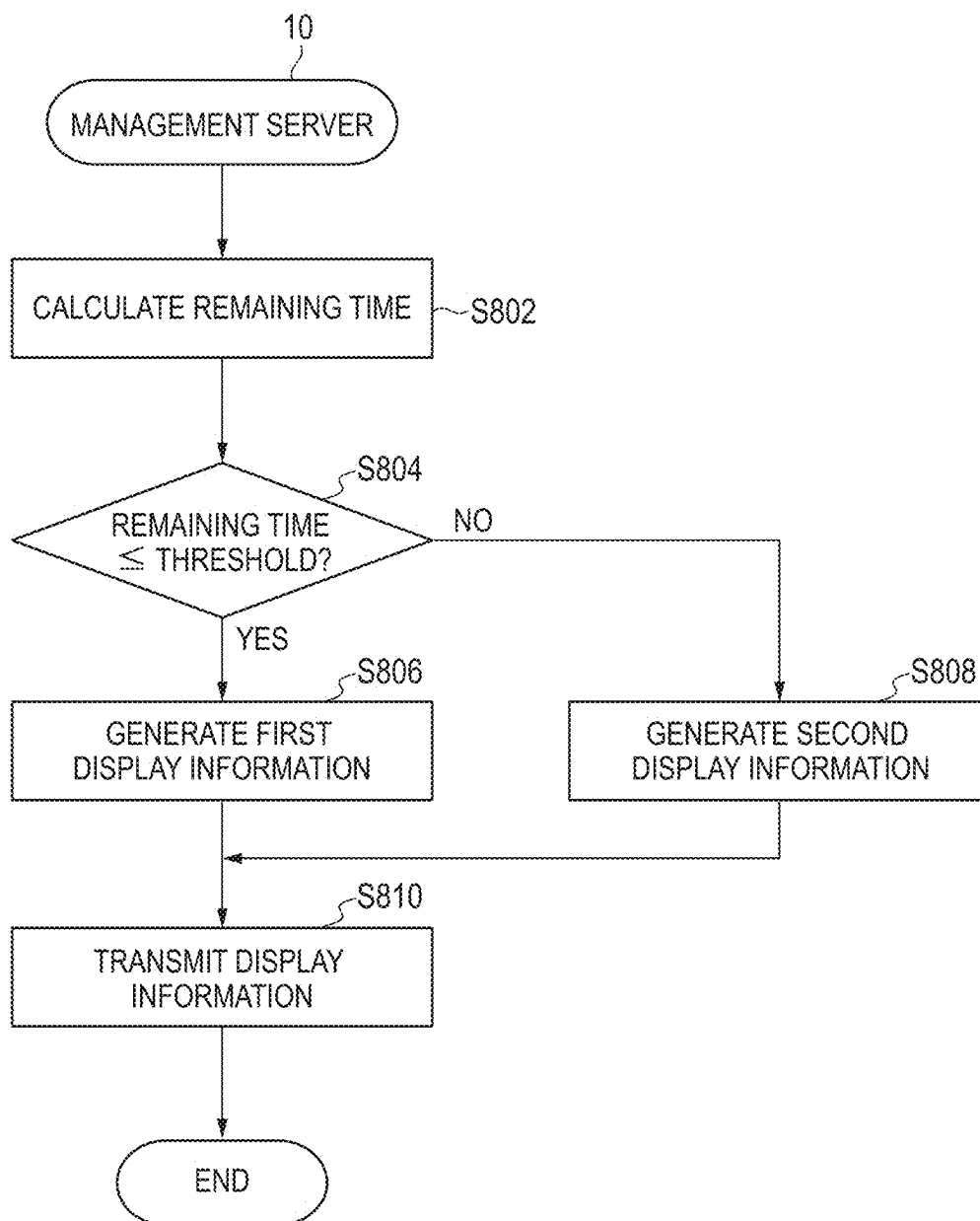




FIG. 22

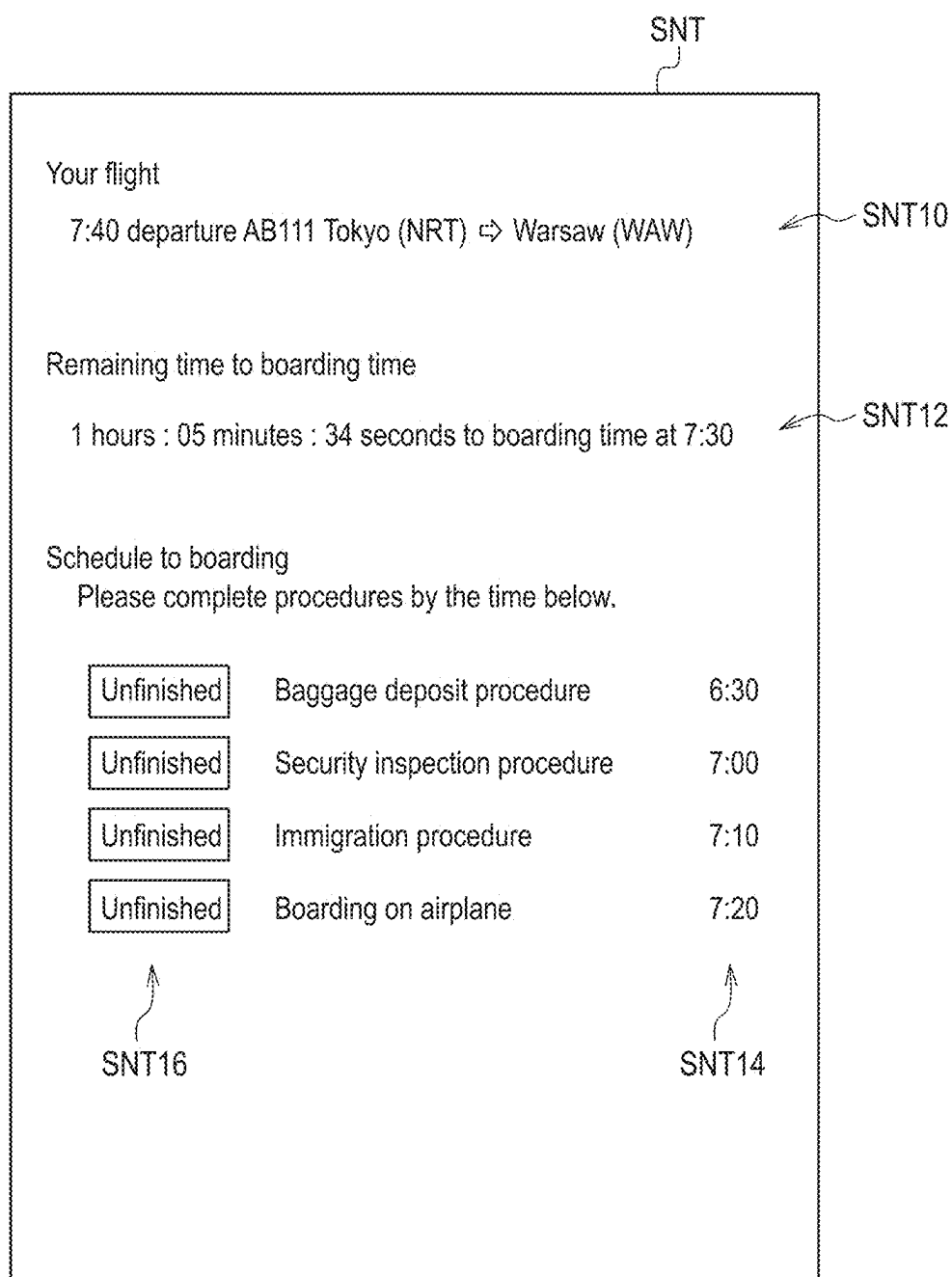


FIG. 23

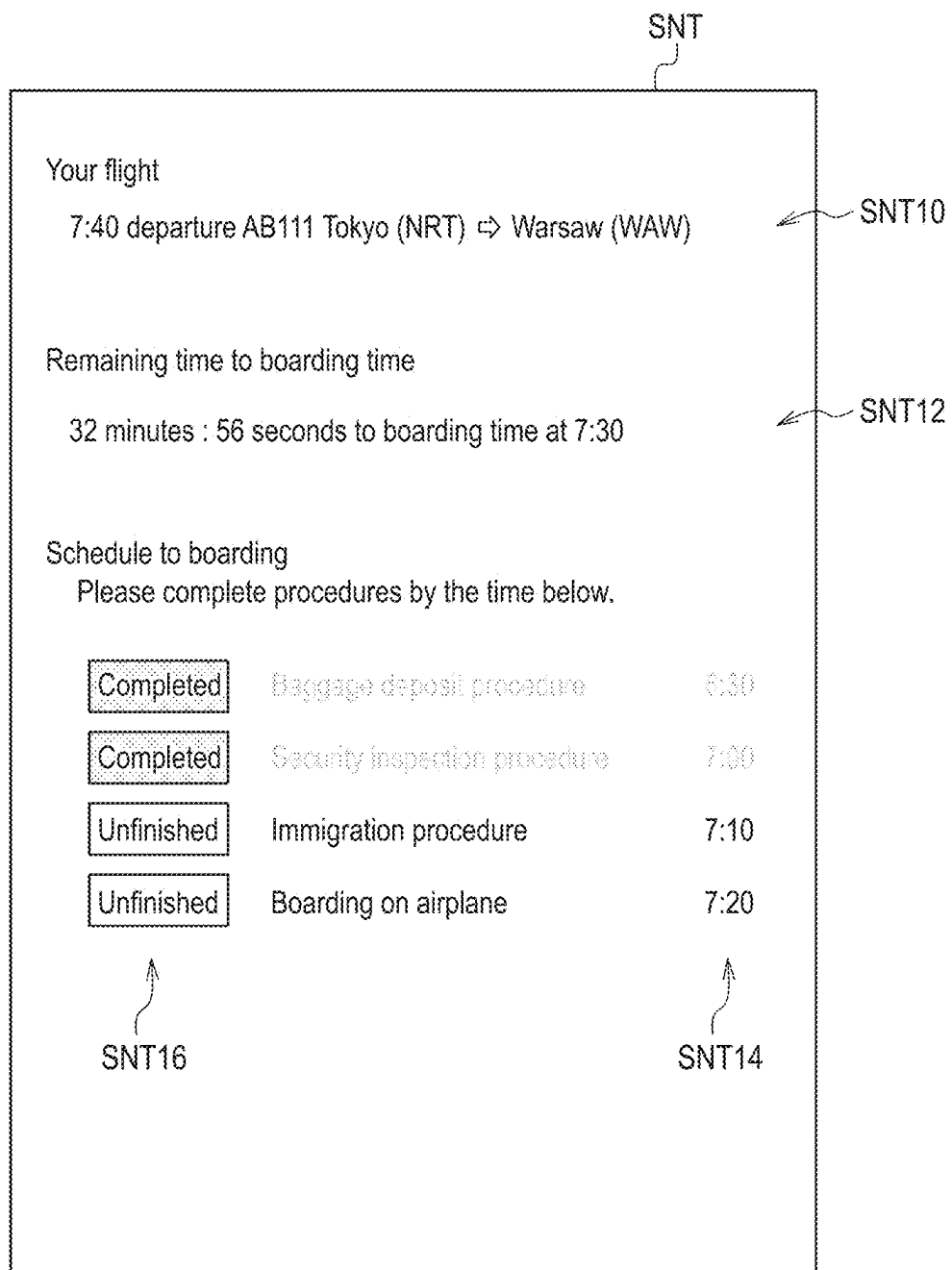


FIG. 24

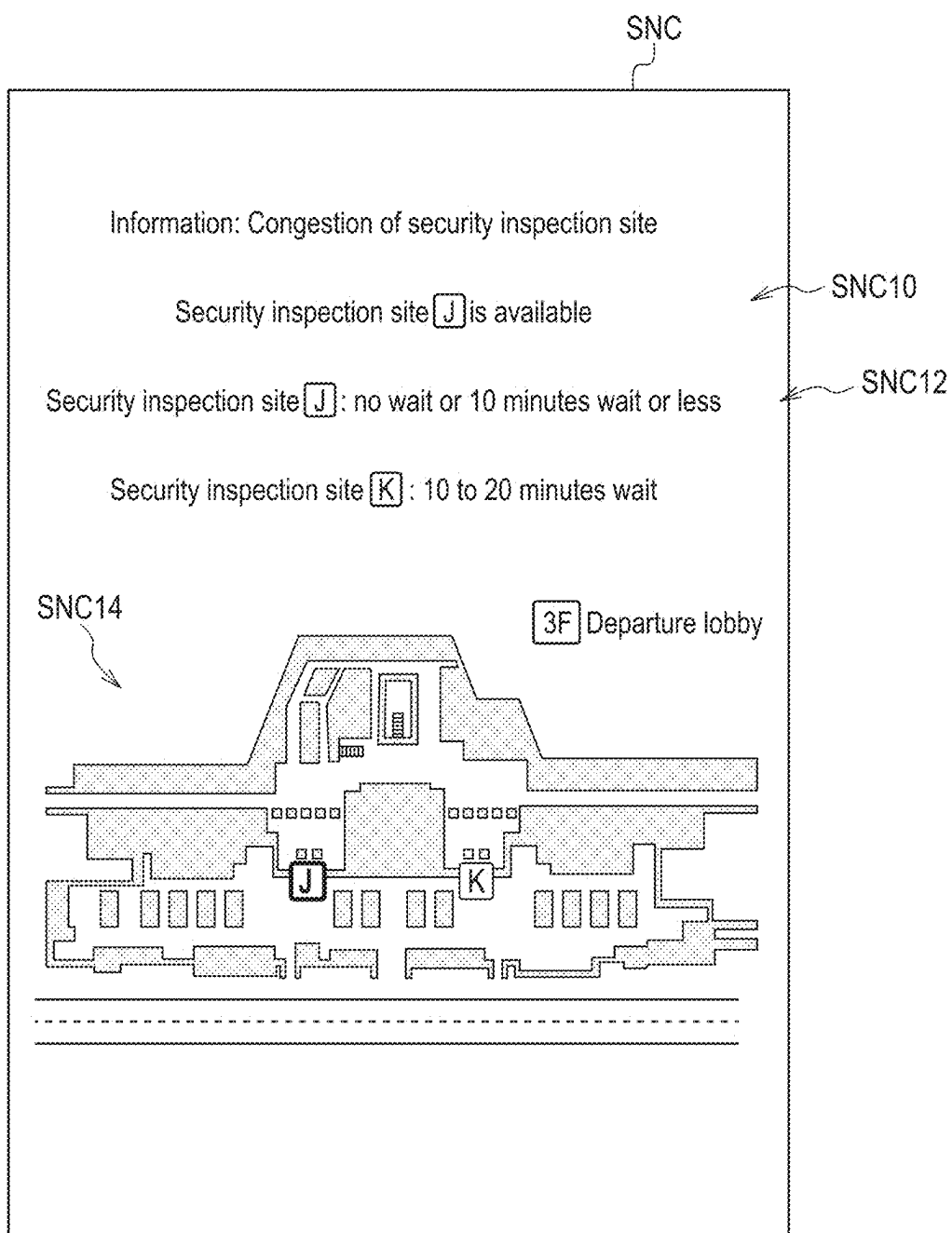


FIG. 25

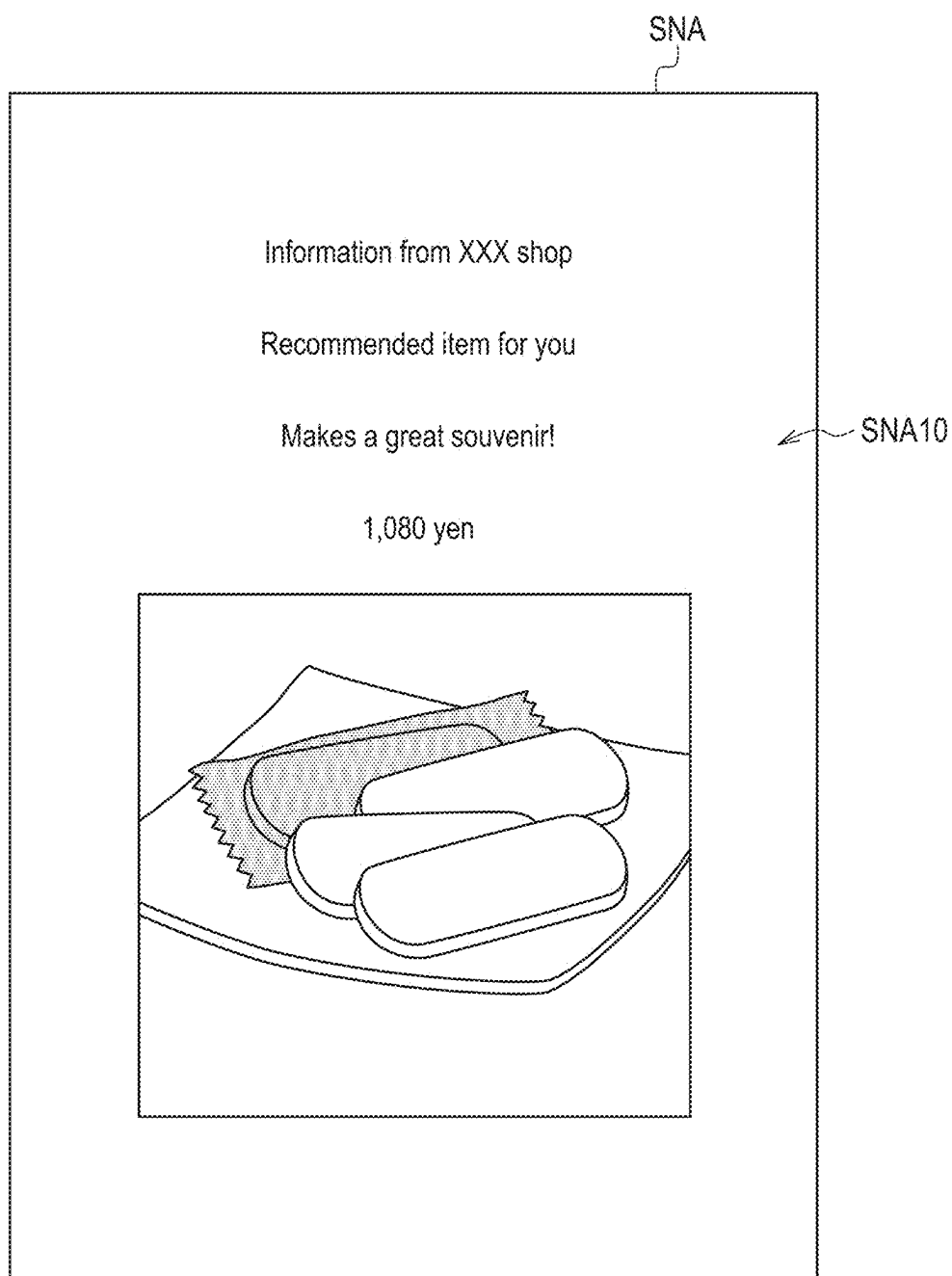


FIG. 26

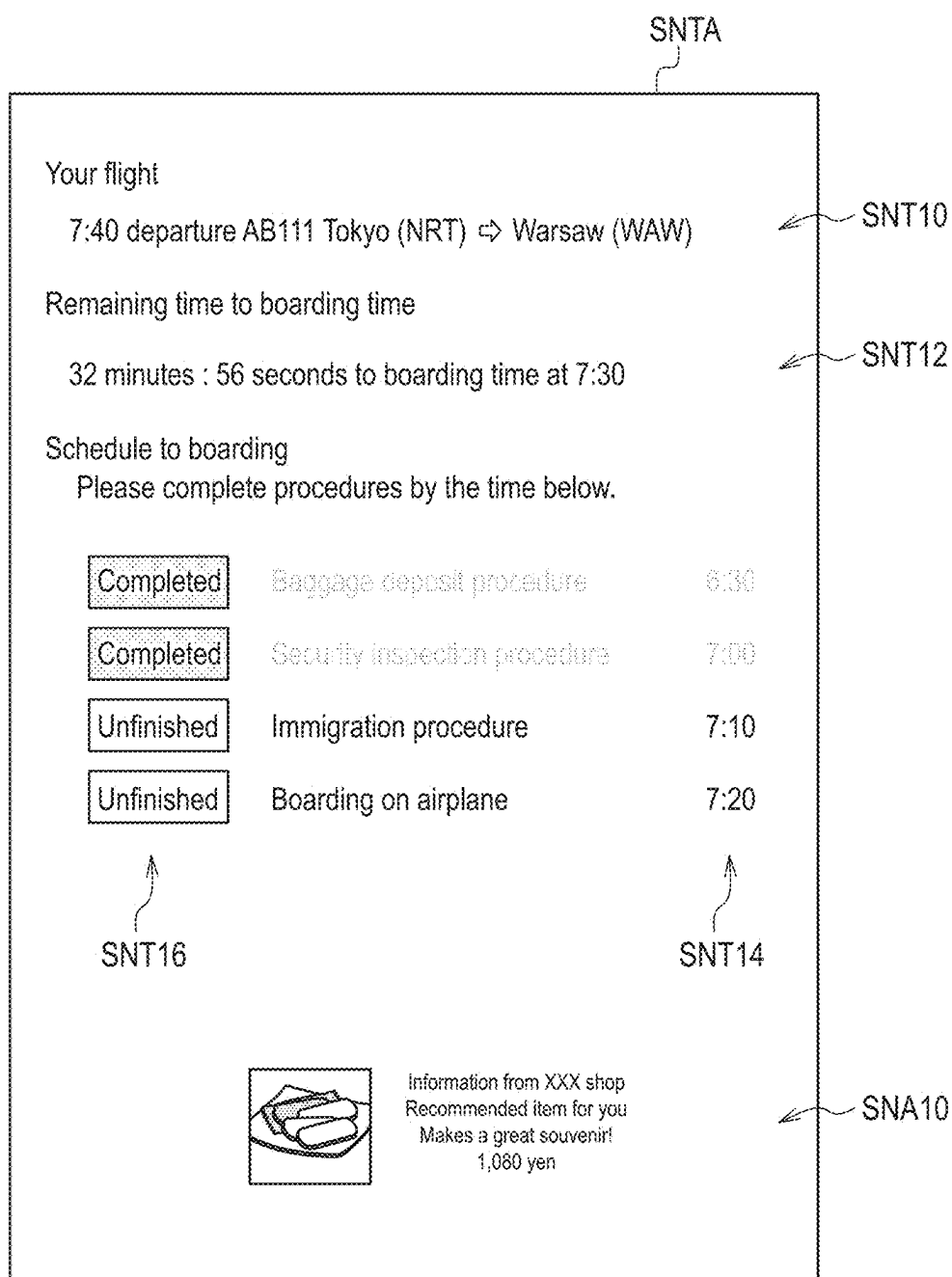


FIG. 27

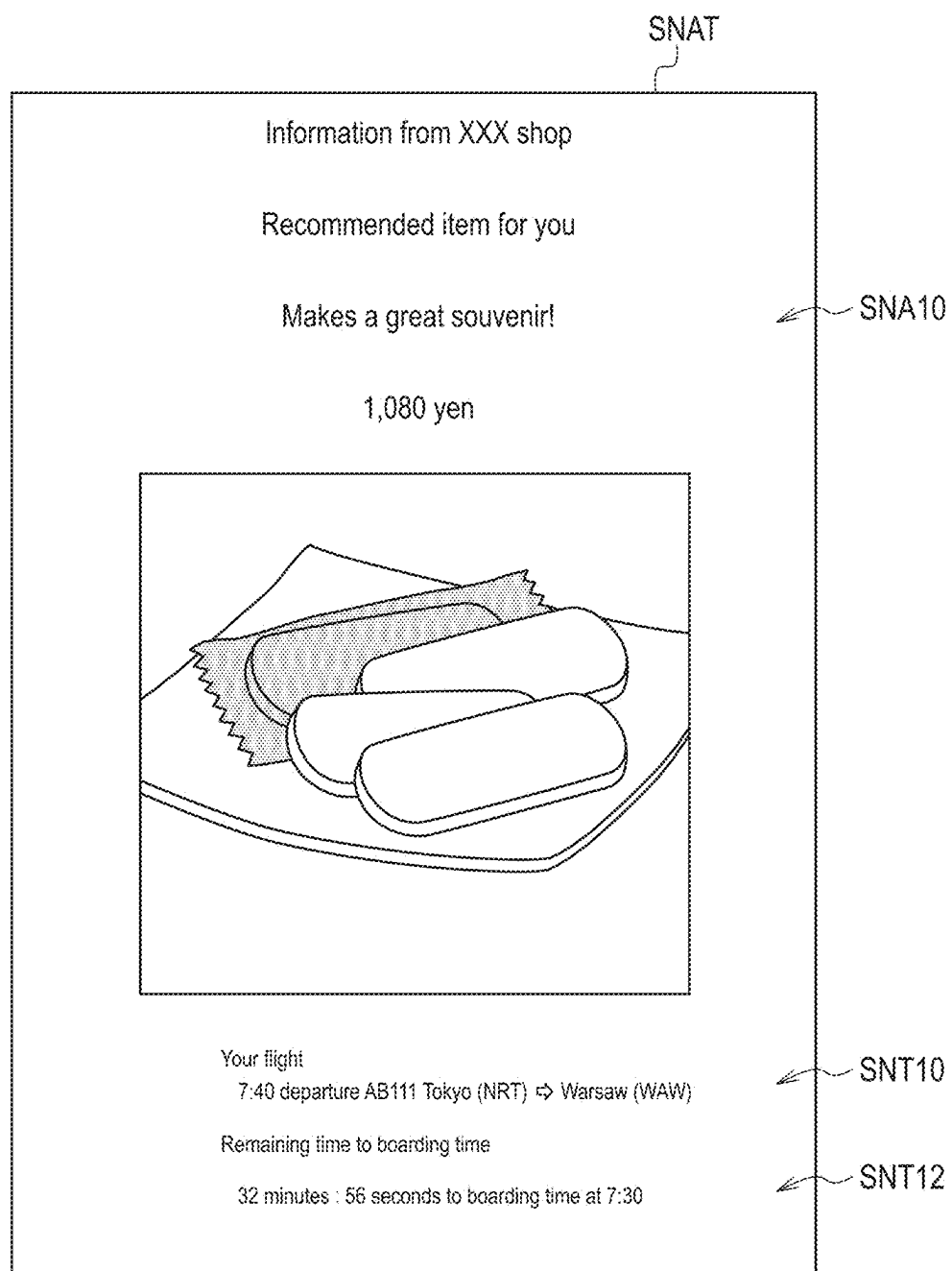


FIG. 28

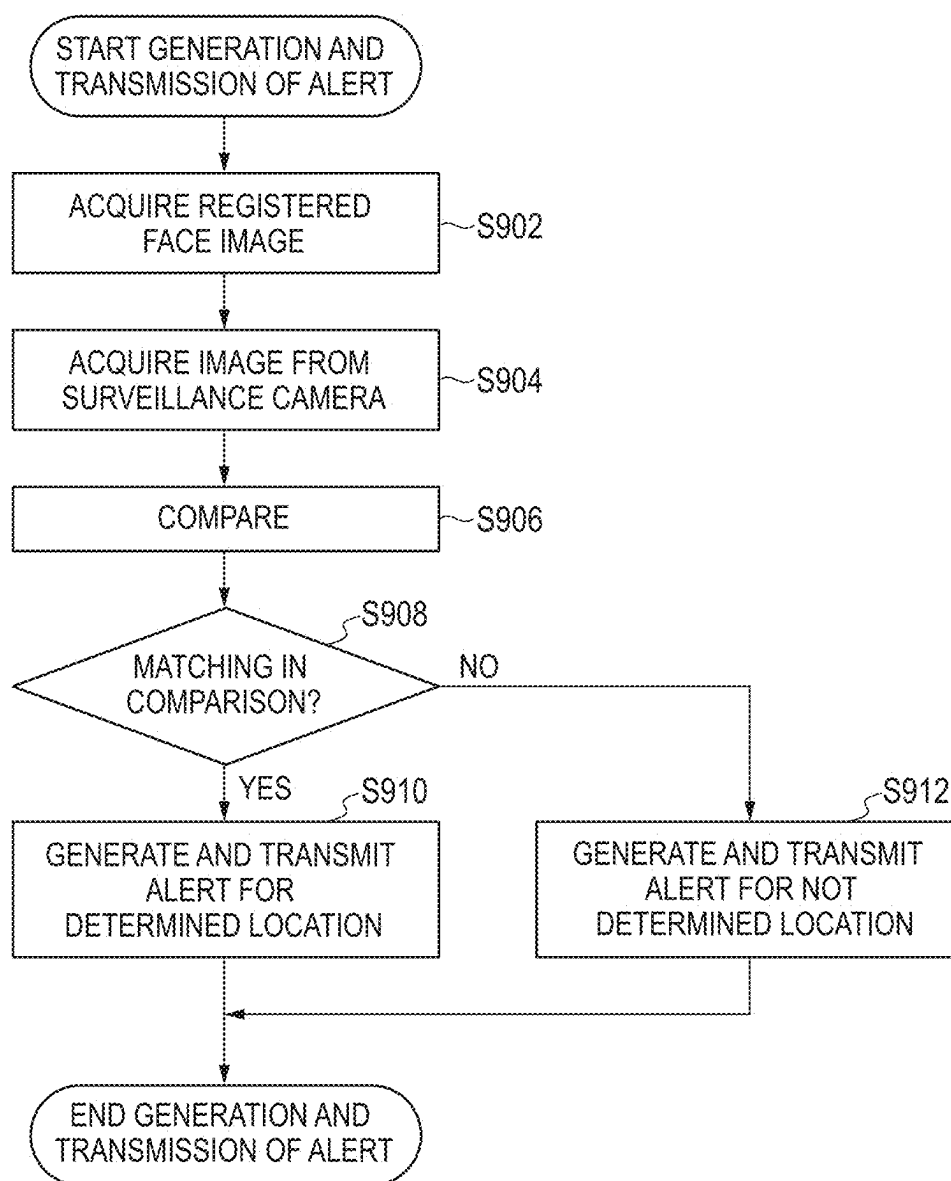


FIG. 29

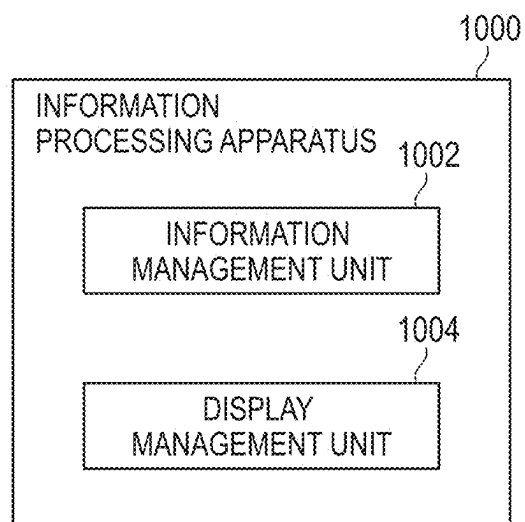
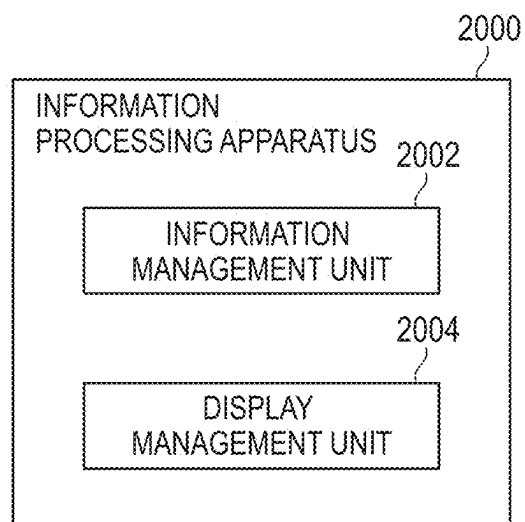




FIG. 30



# INFORMATION PROCESSING APPARATUS, INFORMATION PROCESSING METHOD, AND STORAGE MEDIUM

## INCORPORATION BY REFERENCE

[0001] This application is a continuation of U.S. application Ser. No. 17/225,234 filed on Apr. 8, 2021, which is a continuation of U.S. application Ser. No. 16/846,950 filed on Apr. 13, 2020, which issue as U.S. Pat. No. 11,113,912, which is a continuation of U.S. application Ser. No. 16/390,238 filed on Apr. 22, 2019, which issue as U.S. Pat. No. 10,657,740, which claims priority from National Stage International of Application No. PCT/JP2018/038000, filed on Oct. 11, 2018, the entire disclosure of which is incorporated herein by reference.

## TECHNICAL FIELD

[0002] The present invention relates to an information processing apparatus, an information processing method, and a storage medium.

## BACKGROUND ART

[0003] Japanese Patent Application Publication No. 2018-017924 discloses an information display system including an information display apparatus used to present service information such as flight information. In the information display system disclosed in Japanese Patent Application Publication No. 2018-017924, the information display apparatus transmits face images of nearby users to a server associated with a user terminal, performs user authentication by face authentication, acquires user information acquired from the user terminal, and generates and displays an associated individual guide window.

## SUMMARY

[0004] In the technology disclosed in Japanese Patent Application Publication No. 2018-017924, however, since the user is required to operate the user terminal by himself/herself and register a user's face image captured in advance for user authentication to the server, a burden on the user is large in using the system.

[0005] In view of the problem described above, the example object of the present invention is to provide an information processing apparatus, an information processing method, and a storage medium that can realize display directed to a user in front of a display terminal in accordance with a situation without involving a large burden on the user scheduled for boarding.

[0006] According to one example aspect of the present invention, provided is an information processing apparatus including: an information management unit that registers boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and a display management unit that causes a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and registered biometrics information, which is the biometrics information registered by the information management unit.

[0007] According to another example aspect of the present invention, provided is an information processing apparatus including: an information management unit that registers boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and a display management unit that causes a display terminal to display a deadline for a procedure of the user up to the boarding in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and registered biometrics information, which is the biometrics information registered by the information management unit.

[0008] According to another example aspect of the present invention, provided is an information processing method including: registering boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and causing a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and the registered biometrics information.

[0009] According to yet another example aspect of the present invention, provided is a non-transitory storage medium storing a program that causes a computer to perform: registering boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and causing a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and the registered biometrics information.

## BRIEF DESCRIPTION OF THE DRAWINGS

[0010] FIG. 1 is a schematic diagram illustrating the entire configuration of an information processing system according to a first example embodiment of the present invention.

[0011] FIG. 2 is a block diagram illustrating one example of the hardware configuration of a check-in terminal according to the first example embodiment of the present invention.

[0012] FIG. 3 is a block diagram illustrating one example of the hardware configuration of an automatic baggage deposit machine according to the first example embodiment of the present invention.

[0013] FIG. 4 is a block diagram illustrating one example of the hardware configuration of a security inspection apparatus according to the first example embodiment of the present invention.

[0014] FIG. 5 is a block diagram illustrating one example of the hardware configuration of an automated gate apparatus according to the first example embodiment of the present invention.

[0015] FIG. 6 is a block diagram illustrating one example of the hardware configuration of a signage terminal according to the first example embodiment of the present invention.

[0016] FIG. 7 is a block diagram illustrating one example of the hardware configuration of a boarding gate apparatus according to the first example embodiment of the present invention.

[0017] FIG. 8 is a block diagram illustrating one example of the hardware configuration of a management server according to the first example embodiment of the present invention.

[0018] FIG. 9 is a block diagram illustrating one example of the hardware configuration of an operation terminal according to the first example embodiment of the present invention.

[0019] FIG. 10 is a sequence diagram illustrating the operation performed by the check-in terminal and the management server in the information processing system according to the first example embodiment of the present invention.

[0020] FIG. 11 is a sequence diagram illustrating the operation performed by the automatic baggage deposit machine and the management server in the information processing system according to the first example embodiment of the present invention.

[0021] FIG. 12 is a sequence diagram illustrating the operation performed by the security inspection apparatus and the management server in the information processing system according to the first example embodiment of the present invention.

[0022] FIG. 13 is a sequence diagram illustrating the operation performed by the automated gate apparatus and the management server in the information processing system according to the first example embodiment of the present invention.

[0023] FIG. 14 is a sequence diagram illustrating the operation performed by the signage terminal and the management server in the information processing system according to the first example embodiment of the present invention.

[0024] FIG. 15 is a sequence diagram illustrating the operation performed by the boarding gate apparatus and the management server in the information processing system according to the first example embodiment of the present invention.

[0025] FIG. 16 is a schematic diagram illustrating one example of a user information database of the management server according to the first example embodiment of the present invention.

[0026] FIG. 17 is a schematic diagram illustrating an example of deadline time for each procedure set by the management server according to the first example embodiment of the present invention.

[0027] FIG. 18 is a flowchart (part 1) illustrating the operation including a status management operation and an alert operation performed by the management server in the information processing system according to the first example embodiment of the present invention.

[0028] FIG. 19 is a flowchart (part 2) illustrating the operation including a status management operation and an alert operation performed by the management server in the information processing system according to the first example embodiment of the present invention.

[0029] FIG. 20A is a schematic diagram illustrating one example of an alert displayed on the operation terminal according to the first example embodiment of the present invention.

[0030] FIG. 20B is a schematic diagram illustrating one example of a status review window displayed on the operation terminal according to the first example embodiment of the present invention.

[0031] FIG. 20C is a schematic diagram illustrating one example of a status review window displayed on the operation terminal according to the first example embodiment of the present invention.

[0032] FIG. 21 is a flowchart illustrating a display management operation performed by the management server in the information processing system according to the first example embodiment of the present invention.

[0033] FIG. 22 is a schematic diagram illustrating one example of a display window of first display information in the signage terminal according to the first example embodiment of the present invention.

[0034] FIG. 23 is a schematic diagram illustrating one example of a display window of first display information in the signage terminal according to the first example embodiment of the present invention.

[0035] FIG. 24 is a schematic diagram illustrating one example of a display window of first display information in the signage terminal according to the first example embodiment of the present invention.

[0036] FIG. 25 is a schematic diagram illustrating one example of a display window of second display information in the signage terminal according to the first example embodiment of the present invention.

[0037] FIG. 26 is a schematic diagram illustrating one example of a display window of first display information and second display information with a difference in the level of the size in the signage terminal according to the first example embodiment of the present invention.

[0038] FIG. 27 is a schematic diagram illustrating one example of a display window of first display information and second display information with a difference in the level of the size in the signage terminal according to the first example embodiment of the present invention.

[0039] FIG. 28 is a flowchart illustrating an alert operation performed by the management server according to a second example embodiment of the present invention.

[0040] FIG. 29 is a block diagram illustrating a configuration of an information processing apparatus according to another example embodiment of the present invention.

[0041] FIG. 30 is a block diagram illustrating a configuration of an information processing apparatus according to yet another example embodiment of the present invention.

## EXAMPLE EMBODIMENT

### First Example Embodiment

[0042] An information processing system, an information processing apparatus, and an information processing method according to a first example embodiment of the present invention will be described by using FIG. 1 to FIG. 25.

[0043] First, the configuration of the information processing system according to the present example embodiment will be described by using FIG. 1. FIG. 1 is a schematic diagram illustrating the entire configuration of the information processing system according to the present example embodiment.

[0044] As illustrated in FIG. 1, the information processing system 1 according to the present example embodiment includes a management server 10, a check-in terminal 20, an

automatic baggage deposit machine 30, a security inspection apparatus 40, an automated gate apparatus 50, signage terminals 60, a boarding gate apparatus 70, and an operation terminal 80. The information processing system 1 according to the present example embodiment is, for example, a tracking system that recognizes and manages the status within an airport A of a user U scheduled to board an airplane at the airport A and tracks the user U. Further, the information processing system 1 according to the present example embodiment is also a display management system that manages information directed to the user U scheduled to board an airplane at the airport A and displayed on the signage terminal 60, which is a display terminal, for example.

[0045] Note that the information processing system 1 may recognize and manage the status of a user within a facility who is scheduled to board a vehicle in the facility used for boarding the vehicle in addition to the user U who is scheduled to board an airplane at the airport A. For example, the information processing system 1 may recognize and manage the status within a seaport of a user who is scheduled to board a ship at the seaport. Further, for example, the information processing system 1 may recognize and manage the status within a station of a user who is scheduled to board a train at the station.

[0046] The management server 10 is installed within a facility such as an airport company, an airline company, or the like run in the airport A, for example. The check-in terminal 20 is installed in a check-in lobby P1 within the airport A. The automatic baggage deposit machine 30 is installed in a baggage counter P2 within the airport A. The security inspection apparatus 40 is installed in a security inspection site P3 within the airport A. The automated gate apparatus 50 is installed in an immigration site P4 within the airport A. The signage terminals 60 are installed anywhere within the airport A. For example, the signage terminals 60 are installed in the check-in lobby P1, the baggage deposit counter P2, the security inspection site P3, or the like. Further, the signage terminal 60 is installed on the passage P5 within the airport A, for example. The passage P5 is a passage connected between the immigration site P4 and a boarding gate P6. The boarding gate apparatus 70 is installed at the boarding gate P6 within the airport A. The user U is able to board an airplane through the boarding gate P6. The operation terminal 80 is carried by a staff such as a staff member of an airline company, a staff member of an airport company, or the like, for example. Further, the operation terminal 80 may be installed in an operation place such as a counter where a staff engages in their operation, for example.

[0047] After arriving at the airport A, the user U scheduled to board an airplane of an international flight goes through the check-in lobby P1, the baggage counter P2, the security inspection site P3, the immigration site P4, and the passage P5 in this order, each of which is a place for each procedure to be performed or a place for passage, and then boards an airplane at the boarding gate P6. The user U performs a check-in procedure (boarding procedure) on the check-in terminal 20 in the check-in lobby P1. Further, the user U carries out a baggage deposit procedure with the automatic baggage deposit machine 30 in the baggage counter P2 if necessary. After the check-in procedure and after the baggage deposit procedure if necessary, the user U is subjected to a security inspection procedure by the security inspection

apparatus 40 at the security inspection site P3. After the security inspection, the user U is subjected to an immigration procedure by the automated gate apparatus 50 in the immigration site P4. The user U who exited the immigration site P4 passes through the passage P5 on which the signage terminal 60 is installed and reaches the boarding gate P6. The user U passes through the boarding gate apparatus 70 in the boarding gate P6 and boards an airplane. Further, the user U may be present in front of the signage terminal 60 or pass by in front of the signage terminal 60 anywhere within the airport A and view various information displayed on the signage terminal 60, which is a display terminal.

[0048] Note that the user U may be not only a person scheduled to board an airplane of an international flight but also a person scheduled to board an airplane of a domestic flight. In a case of a user U scheduled to board an airplane of a domestic flight, the immigration in the immigration site P4 is not carried out.

[0049] A plurality of surveillance cameras 90 are installed in respective places within the airport A. The surveillance cameras 90 are installed in the check-in lobby P1, the baggage counter P2, the security inspection site P3, the immigration site P4, the passage P5, and the boarding gate P6, respectively, for example. The determination of the location of the user U by using the surveillance cameras 90 will be described in a second example embodiment.

[0050] The management server 10, the check-in terminal 20, the automatic baggage deposit machine 30, the security inspection apparatus 40, the automated gate apparatus 50, the signage terminal 60, the boarding gate apparatus 70, and the surveillance cameras 90 are connected to a network NW. The network NW is configured with a Local Area Network (LAN) including a premise communication network of the airport A, a Wide Area Network (WAN), a mobile communication network, or the like. The operation terminal 80 is capable of connecting to the network NW by a wireless scheme or a wired scheme.

[0051] The management server 10 is able to communicate with the check-in terminal 20, the automatic baggage deposit machine 30, the security inspection apparatus 40, the automated gate apparatus 50, the signage terminal 60, the boarding gate apparatus 70, the operation terminal 80, and the surveillance cameras 90, respectively, via the network NW.

[0052] Next, each component of the information processing system 1 according to the present example embodiment will be further described by using FIG. 2 to FIG. 9.

[0053] First, the configuration of the check-in terminal 20 will be described by using FIG. 2. FIG. 2 is a block diagram illustrating the hardware configuration of the check-in terminal 20.

[0054] The check-in terminal 20 is an automatic check-in terminal which the user U operates by himself/herself to carry out a check-in procedure (boarding procedure). Note that the check-in terminal 20 may be a check-in terminal operated by a staff member of an airline company and installed at a manned counter. A check-in procedure (boarding procedure) is a procedure to input at least information used for identifying the user U scheduled to board an airplane for registration. Specifically, the check-in procedure is a procedure to input user information including identity information, face information, and boarding information on the user U for registration as described below, for example.

[0055] As illustrated in FIG. 2, the check-in terminal 20 has a central processing unit (CPU) 202, a random access memory (RAM) 204, a storage device 206, an input device 208, and a display 210. Furthermore, the check-in terminal 20 has a medium reading device 212, a biometrics information acquisition device 214, and a communication unit 216. The CPU 202, the RAM 204, the storage device 206, the input device 208, the display 210, the medium reading device 212, the biometrics information acquisition device 214, and the communication unit 216 are connected to a bus line 218.

[0056] The CPU 202 functions as a control unit that operates by executing a program stored in the storage device 206 and controls the operation of the entire check-in terminal 20. Further, the CPU 202 executes an application program stored in the storage device 206 to perform various processes as the check-in terminal 20. The RAM 204 provides a memory field necessary for the operation of the CPU 202.

[0057] More specifically, the CPU 202 functions as a comparison unit that compares a face image of the user U who utilizes the check-in terminal 20 to carry out a check-in procedure. The CPU 202 as the comparison unit compares a passport face image, which is a face image acquired from a passport of the user U by the medium reading device 212, with a captured face image, which is a face image captured by the biometrics information acquisition device 214 in a check-in procedure. In the comparison between the passport face image and the captured face image, the CPU 202 can compare both face images by comparing a face feature amount extracted from the passport face image with a face feature amount extracted from a captured face image.

[0058] Note that, instead of functioning as a comparison unit, the CPU 202 may also be configured to request an external server such as the management server 10 for comparison of a face image of the user U described above and acquire a comparison result from the external server. In such a case, the CPU 202 can transmit a passport face image and a captured image of the user U to the external server to request comparison of the face image.

[0059] Further, the CPU 202 functions as a user information transmission unit that transmits user information on the user U to the management server 10. When there is a matching in the comparison between a passport face image and a captured face image of the user U and identity verification of the user U is successful, the CPU 202 as a user information transmission unit transmits the user information on the user U to the management server 10.

[0060] The user information on the user U includes identity information, face information, and boarding information on the user U associated with each other.

[0061] Identity information is information on an individual user U, such as the name, the nationality, the sexuality, the birthday, or the like of the user U. Identity information is acquired from a passport or a flight ticket medium by the medium reading device 212. Further, identity information can include passport information such as a passport number, a passport issuance country, or the like acquired from the passport of the user U by the medium reading device 212.

[0062] Face information corresponds to a captured face image or a passport face image of the user U obtained after the comparison by the CPU 202. Further, face information may correspond to either one of a captured face image and

a passport face image or may correspond to both the images. Further, face information may not correspond to a captured face image or a passport face image itself and may correspond to a face feature amount extracted from a captured face image or a passport face image.

[0063] Boarding information is information on a flight which the user U intends to use, such as an airline company, a flight number, an origin, a destination, a departure date and time, an arrival date and time of an airplane which the user U is scheduled to board. Further, boarding information may include boarding time when boarding on an airplane performed by the user U is closed. Boarding information is acquired from a flight ticket medium of the user U by the medium reading device 212.

[0064] Note that the user information on the user U transmitted to the management server 10 is registered to a user information database (DB) 106a in the management server 10 as described later.

[0065] Further, the CPU 202 functions as a status information transmission unit that transmits status information, which is information on the status of the user U. After transmitting the user information on the user U to complete a check-in procedure, the CPU 202 as the status information transmission unit transmits, to the management server 10, status information on the user U indicating that the check-in procedure is completed. The status information is associated with at least any of user information on the user U or face information, passport information, and boarding information included in the user information in the management server 10.

[0066] Note that the CPU 202 may be any unit that at least transmits information used for identifying the user U scheduled to board an airplane and status information on a check-in procedure. In such a case, boarding information on the user U may be registered in advance in the DB of a server such as the management server 10.

[0067] Further, the CPU 202 functions as a display control unit that causes the display 210 to display a window such as guidance, notification, or the like to the user U. For example, the CPU 202 as the display control unit causes the display 210 to display a guidance window showing how to use the check-in terminal 20. Further, when there is no matching in the comparison between a passport face image and a captured face image of the user U and identity verification of the user U fails, the CPU 202 causes the display 210 to display a notification window that notifies the user U that the identity verification failed.

[0068] The storage device 206 is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device 206 stores a program executed by the CPU 202, data referenced by the CPU 202 when the program is executed, or the like.

[0069] The input device 208 is a touchscreen embedded in the display 210, for example. The input device 208 functions as an input unit that accepts input of an instruction from the user U. The user U is able to input various information or input an instruction of execution of a process to the check-in terminal 20 via the input device 208.

[0070] The display 210 functions as a display unit that displays various windows to the user U who uses the check-in terminal 20. For example, the display 210 displays a guidance window showing how to use the check-in terminal 20 or a notification window to the user U.

[0071] The medium reading device 212 functions as an information acquisition unit that reads a passport or a flight ticket medium of the user U and acquires information recorded on the passport or the flight ticket. A flight ticket medium may be, for example, a flight ticket sheet, a mobile terminal displaying a duplicate of an e-ticket, or the like. The medium reading device 212 is formed of a code reader, an image scanner, a contactless integrated circuit (IC) reader, an optical character reader (OCR) device, or the like, for example, and acquires information from various media held over the reading unit thereof.

[0072] For example, the medium reading device 212 reads and acquires identity information and passport information on the user U indicated on the passport surface by using an OCR device. Further, for example, the medium reading device 212 reads and acquires a passport face image of the user U indicated on the passport surface by using an image scanner. Further, in the case of an IC passport, the medium reading device 212 reads and acquires identity information on the user U, passport information, a passport face image, or the like stored in the IC chip of the IC passport by using a contactless IC reader. Further, for example, the medium reading device 212 reads a barcode or a QR code (registered trademark) indicated on a flight ticket medium and acquires identity information and boarding information on the user U from the flight ticket medium by using a code reader.

[0073] Note that the check-in terminal 20 can not only acquire identity information and boarding information from a flight ticket medium but also acquire identity information and boarding information by using another scheme. For example, the CPU 202 may inquire an airline company server for reservation purchase information on a flight ticket based on a credit card number, a reservation number, a confirmation number, or the like of the user U and acquire identity information and boarding information on the user U from the airline company server.

[0074] The biometrics information acquisition device 214 functions as a biometrics information acquisition unit that acquires a face image of the user U as biometrics information on the user U carrying out a check-in procedure. The biometrics information acquisition device 214 is a digital camera that captures a face of the user U standing in front of the check-in terminal 20, for example, and captures the face of the user U and acquires the face image thereof.

[0075] The communication unit 216 is connected to the network NW and transmits and receives data via the network NW. The communication unit 216 communicates with the management server 10 or the like under the control of the CPU 202.

[0076] In such a way, the check-in terminal 20 is configured.

[0077] Next, the configuration of the automatic baggage deposit machine 30 will be described by using FIG. 3. FIG. 3 is a block diagram illustrating an example of the hardware configuration of the automatic baggage deposit machine 30. The automatic baggage deposit machine 30 is an apparatus that the user U operates by himself/herself to deposit baggage which is not brought in the cabin of an airplane. The user U is able to carry out a procedure to deposit baggage by using the automatic baggage deposit machine 30 if necessary.

[0078] As illustrated in FIG. 3, the automatic baggage deposit machine 30 has a CPU 302, a RAM 304, a storage device 306, an input device 308, and a display 310. Fur-

thermore, the automatic baggage deposit machine 30 has a biometrics information acquisition device 312, a baggage transport apparatus 314, an output device 316, and a communication unit 318. The CPU 302, the RAM 304, the storage device 306, the input device 308, the display 310, the biometrics information acquisition device 312, the baggage transport apparatus 314, the output device 316, and the communication unit 318 are connected to a bus line 320.

[0079] The CPU 302 functions as a control unit that operates by executing a program stored in the storage device 306 and controls the operation of the entire automatic baggage deposit machine 30. Further, the CPU 302 executes an application program stored in the storage device 306 to perform various processes as the automatic baggage deposit machine 30. The RAM 304 provides a memory field necessary for operation of the CPU 302.

[0080] More specifically, the CPU 302 functions as a comparison request unit that requests comparison of a face image of the user U performing a baggage deposit procedure by using the automatic baggage deposit machine 30. The CPU 302 as the comparison request unit requests the management server 10 to compare, at 1:N, a target face image, which is a face image of the user U captured by the biometrics information acquisition device 312, with a plurality of registered face images registered in the user information DB 106a of the management server 10. Thus, the CPU 302 transmits a target face image captured by the biometrics information acquisition device 312 to the management server 10 as target biometrics information together a comparison with request. Note that, instead of transmitting a target face image as target biometrics information, the CPU 302 may transmit a face feature amount, which is a feature amount extracted from a target face image, and request the comparison.

[0081] Note that, instead of requesting the comparison at 1:N described above, the CPU 302 may request the management server 10 to compare a target face image of the user U with a registered face image associated with information on a passport, a boarding pass, or a QR code of the user U. In such a case, when registering a captured face image or a passport face image of the user U acquired by the check-in terminal 20 to the user information DB 106a as a registered face image, the management server 10 performs registration in association with information on a passport, a boarding pass, or a QR code of the user U. The CPU 302 reads information on a passport, a boarding pass, or a QR code of the user U by using a medium reading device (not illustrated) and transmits the read information to the management server 10 together with a target face image. The management server 10 acquires, from the user information DB 106a, a registered face image associated with information on a passport, a boarding pass, or a QR code which matches information on a passport, a boarding pass, or a QR code transmitted from the automatic baggage deposit machine 30. The management server 10 compares, at 1:1, a target face image transmitted from the automatic baggage deposit machine 30 with a registered information face image acquired from the user information DB 106a.

[0082] Further, the CPU 302 functions as a comparison result acquisition unit that acquires, from the management server 10, comparison result information indicating a result of comparison requested from the management server 10. In a case of 1:N comparison, comparison result information indicates that a registered face image matching a target face

image was found and the identity verification of the user U performing a baggage deposit procedure succeeded or that no matching registered face image was found and the identity verification of the user U failed, as a result of the comparison. In a case of 1:1 comparison, comparison result information indicates that a target face image matches a registered face image and the identity verification of the user U performing a baggage deposit procedure succeeded or that no matching is obtained and the identity verification of the user U failed, as a result of the comparison.

[0083] Note that, instead of functioning as the comparison request unit and the comparison result acquisition unit, the CPU 302 may be configured to function as a comparison unit that compares a target face image with one or more registered face images in the user information DB 106a at 1:N or 1:1 as described above and outputs comparison result information. In such a case, the CPU 302 can compare a face image by referencing the user information DB 106a of the management server 10 or referencing the user information DB stored in the storage device 306 in synchronization with the user information DB 106a.

[0084] Further, when the comparison result information indicates that the identity verification of the user U succeeded, the CPU 302 controls the baggage transport apparatus 314 to perform reception and transportation of baggage of the user U.

[0085] Further, the CPU 302 functions as a status information transmission unit that transmits status information, which is information on the status of the user U. When reception and transportation of baggage is performed by the baggage transport apparatus 314 completed and the baggage deposit procedure is completed, the CPU 302 as the status information transmission unit transmits, to the management server 10, server status information on the user U indicating that the baggage deposit procedure is completed. The status information is associated with at least any of user information on the user U or face information, passport information, and boarding information included in the user information in the management server 10.

[0086] Further, the CPU 302 functions as a display control unit that causes the display 310 to display a window such as guidance, notification, or the like to the user U. For example, the CPU 302 as the display control unit causes the display 310 to display a guidance window showing how to use the automatic baggage deposit machine 30. Further, when comparison result information indicates that the identity verification of the user U failed, the CPU 302 causes the display 310 to display a notification window that notifies the user U that the identity verification failed.

[0087] The storage device 306 is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device 306 stores a program executed by the CPU 302, data referenced by the CPU 302 when the program is executed, or the like.

[0088] The input device 308 is a touchscreen embedded in the display 310, for example. The input device 308 functions as an input unit that accepts input of an instruction from the user U. The user U is able to input various information or input an instruction of execution of a process to the automatic baggage deposit machine 30 via the input device 308.

[0089] The display 310 functions as a display unit that displays various windows to the user U who uses the automatic baggage deposit machine 30. For example, the

display 310 displays a guidance window showing how to use the automatic baggage deposit machine 30 or a notification window to the user U.

[0090] The biometrics information acquisition device 312 functions as a biometrics information acquisition unit that acquires a face image of the user U as biometrics information on the user U performing a baggage deposit procedure. For example, the biometrics information acquisition device 312 is a digital camera that captures a front area of the automatic baggage deposit machine 30 and, when detecting a face of the user U standing in front of the automatic baggage deposit machine 30 in an image captured continuously or periodically, captures the face of the user U and acquires the face image thereof.

[0091] The baggage transport apparatus 314 transports baggage to load the baggage on an airplane which the user U boards when identity verification of the user U is successful. The baggage transport apparatus 314 transports baggage placed on a receiving part by the user U and attached with a baggage tag toward a cargo sorting site.

[0092] The output device 316 outputs a baggage tag to be attached to deposited baggage and outputs a baggage claim tag necessary for claiming baggage after arrival. The user U attaches the baggage tag output from the output device 316 to the baggage to be deposited. Further, for receiving the baggage after arrival, the user U receives a baggage claim tag output from the output device 316. The baggage tag is associated with at least any of user information on the user U or face information, passport information, and boarding information included in the user information in the management server 10.

[0093] The communication unit 318 is connected to the network NW and transmits and receives data via the network NW. The communication unit 318 communicates with the management server 10 or the like under the control of the CPU 302.

[0094] In such a way, the automatic baggage deposit machine 30 is configured.

[0095] Next, the configuration of the security inspection apparatus 40 will be described by using FIG. 4. FIG. 4 is a block diagram illustrating one example of the hardware configuration of the security inspection apparatus 40. The security inspection apparatus 40 is an apparatus that checks whether the user U wears any metal that may be a dangerous object by using a metal detector gate 414, which is a metal detector. Note that the security inspection apparatus 40 may be an apparatus that checks whether or not there is a dangerous object inside carry-on baggage or the like by using an X-ray inspection device or the like other than a metal detector such as the metal detector gate 414.

[0096] As illustrated in FIG. 4, the security inspection apparatus 40 has a CPU 402, a RAM 404, a storage device 406, an input device 408, a display 410, a biometrics information acquisition device 412, the metal detector gate 414, and a communication unit 416. The CPU 402, the RAM 404, the storage device 406, the input device 408, the display 410, the biometrics information acquisition device 412, the metal detector gate 414, and the communication unit 416 are connected to a bus line 418.

[0097] The CPU 402 functions as a control unit that operates by executing a program stored in the storage device 406 and controls the operation of the entire security inspection apparatus 40. Further, the CPU 402 executes an application program stored in the storage device 406 to perform

various processes as the security inspection apparatus 40. The RAM 404 provides a memory field necessary for the operation of the CPU 402.

[0098] More specifically, the CPU 402 functions as a comparison request unit that requests comparison of a face image of the user U subjected to a security inspection procedure by the security inspection apparatus 40. The CPU 402 as the comparison request unit requests the management server 10 to compare, at 1:N, a target face image, which is a face image of the user U captured by the biometrics information acquisition device 412, with a plurality of registered face images registered in the user information DB 106a of the management server 10. Thus, the CPU 402 transmits a target face image captured by the biometrics information acquisition device 412 to the management server 10 as target biometrics information together with a comparison request. Note that, instead of transmitting a target face image as target biometrics information, the CPU 402 may transmit a face feature amount, which is a feature amount extracted from a target face image, and request the comparison.

[0099] Note that, instead of requesting the comparison at 1:N described above, the CPU 402 may request the management server 10 to compare a target face image of the user U with registered face images associated with information on a passport, a boarding pass, or a code symbol such as a QR code of the user U. In such a case, when registering a captured face image or a passport face image of the user U acquired by the check-in terminal 20 to the user information DB 106a as a registered face image, the management server 10 performs registration in association with information on a passport, a boarding pass, or a code symbol of the user U. The CPU 402 reads information on a passport, a boarding pass, or a code symbol of the user U by using a medium reading device (not illustrated) and transmits the read information to the management server 10 together with a target face image. The management server 10 acquires, from the user information DB 106a, a registered face image associated with information on a passport, a boarding pass, or a code symbol which matches information on a passport, a boarding pass, or a code symbol transmitted from the security inspection apparatus 40. The management server 10 compares, at 1:1, a target face image transmitted from the security inspection apparatus 40 with a registered information face image acquired from the user information DB 106a.

[0100] Further, the CPU 402 functions as a comparison result acquisition unit that acquires, from the management server 10, comparison result information indicating a result of comparison requested from the management server 10. In a case of 1:N comparison, comparison result information indicates that a registered face image matching a target face image was found and the identity verification of the user U subjected to the security inspection procedure succeeded or that no matching registered face image was found and the identity verification of the user U failed, as a result of the comparison. In a case of 1:1 comparison, comparison result information indicates that a target face image matches a registered face image and the identity verification of the user U subjected to the security inspection procedure succeeded or that no matching is obtained and the identity verification of the user U failed, as a result of the comparison.

[0101] Note that, instead of functioning as the comparison request unit and the comparison result acquisition unit, the

CPU 402 may be configured to function as a comparison unit that compares a target face image with one or more registered face images in the user information DB 106a at 1:N or 1:1 as described above and outputs comparison result information. In such a case, the CPU 402 can compare a face image by referencing the user information DB 106a of the management server 10 or referencing the user information DB stored in the storage device 406 in synchronization with the user information DB 106a.

[0102] Further, when the comparison result information indicates that the identity verification of the user U succeeded, the CPU 402 continues inspection performed by the metal detector gate 414.

[0103] Further, the CPU 402 functions as a status information transmission unit that transmits status information, which is information on the status of the user U. When inspection performed by the metal detector gate 414 is completed without an anomaly, the CPU 402 as the status information transmission unit transmits, to the management server 10, status information on the user U indicating that the security inspection procedure is completed. The status information is associated with at least any of user information on the user U or face information, passport information, and boarding information included in the user information in the Further, the CPU 402 functions as a display control unit that causes the display 410 to display a window such as guidance, notification, or the like to the user U. For example, the CPU 402 as the display control unit causes the display 410 to display a window that guides the user U to pass through the metal detector gate 414. Further, when comparison result information indicates that the identity verification of the user U failed, the CPU 402 causes the display 410 to display a notification window that notifies the user U that the identity verification failed. Further, when a metal is detected in the inspection performed by the metal detector gate 414, the CPU 402 causes the display 410 to display a notification window to confirm whether no metal is included in carried belongings and again instruct the user U to pass through the metal detector gate 414.

[0104] The storage device 406 is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device 406 stores a program executed by the CPU 402, data referenced by the CPU 402 when the program is executed, or the like.

[0105] The input device 408 is a touchscreen embedded in the display 410, for example. The input device 408 functions as an input unit that accepts input of an instruction from the user U. The user U is able to input various information or input an instruction of execution of a process to the security inspection apparatus 40 via the input device 408.

[0106] The display 410 functions as a display unit that displays various windows to the user U subjected to a security inspection procedure by the security inspection apparatus 40. For example, the display 410 displays a window that guides the user U to pass through the metal detector gate 414 or a notification window to the user U.

[0107] The biometrics information acquisition device 412 functions as a biometrics information acquisition unit that acquires a face image of the user U as biometrics information on the user U subjected to a security inspection procedure. For example, the biometrics information acquisition device 412 is a digital camera that captures a front area of the metal detector gate 414 and, when detecting a face of the



user U standing in front of the metal detector gate **414** in an image captured continuously or periodically, captures the face of the user U and acquires the face image thereof.

[0108] The metal detector gate **414** is a gate-type metal detector and detects a metal carried by the user U passing through the metal detector gate **414**.

[0109] The communication unit **416** is connected to the network NW and transmits and receives data via the network NW. The communication unit **416** communicates with the management server **10** or the like under the control of the CPU **402**.

[0110] In such a way, the security inspection apparatus **40** is configured.

[0111] Next, the configuration of the automated gate apparatus **50** will be described by using FIG. 5. FIG. 5 is a block diagram illustrating one example of the hardware configuration of the automated gate apparatus **50**. The automated gate apparatus **50** is an apparatus that automatically performs an immigration procedure on the user U. The user U who is able to use an immigration procedure with the automated gate apparatus **50** may be, for example, a passenger who has been examined through pre-application and completed user registration for the automated gate apparatus **50**, a passenger who is allowed to use it based on a travel history such as the number of times of traveling, a passenger who has completed user registration based on bilateral agreement, or the like.

[0112] As illustrated in FIG. 5, the automated gate apparatus **50** has a CPU **502**, a RAM **504**, a storage device **506**, an input device **508**, a display **510**, a biometrics information acquisition device **512**, a gate **514**, and a communication unit **516**. The CPU **502**, the RAM **504**, the storage device **506**, the input device **508**, the display **510**, the biometrics information acquisition device **512**, the gate **514**, and the communication unit **516** are connected to a bus line **518**.

[0113] The CPU **502** functions as a control unit that operates by executing a program stored in the storage device **506** and controls the operation of the entire automated gate apparatus **50**. Further, the CPU **502** executes an application program stored in the storage device **506** to perform various processes as the automated gate apparatus **50**. The RAM **504** provides a memory field necessary for the operation of the CPU **502**.

[0114] More specifically, the CPU **502** functions as a comparison request unit that requests comparison of a face image of the user U subjected to an immigration procedure by using the automated gate apparatus **50**. The CPU **502** as the comparison request unit requests the management server **10** to compare, at 1:N, a target face image, which is a face image of the user U captured by the biometrics information acquisition device **512**, with a plurality of registered face images registered in the user information DB **106a** of the management server **10**. Thus, the CPU **502** transmits a target face image captured by the biometrics information acquisition device **512** to the management server **10** as target biometrics information together with a comparison request. Note that, instead of transmitting a target face image as target biometrics information, the CPU **502** may transmit a face feature amount, which is a feature amount extracted from a target face image, and request the comparison.

[0115] Note that, instead of requesting the comparison at 1:N described above, the CPU **502** may request the management server **10** to compare a target face image of the user U with registered face images associated with information

on a passport, a boarding pass, or a code symbol such as a QR code of the user U. In such a case, when registering a captured face image or a passport face image of the user U acquired by the check-in terminal **20** to the user information DB **106a** as a registered face image, the management server **10** performs registration in association with information on a passport, a boarding pass, or a code symbol of the user U. The CPU **502** reads information on a passport, a boarding pass, or a code symbol of the user U by using a medium reading device (not illustrated) and transmits the read information to the management server **10** together with a target face image. The management server **10** acquires, from the user information DB **106a**, a registered face image associated with information on a passport, a boarding pass, or a code symbol which matches information on a passport, a boarding pass, or a code symbol transmitted from the automated gate apparatus **50**. The management server **10** compares, at 1:1, a target face image transmitted from the automated gate apparatus **50** with a registered information face image acquired from the user information DB **106a**.

[0116] Further, the CPU **502** functions as a comparison result acquisition unit that acquires, from the management server **10**, comparison result information indicating a result of comparison requested from the management server **10**. In a case of 1:N comparison, comparison result information indicates that a registered face image matching a target face image was found and the identity verification of the user U subjected to the immigration procedure succeeded or that no matching registered face image was found and the identity verification of the user U failed, as a result of the comparison. In a case of 1:1 comparison, comparison result information indicates that a target face image matches a registered face image and the identity verification of the user U subjected to the immigration procedure succeeded or that no matching is obtained and the identity verification of the user U failed, as a result of the comparison.

[0117] Note that, instead of functioning as the comparison request unit and the comparison result acquisition unit, the CPU **502** may be configured to function as a comparison unit that compares a target face image with one or more registered face images in the user information DB **106a** at 1:N or 1:1 as described above and outputs comparison result information. In such a case, the CPU **502** can compare a face image by referencing the user information DB **106a** of the management server **10** or referencing the user information DB stored in the storage device **506** in synchronization with the user information DB **106a**.

[0118] Further, when the comparison result information indicates that the identity verification of the user U succeeded, the CPU **502** controls the gate **514** to permit the user U to pass through the gate **514**.

[0119] Further, the CPU **502** functions as a status information transmission unit that transmits status information, which is information on the status of the user U. When permitting the user U to pass through the gate **514** to complete the immigration procedure, the CPU **502** as the status information transmission unit transmits, to the management server **10**, status information on the user U indicating that the immigration procedure is completed. The status information is associated with at least any of user information on the user U or face information, passport information, and boarding information included in the user information in the management server **10**.

[0120] Further, the CPU 502 functions as a display control unit that causes the display 510 to display a window such as guidance, notification, or the like to the user U. For example, the CPU 502 as the display control unit causes the display 510 to display a guidance window showing how to use the automated gate apparatus 50. Further, when comparison result information indicates that the identity verification of the user U failed, the CPU 502 causes the display 510 to display a notification window that notifies the user U that the identity verification failed.

[0121] The storage device 506 is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device 506 stores a program executed by the CPU 502, data referenced by the CPU 502 when the program is executed, or the like.

[0122] The input device 508 is a touchscreen embedded in the display 510, for example. The input device 508 functions as an input unit that accepts input of an instruction from the user U. The user U is able to input various information or input an instruction of execution of a process to the automated gate apparatus 50 via the input device 508.

[0123] The display 510 functions as a display unit that displays various windows to the user U subjected to an immigration procedure by using the automated gate apparatus 50. For example, the display 510 displays a guidance window showing how to use the automated gate apparatus 50, a notification window to the user U, or the like.

[0124] The biometrics information acquisition device 512 functions as a biometrics information acquisition unit that acquires a face image of the user U as biometrics information on the user U subjected to an immigration procedure. For example, the biometrics information acquisition device 512 is a digital camera that captures a front area of the automated gate apparatus 50 and, when detecting a face of the user U standing in front of the automated gate apparatus 50 in an image captured continuously or periodically, captures the face of the user U and acquires the face image thereof.

[0125] When identity verification of the user U is successful in the automated gate apparatus 50 and the user U passes through an immigration, the gate 514 transfers from a closed state for a standby to block passage of the user U to an opened state to permit passage of the user U under the control of the CPU 502. The scheme of the gate 514 is not particularly limited and may be, for example, a flapper gate in which one or more flappers provided on one side or both side of the passage are opened or closed, a turn-style gate in which three bars are revolved, or the like.

[0126] The communication unit 516 is connected to the network NW and transmits and receives data via the network NW. The communication unit 516 communicates with the management server 10 or the like under the control of the CPU 502.

[0127] In such a way, the automated gate apparatus 50 is configured.

[0128] Next, the configuration of the signage terminal 60 will be described by using FIG. 6. FIG. 6 is a block diagram illustrating one example of the hardware configuration of the signage terminal 60. The signage terminal 60 is a terminal that displays a guide showing the direction to the boarding gate P6, an advertisement, or the like.

[0129] As illustrated in FIG. 6, the signage terminal 60 has a CPU 602, a RAM 604, a storage device 606, a display 608,

a biometrics information acquisition device 610, and a communication unit 612. The CPU 602, the RAM 604, the storage device 606, the display 608, the biometrics information acquisition device 610, and the communication unit 612 are connected to a bus line 614.

[0130] The CPU 602 functions as a control unit that operates by executing a program stored in the storage device 606 and controls the operation of the entire signage terminal 60. Further, the CPU 602 executes an application program stored in the storage device 606 to perform various processes as the signage terminal 60. The RAM 604 provides a memory field necessary for the operation of the CPU 602.

[0131] More specifically, the CPU 602 functions as a face information transmission unit that transmits a face image of the user U in front of the signage terminal 60 to the management server 10. The user U in front of the signage terminal 60 may be, for example, a user who is present in front of the signage terminal 60 and confirms a guidance displayed on the signage terminal 60, the user U who passes by in front of the signage terminal 60, or the like. The CPU 602 as a face information transmission unit transmits a target face image, which is a face image of the user U captured by the biometrics information acquisition device 610, to the management server 10 as target biometrics information. Note that, instead of transmitting a target face image as target biometrics information, the CPU 602 may transmit a face feature amount, which is a feature amount extracted from a target face image. The management server 10 that has received a target face image from the signage terminal 60 compares, at 1:N, the received target face image with a plurality of registered face images registered in the user information DB 106a.

[0132] Further, the CPU 602 functions as a display control unit that causes the display 608 to display a window such as a guide showing the direction to the boarding gate P6, an advertisement, or the like.

[0133] Note that the CPU 602 of the signage terminal 60 can also function as a comparison request unit, a comparison result acquisition unit, and a status information transmission unit as described below in a similar manner to the CPU 302 of the automatic baggage deposit machine 30 or the like.

[0134] More specifically, the CPU 602 functions as a comparison result request unit that requests comparison of a face image of the user U who is present in front of the signage terminal 60 or passes by in front of the signage terminal 60. The CPU 602 as the comparison request unit requests the management server 10 to compare, at 1:N, a target face image, which is a face image of the user U captured by the biometrics information acquisition device 610, with a plurality of registered face images registered in the user information DB 106a of the management server 10. Thus, the CPU 602 transmits a target face image captured by the biometrics information acquisition device 610 to the management server 10 as target biometrics information together with a comparison request. Note that, instead of transmitting a target face image as target biometrics information, the CPU 602 may transmit a face feature amount, which is a feature amount extracted from a target face image, to request the comparison.

[0135] Note that, instead of requesting the comparison at 1:N described above, the CPU 602 may request the management server 10 to compare a target face image of the user U with registered face images associated with information on a passport, a boarding pass, or a code symbol such as a

QR code of the user U. In such a case, when registering a captured face image or a passport face image of the user U acquired by the check-in terminal 20 to the user information DB 106a as a registered face image, the management server 10 performs registration in association with information on a passport, a boarding pass, or a code symbol of the user U. The CPU 602 reads information on a passport, a boarding pass, or a code symbol of the user U by using a medium reading device (not illustrated) and transmits the read information to the management server 10 together with a target face image. The management server 10 acquires, from the user information DB 106a, a registered face image associated with information on a passport, a boarding pass, or a code symbol which matches information a passport, a boarding pass, or a code symbol on transmitted from the signage terminal 60. The management server 10 compares, at 1:1, a target face image transmitted from the signage terminal 60 with a registered information face image acquired from the user information DB 106a.

[0136] Further, the CPU 602 functions as a comparison result acquisition unit that acquires, from the management server 10, comparison result information indicating a result of comparison requested from the management server 10. In a case of 1:N comparison, comparison result information indicates that a registered face image matching a target face image was found and the identity verification of the user U in front of the signage terminal 60 succeeded or that no matching registered face image was found and the identity verification of the user U failed as a result of the comparison. In a case of 1:1 comparison, comparison result information indicates that a target face image matches a registered face image and the identity verification of the user U in front of the signage terminal 60 succeeded or that no matching is obtained and the identity verification of the user U failed, as a result of the comparison.

[0137] Note that, instead of functioning as the comparison request unit and the comparison result acquisition unit, the CPU 602 may be configured to function as a comparison unit that compares a target face image with one or more registered face images in the user information DB 106a at 1:N or 1:1 as described above and outputs comparison result information. In such a case, the CPU 602 can compare a face image by referencing the user information DB 106a of the management server 10 or referencing the user information DB stored in the storage device 606 in synchronization with the user information DB 106a.

[0138] Further, the CPU 602 functions as a status information transmission unit that transmits status information, which is information regarding the status of the user U. The CPU 602 as the status information transmission unit transmits, to the management server 10, status information indicating that presence or passage of the user U in front of the signage terminal 60 is completed when identity verification of the user U in front of the signage terminal 60 is successful. The status information is associated with user information on the user U or at least any one of face information, passport information, and boarding information included in the user information in the management server 10.

[0139] Further, the CPU 602 functions as display control unit that controls display on the display 608 based on display information transmitted from the management server 10 as described later. The CPU 602 as the display control unit causes the display 608 to display different display information in accordance with remaining time to the boarding time

for the user U in front of the signage terminal 60 who succeeded in identity verification and identified by the comparison with a target face image.

[0140] The storage device 606 is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device 606 stores a program executed by the CPU 602, data referenced by the CPU 602 when the program is executed, or the like.

[0141] The display 608 functions as a display unit that displays various windows. For example, the display 608 displays a window of a guide showing the direction to the boarding gate P6, an advertisement, or the like. The guide showing directions, the advertisement or the like displayed on the display 608 corresponds to an individual identified as the user U in front of the signage terminal 60, for example.

[0142] The biometrics information acquisition device 610 functions as a biometrics information acquisition unit that acquires a face image of the user U as biometrics information on the user U in front of the signage terminal 60. For example, the biometrics information acquisition device 610 is a digital camera that captures a front area of the signage terminal 60 and, when detecting a face of the user U in front of the signage terminal 60 in an image captured continuously or periodically, captures the face of the user U and acquires the face image thereof.

[0143] The communication unit 612 is connected to the network NW and transmits and receives data via the network NW. The communication unit 612 communicates with the management server 10 or the like under the control of the CPU 602.

[0144] In such way, a the signage terminal 60 is configured.

[0145] Next, the configuration of the boarding gate apparatus 70 will be described by using FIG. 7. FIG. 7 is a block diagram illustrating one example of the hardware configuration of the boarding gate apparatus 70. The boarding gate apparatus 70 is an apparatus that confirms that the user U is a passenger of an airplane that is boardable from the boarding gate P6 to which the boarding gate apparatus 70 is installed.

[0146] As illustrated in FIG. 7, the boarding gate apparatus 70 has a CPU 702, a RAM 704, a storage device 706, an input device 708, a display 710, a biometrics information acquisition device 712, a gate 714, and a communication unit 716. The CPU 702, the RAM 704, the storage device 706, the input device 708, the display 710, the biometrics information acquisition device 712, the gate 714, and the communication unit 716 are connected to a bus line 718.

[0147] The CPU 702 functions as a control unit that operates by executing a program stored in the storage device 706 and controls the operation of the entire boarding gate apparatus 70. Further, the CPU 702 executes an application program stored in the storage device 706 to perform various processes as the boarding gate apparatus 70. The RAM 704 provides a memory field necessary for the operation of the CPU 702.

[0148] More specifically, the CPU 702 functions as a comparison request unit that requests comparison of a face image and confirmation of boarding information on the user U boarding an airplane from the boarding gate P6.

[0149] The CPU 702 as the comparison request unit requests the management server 10 to compare, at 1:N, a target face image, which is a face image of the user U

captured by the biometrics information acquisition device 712, with a plurality of registered face images registered in the user information DB 106a of the management server 10. Thus, the CPU 702 transmits a target face image captured by the biometrics information acquisition device 712 to the management server 10 as target biometrics information together with a comparison request. Note that, instead of transmitting a target face image as target biometrics information, the CPU 702 may transmit a face feature amount, which is a feature amount extracted from a target face image, and request the comparison.

[0150] Note that, instead of requesting the comparison at 1:N described above, the CPU 702 may request the management server 10 to compare a target face image of the user U with registered face images associated with information on a passport, a boarding pass, or a code symbol such as a QR code of the user U. In such a case, when registering a captured face image or a passport face image of the user U acquired by the check-in terminal 20 to the user information DB 106a as a registered face image, the management server 10 performs registration in association with information on a passport, a boarding pass, or a code symbol of the user U. The CPU 702 reads information on a passport, a boarding pass, or a code symbol of the user U by using a medium reading device (not illustrated) and transmits the read information to the management server 10 together with a target face image. The management server 10 acquires, from the user information DB 106a, a registered face image associated with information on a passport, a boarding pass, or a code symbol which matches information on a passport, a boarding pass, or a code symbol transmitted from the boarding gate apparatus 70. The management server 10 compares, at 1:1, a target face image transmitted from the boarding gate apparatus 70 with a registered information face image acquired from the user information DB 106a.

[0151] Further, the CPU 702 as the comparison request unit requests the management server 10 to confirm that boarding information associated with a registered face image matching a target face image is associated with an airplane boardable from the boarding gate P6.

[0152] Further, the CPU 702 functions as a comparison result acquisition unit that acquires, from the management server 10, comparison result information indicating a result of comparison requested from the management server 10. In a case of 1:N comparison, comparison result information indicates that a registered face image matching a target face image was found and the identity verification of the user U boarding an airplane succeeded or that no matching registered face image was found and the identity verification of the user U failed, as a result of the comparison. Further, comparison result information indicates that the boarding information is confirmed to be related to an airplane and the confirmation of the boarding information succeeded or that the boarding information is not confirmed to be related to the airplane and the confirmation of the boarding information failed. In a case of 1:1 comparison, comparison result information indicates that a target face image matches a registered face image and the identity verification of the user U boarding an airplane succeeded or that no matching is obtained and the identity verification of the user U failed, as a result of the comparison.

[0153] Note that, instead of functioning as the comparison request unit and the comparison result acquisition unit, the CPU 702 may be configured to function as a comparison unit

that compares a target face image with one or more registered face images in the user information DB 106a at 1:N or 1:1 as described above and outputs comparison result information. Further, the CPU 702 may be configured to confirm that boarding information associated with a registered face image matching a target face image is associated with an airplane boardable from the boarding gate P6. In such cases, the CPU 702 can compare a face image and confirm boarding information by referencing the user information DB 106a of the management server 10 or referencing the user information DB stored in the storage device 706 in synchronization with the user information DB 106a.

[0154] Further, when the comparison result information indicates that the identity verification of the user U succeeded and the confirmation of boarding information succeeded, the CPU 702 controls the gate 714 to permit the user U to pass through the gate 714.

[0155] Further, the CPU 702 functions as a status information transmission unit that transmits status information, which is information on the status of the user U. When permitting the user U to pass through the gate 714 to complete the boarding on the airplane, the CPU 702 as the status information transmission unit transmits, to the management server 10, status information on the user U indicating that the boarding on an airplane is completed. The status information is associated with at least any of user information on the user U or face information, passport information, and boarding information included in the user information in the management server 10.

[0156] Further, the CPU 702 functions as a display control unit that causes the display 710 to display a window such as guidance, notification, or the like to the user U. For example, the CPU 702 as the display control unit causes the display 710 to display a guidance window showing how to use the boarding gate apparatus 70. Further, when comparison result information indicates that the identity verification of the user U failed, the CPU 702 causes the display 710 to display a notification window that notifies the user U that the identity verification failed. Further, when comparison result information indicates that the confirmation of boarding information failed, the CPU 702 causes the display 710 to display a notification window that notifies the user U that the confirmation of boarding information failed.

[0157] The storage device 706 is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device 706 stores a program executed by the CPU 702, data referenced by the CPU 702 when the program is executed, or the like.

[0158] The input device 708 is a touchscreen embedded in the display 710, for example. The input device 708 functions as an input unit that accepts input of an instruction from the user U. The user U is able to input various information or input an instruction of execution of a process to the boarding gate apparatus 70 via the input device 708.

[0159] The display 710 functions as a display unit that displays various windows to the user U boarding an airplane from the boarding gate P6. For example, the display 710 displays a guidance window showing how to use the boarding gate apparatus 70, a notification window to the user U, or the like.

[0160] The biometrics information acquisition device 712 functions as a biometrics information acquisition unit that acquires a face image of the user U as biometrics informa-

tion on the user U boarding an airplane from the boarding gate P6. For example, the biometrics information acquisition device 712 is a digital camera that captures a front area of the boarding gate apparatus 70 and, when detecting a face of the user U standing in front of the boarding gate apparatus 70 in an image captured continuously or periodically, captures the face of the user U and acquires the face image thereof.

[0161] When identity verification of the user U and confirmation of boarding information are successful in the boarding gate apparatus 70, the gate 714 transfers from a closed state for a standby to block passage of the user U to an opened state to permit passage of the user U under the control of the CPU 702. The scheme of the gate 714 is not particularly limited and may be, for example, a flapper gate in which one or more flappers provided on one side or both sides of the passage are opened or closed, a turn-style gate in which three bars are revolved, or the like.

[0162] The communication unit 716 is connected to the network NW and transmits and receives data via the network NW. The communication unit 716 communicates with the management server 10 or the like under the control of the CPU 702.

[0163] In such a way, the boarding gate apparatus 70 is configured.

[0164] Next, the configuration of the management server 10 will be described by using FIG. 8. FIG. 8 is a block diagram illustrating example of the hardware configuration of the management server 10. The management server 10 is an information processing apparatus that manages user information and the status on the user U and compares a target face image with a plurality of registered face images in user information DB 106a in response to a face image comparison request. Note that the function of the management server 10 may be implemented by a single server or may be implemented by a plurality of server apparatuses.

[0165] As illustrated in FIG. 8, the management server 10 has a CPU 102, a RAM 104, a storage device 106, and a communication unit 108. The CPU 102, the RAM 104, the storage device 106, and the communication unit 108 are connected to a bus line 110.

[0166] The CPU 102 functions as a control unit that operates by executing a program stored in the storage device 106 and controls the operation of the entire management server 10. Further, the CPU 102 executes an application program stored in the storage device 106 to perform various processes as the management server 10. The RAM 104 provides a memory field necessary for the operation of the CPU 102.

[0167] More specifically, the CPU 102 functions as an information management unit that stores user information on the user U received from the check-in terminal 20 in the storage device 106 and manages the stored user information. The CPU 102 as the information management unit registers user information received from the check-in terminal 20 to the user information DB 106a stored in the storage device 106 and manages the registered user information. The CPU 102 registers the received user information to the user information DB 106a every time user information is received from the check-in terminal 20. As described above, user information on the user U includes identity information, face information, and boarding information on the user U associated with each other. Face information corresponds to a captured face image or a passport face image acquired by

the check-in terminal 20. A registered face image, which is a captured face image or a passport face image registered in the user information DB 106a, is used for comparison of a face image used for identity verification of the user U in the automatic baggage deposit machine 30, the security inspection apparatus 40, the automated gate apparatus 50, and the boarding gate apparatus 70.

[0168] Further, the CPU 102 functions as a status management unit that stores the status of the user U regarding a plurality of procedures up to the boarding on an airplane by the user U in the storage device 106 and manages the stored status. The CPU 102 as the status management unit stores and manages the status of the user U in the user information DB 106a in association with the user information on the user U. The CPU 102 manages the status of the user U regarding each procedure based on status information received from the check-in terminal 20, the automatic baggage deposit machine 30, the security inspection apparatus 40, the automated gate apparatus 50, and the boarding gate apparatus 70, which are external apparatuses. Further, the CPU 102 manages the status of the user U regarding presence or passage in front of the signage terminal 60, which is a procedure performed by the user U, based on a comparison result of a target face image received from the signage terminal 60. Furthermore, the CPU 102 manages the status regarding each procedure of the user U together with deadline time, which is a deadline set as described later for a plurality of procedures up to boarding on an airplane by the user U.

[0169] Specifically, the CPU 102 manages the status of the user U regarding a check-in procedure. That is, in response to receiving status information on the user U from the check-in terminal 20 that indicates the completion of the check-in procedure, the CPU 102 updates the status of the user U of interest regarding the check-in procedure in the user information DB 106a from an unfinished state to a completed state.

[0170] Further, the CPU 102 manages the status of the user U regarding a baggage deposit procedure. That is, in response to receiving status information on the user U from the automatic baggage deposit machine 30 that indicates the completion of the baggage deposit procedure, the CPU 102 updates the status of the user U of interest regarding the baggage deposit procedure in the user information DB 106a from an unfinished state to a completed state. Note that a baggage deposit procedure by the user U is an optional procedure performed when it is necessary to deposit baggage. Thus, the status of the user U regarding a baggage deposit procedure may remain unfinished. This is intended for a case where some users U have less baggage and do not deposit their baggage.

[0171] Further, the CPU 102 manages the status of the user U regarding a security inspection procedure. That is, in response to receiving status information on the user U from the security inspection apparatus 40 that indicates the completion of the security inspection procedure, the CPU 102 updates the status of the user U of interest regarding the security inspection procedure in the user information DB 106a from an unfinished state to a completed state.

[0172] Further, the CPU 102 manages the status of the user U regarding an immigration procedure. That is, in response to receiving status information on the user U from the automated gate apparatus 50 that indicates the completion of the immigration procedure, the CPU 102 updates the status

of the user U of interest regarding the immigration procedure in the user information DB 106a from an unfinished state to a completed state.

[0173] Further, the CPU 102 manages the status of the user U regarding boarding on an airplane. That is, in response to receiving status information on the user U from the boarding gate apparatus 70 that indicates the completion of the boarding on the airplane, the CPU 102 updates the status of the user U of interest regarding the boarding on the airplane in the user information DB 106a from an unfinished state to a completed state.

[0174] Furthermore, the CPU 102 also manages the status of the user U regarding the presence or passage in front of the signage terminal 60. In such a case, in response to receiving a target face image from the signage terminal 60, the CPU 102 compares the received target face image with a plurality of registered face images in the user information DB 106a. When a matching registered face image is found as a result of the comparison, the CPU 102 updates the status of the user U of interest regarding the presence or passage in front of the signage terminal 60 in the user information DB 106a from an unfinished state to a completed state.

[0175] In such a way, the CPU 102 as the status management unit manages the status regarding a check-in procedure, a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal 60, and boarding on the airplane, which are a plurality of procedures of the user U up to the boarding. The CPU 102 manages the status of the user U regarding each procedure based on a result of the comparison between a target face image acquired in each procedure and one or more registered face images. That is, when there is a matching in comparison between a target face image acquired in each procedure and a registered face image, the CPU 102 updates the status of the user U of interest regarding each procedure from an unfinished state to a completed state. One procedure included in the plurality of procedures in the airport A and another procedure subsequent to the one procedure mean two procedures adjacent in the order of the plurality of procedures from a check-in procedure to boarding on an airplane, for example, and another procedure subsequent to one procedure means the next procedure to the one procedure. For example, specifically, when the security inspection procedure is one procedure, the immigration procedure is another procedure subsequent to the one procedure.

[0176] Note that the CPU 102 is not necessarily required to manage the status of the user U regarding all the above plurality of procedures and may manage the status of the user U regarding any one or more procedures out of the above procedures.

[0177] Further, the CPU 102 functions as a deadline setting unit that, based on boarding information on the user U, sets deadline time, which is predetermined time by which the user U has to complete a plurality of procedures of the user U including each procedure in the airport A. The procedures for which deadline time is set are a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane after a check-in procedure. Further, the CPU 102 handles the presence or passage of the user U in front of the signage terminal 60 as a procedure for which deadline time is set. The CPU 102 sets deadline time for the plurality of procedures up to the boarding on an airplane by the user U.

[0178] Note that the deadline time by which each procedure has to be completed may be ideal deadline time or may be punctual deadline time. The ideal deadline time is the time by which a procedure is ideally completed, that is, the time by which a procedure has to be completed in terms of smooth progress of the procedure but after which the procedure can be performed. The punctual deadline time is the time which is later than the ideal deadline time and by which a procedure is required to be completed, that is, the time on and after which the procedure is closed. The CPU 102 as the deadline setting unit may set either one of the ideal deadline time and the punctual deadline time or may set deadline time stepwise to set the ideal deadline time as first deadline time and set the punctual deadline time as second deadline time. Further, the CPU 102 may set multiple sets of time stepwise as the ideal deadline time.

[0179] The CPU 102 as the deadline setting unit calculates, from the departure time in the departure date and time included in boarding information on the user U, the boarding time when boarding on the airplane by the user U is closed. For example, the CPU 102 can calculate the time of 10 minutes before the departure time of an airplane as the boarding time. Further, the CPU 102 can acquire the boarding time included in the boarding information itself on the user U J from the boarding information.

[0180] From the calculated or acquired boarding time, the CPU 102 as the deadline setting unit calculates, for each procedure each deadline time by which the user U has to complete each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal 60, and boarding on the airplane. The CPU 102 can set respective deadlines for a plurality of procedures based on the order of the plurality of procedures up to the boarding on the airplane by the user U and the boarding information. That is, the CPU 102 can set a later deadline of the procedure of interest for the order of the plurality of procedures up to the boarding on the airplane by the user U is closer to the boarding.

[0181] Note that, typically, the user U may not necessarily be present in front of all the signage terminals 60 installed at respective places within the airport A or pass by in front of all the signage terminals 60. Thus, the CPU 102 is not required to set deadline time for presence or passage in front of respective signage terminals 60 installed at respective places within the airport A. The CPU 102 may set deadline time for presence or passage in front of the signage terminal 60 installed at a particular place where the user U necessarily passes or is highly likely to pass, such as the passage P5, for example.

[0182] For example, the CPU 102 can calculate the time of 60 minutes before the boarding time as the deadline time for the baggage deposit procedure. Further, for example, the CPU 102 can calculate the time of 30 minutes before the boarding time as the deadline time for the security inspection procedure. Further, for example, the CPU 102 can calculate the time of 20 minutes before the boarding time as the deadline time for the immigration procedure. Further, for example, the CPU 102 can calculate the time of 15 minutes before the boarding time as the deadline time for the presence or passage in front of the signage terminal 60 on the passage P5. Further, for example, the CPU 102 can calculate the time of 10 minutes before the boarding time as the deadline time for the boarding on the airplane. Note that the CPU 102 can calculate each deadline time for each

procedure from the departure time itself included in the boarding information. Further, the CPU 102 can acquire the boarding time or the departure time with reference to a database or the like regarding the flight schedule from the flight number of the airplane included in the boarding information and calculate the deadline time for each procedure.

[0183] Note that the CPU 102 as the deadline setting unit may set deadline time in accordance with the number of users U who have performed the check-in procedure. That is, the CPU 102 may set a time period between the deadline time and the boarding time to be shorter when the number of users U who have performed the check-in procedure is larger.

[0184] Further, the CPU 102 as the deadline setting unit may appropriately change and set the deadline time for each procedure described above in accordance with the degree of congestion in the airport A, which is a facility in which the user U boards on an airplane, in other words, in accordance with the number of users U of the airport A. For example, the CPU 102 can determine the degree of congestion in the airport A based on the number of pieces of user information registered to the user information DB 106a per unit time. That is, when the number of pieces of user information registered to the user information DB 106a per unit time is larger, it can be determined that the airport A is more congested. When the airport A is more congested, the CPU 102 may set the deadline time to earlier time.

[0185] Further, the CPU 102 as the deadline setting unit stores in the user information DB 106a and manages the set deadline time for each procedure in association with the user information and the status on the user U in the user information DB 106a, for example. That is, the CPU 102 as the status management unit can manage the status of the user U regarding each procedure together with the deadline time that is a set deadline.

[0186] The CPU 102 as the deadline setting unit sets deadline time as deadlines for a check-in procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal 60, and boarding on an airplane, which are a plurality of procedures of the user U up to boarding, as described above. Note that the CPU 102 is not necessarily required to set deadline time for all the above plurality of procedures and may set deadline time for any one or more procedures out of the above procedures.

[0187] Further, the CPU 102 functions as a determination unit that, for each procedure of the user U up to boarding for which deadline time is set as described above, determines whether or not the deadline time has expired with the unfinished status. The CPU 102 as the determination unit can determine whether or not the deadline time has expired for an unfinished procedure regularly or irregularly, for example.

[0188] Further, the CPU 102 functions as an alert unit to generate and transmit an alert indicating that a procedure of the user U for which the deadline time is determined to expire with the unfinished status after the deadline time has expired. The CPU 102 as the alert unit may be transmitted to the operation terminal 80 carried by a staff member S. When the location of the user U is determined as described in a second example embodiment, for example, the CPU 102 may transmit an alert to the operation terminal 80 carried by the staff member S near the place. Furthermore, the CPU 102

may notify the operation terminals 80 carried by the nearby staff member S and other staff members S of the staff member S who is near the determined location of the user U or display the staff member S who is near the determined location of the user U on the operation terminals 80 carried by the nearby staff member S and other staff members S. Thereby, the CPU 102 can teach which staff member S is appropriate to address the alert.

[0189] For example, an alert may include a part or whole of the user information on the user U who has not yet completed the procedure, a part or whole of the boarding information, or the status regarding each procedure. An alert may include a registered face image of the user U, preferably, a registered face image which is a captured face image of the user U captured by the check-in terminal 20. The staff member S who has received an alert by the operation terminal 80 is able to rely on a registered face image included in the alert to search for the user U who has not yet completed the procedure. Note that, while a registered face image included in an alert may be a passport face image of the user U, a registered face image included in an alert is preferably a captured face image captured by the check-in terminal 20 that indicates the recent face of the user U, namely, the face captured at the time of check-in procedure.

[0190] Further, the CPU 102 functions as a comparison unit that compares a face image of the user U when receiving a target face image and a comparison request or a target face image from the automatic baggage deposit machine 30, the security inspection apparatus 40, the automated gate apparatus 50, the signage terminal 60, and the boarding gate apparatus 70. The CPU 102 as the comparison unit compares, at 1:N, a target face image corresponding to target biometrics information with a plurality of registered face images corresponding to registered biometrics information registered in the user information DB 106a. Note that user information on a plurality of users U who have performed the check-in procedure by using the check-in terminal 20 is registered in the user information DB 106a. When comparing a target face image with a registered face image, the CPU 102 can compare a face feature amount extracted from the target face image corresponding to target biometrics information with a face feature amount extracted from the registered face image corresponding to registered biometrics information and thereby compare both face images.

[0191] The CPU 102 as the comparison unit compares a target face image with a plurality of registered face images and attempts to find a registered face image matching the target face image out of the plurality of registered face images. The CPU 102 also functions as a transmission unit to transmit comparison result information indicating a result of comparison to each apparatus of the automatic baggage deposit machine 30, the security inspection apparatus 40, the automated gate apparatus 50, and the boarding gate apparatus 70 which have transmitted a comparison request. The comparison result information indicates that there is a matching in the comparison or there is no matching in the comparison. That is, the comparison result information indicates that a registered face image matching a target face image was found and identity verification of the user U performing a baggage deposit procedure succeeded or that no matching registered face image was found and identity verification of the user U failed as a result of the comparison. Note that the CPU 102 may compare a target face image with a single registered face image.

[0192] Further, instead of the CPU 202 of the check-in terminal 20, the CPU 102 may also compare, at 1:1, a captured face image captured by the biometrics information acquisition device 214 with a passport face image acquired by the medium reading device 212. In such a case, the CPU 102 receives a captured face image and a passport face image to be compared or a face feature amount thereof from the check-in terminal 20. Further, the CPU 102 transmits comparison result information indicating a result of the comparison between the captured face image and the passport face image to the check-in terminal 20.

[0193] Further, the CPU 102 functions as a display management unit that causes the signage terminal 60 to display different display information in accordance with remaining time to the boarding time for the user U in front of the signage terminal 60 who succeeded in identity verification by the comparison with a target face image, as described below.

[0194] First, the CPU 102 functions as a time calculation unit that calculates remaining time to the boarding time for the user in front of the signage terminal 60 who succeeded in identity verification by the comparison with a target face image transmitted from the signage terminal 60 and identified. The CPU 102 acquires boarding information on the identified user U by referencing the user information DB 106a to calculate the remaining time. As described above, the CPU 102 calculates the boarding time from the departure time included in the boarding information or acquires the boarding time included in the boarding information. Furthermore, the CPU 102 calculates remaining time to the boarding time based on the boarding information calculated or acquired from the boarding information.

[0195] Further, the CPU 102 functions as a display information generation unit that generates display information to be displayed on the signage terminal 60 in accordance with the calculated remaining time to the boarding time. The CPU 102 as the display information generation unit generates different display information in accordance with the length of the remaining time, that is, whether little time remains before the boarding time or enough time remains before the boarding time.

[0196] To generate display information, the CPU 102 compares the remaining time to the boarding time based on boarding information with a predetermined threshold and determines the length of the remaining time, that is, whether little time remains before the boarding time or enough time remains before the boarding time. When the remaining time is less than or equal to the predetermined threshold, the CPU 102 determines that the remaining time is short, that is, little time remains before the boarding time. On the other hand, when the remaining time exceeds the predetermined threshold, the CPU 102 determines that the remaining time is long, that is, enough time remains before the boarding time. The CPU 102 may set change as appropriate and sets the threshold that is a criterion of determination in accordance with the degree of congestion of the airport A, in other words, the number of users U of the airport A. The CPU 102 may set the threshold that is a criterion of determination to be longer when the airport A is more congested. The degree of congestion of the airport A is determined as described above.

[0197] When determining that little time remains before the boarding time, the CPU 102 generates first display information as display information to be displayed on the

signage terminal 60 to the user U identified in front of the signage terminal 60. The first display information includes the boarding information on the user U, the remaining time to the boarding time, the deadline time for each procedure, the status of each procedure, or the like, for example. The boarding information includes boarding information such as the departure time, the flight number, the origin, the destination, or the like of the boarding flight which the user U boards, for example. The deadline time is the deadline time for a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane, for example. Deadline information included in the first display information may be information on an ideal deadline or may be the punctual deadline time. Further, progress information indicates a progress status as to whether or not each procedure is completed, for example.

[0198] Note that the CPU 102 may exclude the deadline time for an already completed procedure from the first display information by referencing the status registered in the user information DB 106a. That is, the CPU 102 may include only the deadline time for one or more unfinished procedures in the first display information out of respective procedures. In such a case, the CPU 102 may additionally exclude the progress information on the one or more already completed procedures from the first display information.

[0199] Further, the first display information may include the congestion information, which is information indicating the degree of congestion of the security inspection site P3, the immigration site P4, or the like, which are procedure sites, for example. The congestion information includes availability information that identifies a relatively less congested procedure site or the least congested procedure site out of a plurality of procedure sites where the same procedure is performed, for example. Further, the congestion information includes waiting time required before completion of a procedure at a procedure site such as the security inspection site P3, the immigration site P4, or the like or the number of waiting persons who are waiting for a procedure, such as standing in a line for a procedure site, for example. The waiting time is expected waiting time automatically calculated based on the number of persons counted by a human detection sensor, a camera, or the like installed in the security inspection site P3, the immigration site P4, or the like, for example. Note that congestion information such as availability information, waiting time, the number of waiting persons in a line, or the like may be calculated by using the number of faces detected by a camera installed to a security inspection apparatus or an immigration booth, a security camera installed in each room, or the like other than the number of persons counted by a human detection sensor. For example, congestion information is transmitted regularly or irregularly to the management server 10 from an information terminal installed in the security inspection site P3, the immigration site P4, or the like via the network NW and can be referenced by the CPU 102.

[0200] Note that the first display information is not necessarily required to include all of the boarding information on the user U, the remaining time to the boarding time, the deadline time for each procedure, the status of each procedure, and congestion information described above and may include one or multiple pieces of the above information.

[0201] On the other hand, when determining that enough time remains before the boarding time, the CPU 102 generates second display information as display information to



be displayed on the signage terminal **60** to the identified user U in front of the signage terminal **60**. The display second information includes advertisement information or the like, for example. Advertisement information may be a general advertisement whose content is not different for the user U or may be a targeting advertisement whose content is different for the user U in accordance with the user U.

[0202] The CPU **102** can acquire advertisement information by referencing an advertisement information DB **106b** stored in the storage device **106**, an advertisement DB in an external server, or the like and generate the second display information including the acquired advertisement information.

[0203] In the case of a targeting advertisement, the CPU **102** acquires attribute information on the user U, such as an age, a sexuality, or the like from user information on the user U registered in the user information DB **106a**, for example. Further, the CPU **102** can perform age estimation, sexuality estimation, or the like by image processing on a target face image of the user U transmitted from the signage terminal **60** to acquire attribute information on the user U, such as an age, a sexuality, or the like, for example. The CPU **102** can generate advertisement information in accordance with attribute information on the acquired user U. The CPU **102** can select and acquire advertisement information in accordance with attribute information on the user U from the advertisement information DB **106b**, an advertisement DB in an external server, or the like.

[0204] Further, for example, if it is possible to acquire purchase history information on the user U recorded in an external server based on credit card information or the like used when the user U purchased the boarding pass, the CPU **102** can also acquire advertisement information in accordance with purchase history information acquired from an external server. In such a case, the user U may provide allowance in advance for acquisition of purchase history information by the management server **10**. The CPU **102** can select and acquire advertisement information in accordance with purchase history information on the user U from the advertisement information DB **106b**, an advertisement DB in an external server, or the like.

[0205] The CPU **102** transmits the first display information or the second display information generated as described above via the network NW to the signage terminal **60** in front of which the identified user U is present. The signage terminal **60** causes the display **608** to display the received first display information or second display information.

[0206] Note that advertisement information displayed on the signage terminal **60** may be pre-stored in the storage device **606** of the signage terminal **60** other than network-delivery advertisement information transmitted from the management server **10** as described above. In such a case, the CPU **102** transmits advertisement designation information, which is information that designates advertisement information to be displayed, as the second display information to the signage terminal **60**. The signage terminal **60** can acquire the advertisement information designated by the received advertisement designation information from the storage device **606** to display the acquired designated advertisement information on the display **608**.

[0207] As described above, the CPU **102** causes the signage terminal **60** to display different display information in accordance with the remaining time to the boarding time

based on boarding information on the user U identified by the comparison between target biometrics information transmitted from the signage terminal **60** and registered biometrics information.

[0208] The storage device **106** is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device **106** stores a program executed by the CPU **102**, data referenced by the CPU **102** when the program is executed, or the like.

[0209] Further, the storage device **106** stores the user information DB **106a**. User information on a plurality of users U who have performed the check-in procedure by using the check-in terminal **20** is registered in the user information DB **106a**. Further, the status of the user U is registered in the user information DB **106a** in association with the user information.

[0210] Further, the storage device **106** stores the advertisement information DB **106b** therein. Various pieces of advertisement information displayed on the signage terminal **60** are registered in the advertisement information DB **106b**. The CPU **102** can select and acquire advertisement information in accordance with the user U from the advertisement information DB **106b**.

[0211] The communication unit **108** is connected to the network NW and transmits and receives data via the network NW. The communication unit **108** communicates with the check-in terminal **20**, the automatic baggage deposit machine **30**, the security inspection apparatus **40**, the automated gate apparatus **50**, the signage terminal **60**, the boarding gate apparatus **70**, or the like under the control of the CPU **102**.

[0212] In such a way, the management server **10** is configured.

[0213] Next, the configuration of the operation terminal **80** will be described by using FIG. 9. FIG. 9 is a block diagram illustrating one example of the hardware configuration of the operation terminal **80**. The operation terminal **80** is a terminal carried by the staff member S such as a staff of the airline company, a staff of an airport company, or the like to use for their operation and may be, for example, a tablet terminal, a smartphone, a mobile phone, or the like. Note that the operation terminal **80** may be, for example, a stationary type information processing terminal installed in an operation place such as a counter where a staff engages in their operation.

[0214] As illustrated in FIG. 9, the operation terminal **80** has a CPU **802**, a RAM **804**, a storage device **806**, an input device **808**, a display **810**, a mobile wireless communication unit **812**, and a short-range wireless communication unit **814**. The CPU **802**, the RAM **804**, the storage device **806**, the input device **808**, the display **810**, the mobile wireless communication unit **812**, and the short-range wireless communication unit **814** are connected to a bus line **816**.

[0215] The CPU **802** functions as a control unit that operates by executing a program stored in the storage device **806** and controls the operation of the entire operation terminal **80**. Further, the CPU **802** executes an application program stored in the storage device **806** to perform various processes as the operation terminal **80**. The RAM **804** provides a memory field necessary for the operation of the CPU **802**.

[0216] More specifically, the CPU **802** functions as an information acquisition unit and can access the user infor-

mation DB **106a** of the management server **10** to acquire the user information and the status on the user **U** stored in the user information DB **106a**. The CPU **802** as the information acquisition unit can cause the display **810** to display the acquired user information and status.

[0217] Further, the CPU **802** function as an alert receiving unit and can receive an alert transmitted from the management server **10**. The CPU **802** as the alert receiving unit can cause the display **810** to display the received alert.

[0218] The storage device **806** is formed of a storage medium such as a non-volatile memory, a hard disk drive, or the like and functions as a storage unit. The storage device **806** stores a program executed by the CPU **802**, data referenced by the CPU **802** when the program is executed, or the like.

[0219] The input device **808** is a touchscreen embedded in the display **810**, for example. The input device **808** functions as an input unit that accepts input of an instruction from the staff member **S**. The staff member **S** is able to input various information or input an instruction of execution of a process to the operation terminal **80** via the input device **808**.

[0220] The display **810** functions as a display unit that displays various windows to the staff member **S**. For example, the display **810** displays a window indicating the content of the user information and the status on the user **U** stored in the user information DB **106a**, a window indicating the contents of an alert received from the management server **10**, or the like.

[0221] The mobile wireless communication unit **812** is connected to the network **NW** via a mobile communication network under the control of the CPU **802**. Without being limited in particular, the communication scheme of the mobile wireless communication unit **812** may be, for example, the third generation mobile communication scheme, the Long Term Evolution (LTE) scheme, the fourth generation mobile communication scheme, or the like.

[0222] The short-range wireless communication unit **814** performs wireless communication with an access point, an external device, or the like under the control of the CPU **802** and is connected to the network **NW** via an access point in a premise of the airport **A**, for example. Without being limited in particular, the communication scheme of the short-range wireless communication unit **814** may be, for example, a wireless LAN scheme such as Wi-Fi (registered trademark), a Bluetooth (registered trademark) communication scheme, a Near Field Communication (NFC) scheme, an infrared communication scheme, or the like.

[0223] In such a way, the operation terminal **80** is configured.

[0224] The information processing system **1** according to the present example embodiment sets deadline time for each procedure up to boarding based on boarding information acquired by the check-in terminal **20** for the user **U** scheduled to board an airplane and manages the status of the user **U** regarding each procedure in the management server **10**. Therefore, according to the information processing system **1** of the present example embodiment, it is possible to recognize and manage in more detail the status of the user **U** scheduled to board an airplane.

[0225] Further, in the information processing system **1** according to the present example embodiment, when deadline time has expired with the status of a procedure still being in an unfinished state, the management server **10** generates an alert indicating the expiration and transmits the

alert to the operation terminal **80**. Further, the alert includes a registered face image of the user **U**. Therefore, according to the information processing system **1** of the present example embodiment, the staff member **S** is able to rely on the registered face image of the user **U** and easily search for the user **U** who has not yet completed the procedure.

[0226] The operation of each component in the information processing system **1** according to the present example embodiment will be further described below by using FIG. **10** to FIG. **20C**.

[0227] First, the operation performed by the check-in terminal **20** will be described by using FIG. **10** together with the operation performed by the management server **10**. FIG. **10** is a sequence diagram illustrating the operation performed by the check-in terminal **20** and the management server **10**.

[0228] The user **U** who arrived at the airport **A** moves to the check-in lobby **P1** and performs a check-in procedure on the check-in terminal **20**. The check-in terminal **20** and the management server **10** perform a process illustrated in FIG. **10** every time each of the plurality of users **U** performs a check-in procedure.

[0229] As illustrated in FIG. **10**, the medium reading device **212** of the check-in terminal **20** determines whether or not a passport of the user **U** is held over the reading unit (step **S102**) and waits until a passport is held over (step **S102**, NO).

[0230] In response to determining that a passport is held over (step **S102**, YES), the medium reading device **212** acquires passport information on the user **U** from the held passport (step **S104**). The acquired passport information includes a passport face image of the user **U**. At this time, the medium reading device **212** can also acquire identity information on the user **U** from the passport.

[0231] Next, the medium reading device **212** determines whether or not a flight ticket medium of the user **U** is held over the reading unit (step **S106**) and waits until a flight ticket medium is held over (step **S106**, NO).

[0232] In response to determining that a flight ticket medium is held over (step **S106**, YES), the medium reading device **212** acquires boarding information on the user **U** from the held flight ticket medium (step **S108**). At this time, the medium reading device **212** can also acquire identity information on the user **U** from the flight ticket medium.

[0233] Note that the order of step **S104** of acquiring passport information, step **S108** of acquiring boarding information, and step **S110** of acquiring a face image is not limited to the above. Some steps of steps **S104**, **S108**, and **S110** may be performed in earlier order, or a combination of all or some of steps **S104**, **S108**, and **S110** may be performed at the same time.

[0234] Next, the biometrics information acquisition device **214** of the check-in terminal **20** captures the face of the user **U** standing in front of the check-in terminal **20** and acquires a captured face image, which is a face image of the user **U** (step **S110**).

[0235] Next, the CPU **202** of the check-in terminal **20** compares, at 1:1, a captured face image captured by the biometrics information acquisition device **214** with the passport face image acquired by the medium reading device **212** (step **S112**).

[0236] As a result of the comparison, if the CPU **202** determines that there is a matching in the comparison between the captured face image and the passport face

image and identity verification is successful (step S114, YES), the CPU 202 transmits the user information on the user U to the management server 10 via the network NW (step S116). The user information includes identity information, face information, and boarding information on the user U associated with each other.

[0237] In response to receiving user information from the check-in terminal 20, the CPU 102 of the management server 10 registers the received user information to the user information DB 106a (step S118). The CPU 102 registers a captured face image or a passport face image included in user information to the user information DB 106a as a registered face image. In such a way, the check-in terminal 20 completes a check-in procedure on the user U. The user U who has completed a check-in procedure moves to the baggage counter P2 if it is necessary to deposit baggage or, if not, moves to the security inspection site P3.

[0238] Furthermore, the CPU 202 of the check-in terminal 20 transmits, to the management server 10 via the network NW, status information on the user U indicating that the check-in procedure is completed after the comparison of a face image after the transmission or along with the transmission of user information (step S120).

[0239] In response to receiving status information from the check-in terminal 20, which is the external apparatus, the CPU 102 of the management server 10 updates the status of the user U of interest regarding the check-in procedure in the user information DB 106a from an unfinished state to a completed state (step S122). In such a way, the CPU 102 manages the status of the user U regarding the check-in procedure based on a comparison result between a captured face image acquired at a check-in procedure and a passport face image. That is, when there is a matching in the comparison between a captured face image acquired at a check-in procedure and a passport face image, the CPU 102 updates the status of the user U of interest regarding a check-in procedure from an unfinished state to a completed state.

[0240] On the other hand, as a result of the comparison, if there is no matching in the comparison between a captured face image and a passport face image and the CPU determines that identity verification fails (step S114, NO), the CPU 202 notifies the user U that the identity verification failed (step S124). At this time, the CPU 202 causes the display 210 to display a notification window notifying that the identity verification failed. Note that the CPU 202 may notify the user U that the identity verification failed by using a voice, for example, in addition to the display of the notification window or instead of the display of the notification window.

[0241] In such a way, a check-in procedure by the user U is performed on the check-in terminal 20.

[0242] Note that the user U may perform a check-in procedure at a manned check-in counter instead of a check-in procedure with the check-in terminal 20. In such a case, a terminal operated by a staff member in the check-in counter can perform the same process as that of the check-in terminal 20 described above. Further, the user U may utilize online check-in performed from a website of an airline company instead of a check-in procedure with the check-in terminal 20. In such a case, for example, the user U may separately register a face image as a registered face image from a dedicated terminal installed in the airport A to the user information DB 106a of the management server 10.

[0243] Next, the operation performed by the automatic baggage deposit machine 30 will be described by using FIG. 11 together with the operation performed by the management server 10. FIG. 11 is a sequence diagram illustrating the operation performed by the automatic baggage deposit machine 30 and the management server 10.

[0244] The user U who has completed a check-in procedure moves to the baggage counter P2 if necessary and performs a baggage deposit procedure with the automatic baggage deposit machine 30. The automatic baggage deposit machine 30 and the management server 10 perform the process illustrated in FIG. 11 every time each of the plurality of users U performs a baggage deposit procedure.

[0245] As illustrated in FIG. 11, the biometrics information acquisition device 312 of the automatic baggage deposit machine 30 continuously or periodically captures a front area of the automatic baggage deposit machine 30 and determines whether or not a face of the user U standing in front of the automatic baggage deposit machine 30 is detected in the captured image (step S202). The biometrics information acquisition device 312 waits until a face of the user U is detected in the image (step S202, NO).

[0246] If the biometrics information acquisition device 312 determines that a face of the user U is detected (step S202, YES), the biometrics information acquisition device 312 captures the face of the user U and acquires the face image of the user U as a target face image (step S204).

[0247] Next, the CPU 302 of the automatic baggage deposit machine 30 transmits the target face image of the user U captured by the biometrics information acquisition device 312 to the management server 10 via the network NW together with a comparison request (step S206). Thereby, the CPU 302 requests the management server 10 to compare, at 1:N, the target face image of the user U captured by the biometrics information acquisition device 312 with a plurality of registered face images registered in the user information DB 106a of the management server 10.

[0248] In response to receiving a target face image and a comparison request from the automatic baggage deposit machine 30, the CPU 102 of the management server 10 performs comparison of the face image of the user U (step S208). The CPU 102 compares, at 1:N, a target face image received from the automatic baggage deposit machine 30 with a plurality of registered face images registered in the user information DB 106a.

[0249] Next, the CPU 102 transmits comparison result information indicating a result of the comparison to the automatic baggage deposit machine 30 via the network NW (step S210).

[0250] The CPU 302 of the automatic baggage deposit machine 30 receives comparison result information from the management server 10. If the comparison result information indicates that a registered face image matching the target face image is found and identity verification of the user U performing the baggage deposit procedure is successful (step S212, YES), the CPU 302 performs a baggage deposit process on the user U (step S214). In the deposit process, the CPU 302 acquires boarding information or passport information associated with the registered face image matching the target face image to use the boarding information or passport information in the deposit process. The CPU 302 controls each unit of the automatic baggage deposit machine 30. Thereby, the output device 316 outputs a baggage tag and a baggage claim tag in association with the boarding

information or the passport information associated with the registered face image. Further, the baggage transport apparatus 314 transports baggage, which is placed on a receiving part by the user U and attached with the baggage tag, to a cargo sorting site. The user U who has deposited baggage moves to the security inspection site P3.

[0251] Next, the CPU 302 transmits, to the management server 10 via the network NW, status information on the user U indicating that the baggage deposit procedure is completed after the comparison of the face image (step S216).

[0252] In response to receiving status information from the automatic baggage deposit machine 30, which is an external apparatus, the CPU 102 of the management server 10 updates the status of the user U of interest regarding the baggage deposit procedure in the user information DB 106a from an unfinished state to a completed state (step S218). In such a way, the CPU 102 manages the status of the user U regarding the baggage deposit procedure based on a result of the comparison between a target face image acquired in the baggage deposit procedure and a registered face image. That is, when there is a matching in the comparison between a target face image acquired in a baggage deposit procedure and registered face images, the CPU 102 updates the status of the user U of interest regarding the baggage deposit procedure from an unfinished state to a completed state.

[0253] On the other hand, if the comparison result information indicates that no matching registered face image is found and identity verification of the user U fails (step S212, NO), the CPU 302 of the automatic baggage deposit machine 30 notifies the user U that the identity verification failed (step S220). At this time, the CPU 302 causes the display 310 to display a notification window notifying that the identity verification failed. Note that the CPU 302 may notify the user U that the identity verification failed by using a voice, for example, in addition to the display of the notification window or instead of the display of the notification window. When identity verification failed in such a way, a baggage deposit process may be performed with a normal process by causing a medium reading device (not illustrated) to read a boarding pass or a passport.

[0254] In such a way, a baggage deposit procedure is performed at the automatic baggage deposit machine 30.

[0255] Next, the operation performed by the security inspection apparatus 40 will be described by using FIG. 12 together with the operation performed by the management server 10. FIG. 12 is a sequence diagram illustrating the operation performed by the security inspection apparatus 40 and the management server 10.

[0256] The user who has completed the check-in procedure and completed the baggage deposit procedure as necessary moves to the security inspection site P3 and is subjected to a security inspection procedure by the security inspection apparatus 40. The security inspection apparatus 40 and the management server 10 perform the process illustrated in FIG. 12 every time each of the plurality of users U is subjected to the security inspection procedure.

[0257] As illustrated in FIG. 12, the biometrics information acquisition device 412 of the security inspection apparatus 40 continuously or periodically captures a front area of the metal detector gate 414 and determines whether or not a face of the user U standing in front of the metal detector gate 414 is detected in the captured image (step S302). The biometrics information acquisition device 412 waits until a face of the user U is detected in the image (step S302, NO).

[0258] If the biometrics information acquisition device 412 determines that a face of the user U is detected (step S302, YES), the biometrics information acquisition device 412 captures the face of the user U and acquires the face image of the user U as a target face image (step S304).

[0259] Next, the CPU 402 of the security inspection apparatus 40 transmits the target face image of the user U captured by the biometrics information acquisition device 412 to the management server 10 via the network NW together with a comparison request (step S306). Thereby, the CPU 402 requests the management server 10 to compare, at 1:N, the target face image of the user U captured by the biometrics information acquisition device 412 with a plurality of registered face images registered in the user information DB 106a of the management server 10.

[0260] In response to receiving a target face image and a comparison request from the security inspection apparatus 40, the CPU 102 of the management server 10 performs comparison of the face image of the user U (step S308). The CPU 102 compares, at 1:N, a target face image received from the security inspection apparatus 40 with a plurality of registered face images registered in the user information DB 106a.

[0261] Next, the CPU 102 transmits comparison result information indicating a result of the comparison to the security inspection apparatus 40 via the network NW (step S310).

[0262] The CPU 402 of the security inspection apparatus 40 receives comparison result information from the management server 10. If the comparison result information indicates that a registered face image matching the target face image is found and identity verification of the user U subjected to the security inspection procedure is successful (step S312, YES), the CPU 402 performs a security inspection process of the user U (step S314). In the security inspection process, the CPU 402 controls each unit of the security inspection apparatus 40. Thereby, the metal detector gate 414 detects a metal worn by the user U who is passing through the metal detector gate 414. The user U who has passed through the metal detector gate 414 moves to the immigration site P4.

[0263] Next, the CPU 402 transmits, to the management server 10 via the network NW, status information on the user U indicating that the security inspection procedure is completed after the comparison of the face image (step S316).

[0264] In response to receiving status information from the security inspection apparatus 40, which is an external apparatus, the CPU 102 of the management server 10 updates the status of the user U of interest regarding the security inspection procedure in the user information DB 106a from an unfinished state to completed state (step S318). In such a way, the CPU 102 manages the status of the user U regarding the security inspection procedure based on a result of the comparison between a target face image acquired in the security inspection procedure and a registered face image. That is, when there is a matching in the comparison between a target face image acquired in a security inspection procedure and registered face images, the CPU 102 updates the status of the user U of interest regarding the security inspection procedure from an unfinished state to a completed state.

[0265] On the other hand, if the comparison result information indicates that no matching registered face image is found and identity verification of the user U fails (step S312,

NO), the CPU 402 of the security inspection apparatus 40 notifies the user U that the identity verification failed (step S320). At this time, the CPU 402 causes the display 410 to display a notification window notifying that the identity verification failed. Note that the CPU 402 may notify the user U that the identity verification failed by using a voice, for example, in addition to the display of the notification window or instead of the display of the notification window.

[0266] In such a way, a security inspection procedure is performed by the security inspection apparatus 40.

[0267] Next, the operation performed by the automated gate apparatus 50 will be described by using FIG. 13 together with the operation performed by the management server 10. FIG. 13 is a sequence diagram illustrating the operation performed by the automated gate apparatus 50 and the management server 10.

[0268] The user U who has completed the security inspection procedure moves to the immigration site P4 and is subjected to an immigration procedure by the automated gate apparatus 50. The automated gate apparatus 50 and the management server 10 perform the process illustrated in FIG. 13 every time each of the plurality of users U is subjected to the immigration procedure.

[0269] As illustrated in FIG. 13, the biometrics information acquisition device 512 of the automated gate apparatus 50 continuously or periodically captures a front area of the automated gate apparatus 50 and determines whether or not a face of the user U standing in front of the automated gate apparatus 50 is detected in the captured image (step S402). The biometrics information acquisition device 512 waits until a face of the user U is detected in the image (step S402, NO).

[0270] If the biometrics information acquisition device 512 determines that a face of the user U is detected (step S402, YES), the biometrics information acquisition device 512 captures the face of the user U and acquires the face image of the user U as a target face image (step S404).

[0271] Next, the CPU 502 of the automated gate apparatus 50 transmits the target face image of the user U captured by the biometrics information acquisition device 512 to the management server 10 via the network NW together with a comparison request (step S406). Thereby, the CPU 502 requests the management server 10 to compare, at 1:N, the target face image of the user U captured by the biometrics information acquisition device 512 with a plurality of registered face images registered in the user information DB 106a of the management server 10.

[0272] In response to receiving a target face image and a comparison request from the automated gate apparatus 50, the CPU 102 of the management server 10 performs comparison of the face image of the user U (step S408). The CPU 102 compares, at 1:N, a target face image received from the automated gate apparatus 50 with a plurality of registered face images registered in the user information DB 106a.

[0273] Next, the CPU 102 transmits comparison result information indicating a result of the comparison to the automated gate apparatus 50 via the network NW (step S410).

[0274] The CPU 502 of the automated gate apparatus 50 receives comparison result information from the management server 10. If the comparison result information indicates that a registered face image matching the target face image is found and identity verification of the user U

subjected to the immigration procedure is successful (step S412, YES), the CPU 502 performs an opening process of the gate 514 (step S414). Thereby, the CPU 502 transfers the gate 514 from a closed state for standby that blocks passage of the user U to an opened state that allows passage of the user U. The user U passes through the opened gate 514 and moves to the boarding gate P6.

[0275] Next, the CPU 502 transmits, to the management server 10 via the network NW, status information indicating that the immigration procedure on the user U is completed after the comparison of the face image (step S416).

[0276] In response to receiving status information from the automated gate apparatus 50, which is an external apparatus, the CPU 102 of the management server 10 updates the status of the user U of interest regarding the immigration procedure in the user information DB 106a from an unfinished state to a completed state (step S418). In such a way, the CPU 102 manages the status of the user U regarding the immigration procedure based on a result of the comparison between a target face image acquired in the immigration procedure and a registered face image. That is, when there is a matching in the comparison between a target face image acquired in an immigration procedure and registered face images, the CPU 102 updates the status of the user U of interest regarding the immigration procedure from an unfinished state to a completed state.

[0277] On the other hand, if the comparison result information indicates that no matching registered face image is found and identity verification of the user U fails (step S412, NO), the CPU 502 of the automated gate apparatus 50 notifies the user U that the identity verification failed (step S420). At this time, the CPU 502 causes the display 510 to display a notification window notifying that the identity verification failed. Note that the CPU 502 may notify the user U that the identity verification failed by using a voice, for example, in addition to the display of the notification window or instead of the display of the notification window.

[0278] In such a way, an immigration procedure is performed by the automated gate apparatus 50.

[0279] Next, the operation performed by the signage terminal 60 will be described by using FIG. 14 together with the operation performed by the management server 10. FIG. 14 is a sequence diagram illustrating the operation performed by the signage terminal 60 and the management server 10.

[0280] The user U who has completed the immigration procedure passes through the passage P5 on which the signage terminal 60 is installed and moves to the boarding gate P6. At this time, the user U passes by in front of the signage terminal 60 installed on the passage P5 or once stops to be present in front of the signage terminal 60 to check the guide showing the direction to the boarding gate P6 displayed on the signage terminal 60. The signage terminal 60 on the passage P5 and the management server 10 perform the process illustrated in FIG. 14 every time each of the plurality of users U passes by in front of the signage terminal 60 or is present in front of the signage terminal 60. Note that the signage terminals 60 installed in other places within the airport A other than the passage P5 can also perform the process illustrated in FIG. 14 in the same manner as the signage terminal 60 installed on the passage P5.

[0281] As illustrated in FIG. 14, the biometrics information acquisition device 610 of the signage terminal 60 installed on the passage P5 continuously or periodically

captures a front area of the signage terminal 60 and determines whether or not a face of the user U in front of the signage terminal 60 is detected in the captured image (step S502). The user U in front of the signage terminal 60 is the user U who passes by the area in front of the signage terminal 60 or is present in front of the signage terminal 60. The biometrics information acquisition device 610 waits until a face of the user U is detected in the image (step S502, NO).

[0282] If the biometrics information acquisition device 610 determines that a face of the user U is detected (step S502, YES), the biometrics information acquisition device 610 captures the face of the user U and acquires the face image of the user U as a target face image (step S504).

[0283] Next, the CPU 602 of the signage terminal 60 transmits the target face image of the user U captured by the biometrics information acquisition device 610 to the management server 10 via the network NW (step S506).

[0284] In response to receiving a target face image from the signage terminal 60, which is an external apparatus, the CPU 102 of the management server 10 performs comparison of the face image of the user U (step S508). The CPU 102 compares, at 1:N, a target face image received from the signage terminal 60 with a plurality of registered face images registered in the user information DB 106a.

[0285] Note that the CPU 602 of the signage terminal 60 can be configured to stream a video of the front area of the signage terminal 60 to the management server 10. In such a case, the CPU 102 of the management server 10 may detect and acquire a target face image of the user U from the video streamed from the signage terminal 60 instead of receiving a target face image from the signage terminal 60.

[0286] As a result of the comparison, if a registered face image matching the target face image is found and the user U in front of the signage terminal 60 is identified (step S510, YES), the CPU 102 updates the status of the user U of interest (step S512). That is, the CPU 102 updates the status of the user U of interest regarding the presence or passage in front of the signage terminal 60 in the user information DB 106a from an unfinished state to a completed state. In such a way, the CPU 102 manages the status of the user U regarding the presence or passage in front of the signage terminal 60 based on a result of the comparison between a target face image acquired in the presence or passage in front of the signage terminal 60 and the registered face images. That is, when there is a matching in the comparison between the target face image acquired in the presence or passage in front of the signage terminal 60 and registered face images, the CPU 102 updates the status of the user U of interest regarding the presence or passage in front of the signage terminal 60 from an unfinished state to a completed state.

[0287] Next, the CPU 102 generates display information in accordance with the length of the remaining time to the boarding time of the identified user U in front of the signage terminal 60 (step S514). Subsequently, the CPU 102 transmits the generated display information to the signage terminal 60 via the network NW (step S516). In such a way, the CPU 102 performs management so as to cause the signage terminal 60 to display the display information in accordance with the length of the remaining time to the boarding time of the identified user U. Note that the timing of the step S514 or S516 is not limited to the above. The CPU 102 can

perform step S514 or S516 before step S512 or can perform step S514 or S516 at the same time as step S512, for example.

[0288] The CPU 602 of the signage terminal 60 causes the display 608 to display the display information received from the management server 10 to the identified user U (step S518).

[0289] On the other hand, as a result of the comparison, if no registered face image matching the target face image is found and the user U in front of the signage terminal 60 is not identified (step S510, NO), the CPU 102 ends the process for the target face image transmitted from the signage terminal 60.

[0290] Note that the management server 10 may omit step S512 regarding update of the status in the process illustrated in FIG. 14 described above for the signage terminal 60 in which the status regarding presence or passage of the user U in front thereof is not required to be managed. As discussed above, the management server 10 can manage display information displayed on the signage terminal 60 without managing the status. The management server 10 can suitably set the signage terminal 60 which manages both the status and the display information and the signage terminal 60 which manages either one of the status and the display information in accordance with the installation place of the signage terminal 60.

[0291] Next, the operation performed by the boarding gate apparatus 70 will be described by using FIG. 15 together with the operation performed by the management server 10. FIG. 15 is a sequence diagram illustrating the operation performed by the boarding gate apparatus 70 and the management server 10.

[0292] The user U who has passed through the passage P5 on which the signage terminal 60 is installed moves to the boarding gate P6 and walks through the boarding gate apparatus 70 to board an airplane. The boarding gate apparatus 70 and the management server 10 perform the process illustrated in FIG. 15 every time each of the plurality of users U boards an airplane.

[0293] As illustrated in FIG. 15, the biometrics information acquisition device 712 of the boarding gate apparatus 70 continuously or periodically captures a front area of the boarding gate apparatus 70 and determines whether or not a face of the user U standing in front of the boarding gate apparatus 70 is detected in the captured image (step S602). The biometrics information acquisition device 712 waits until a face of the user U is detected in the image (step S602, NO).

[0294] If the biometrics information acquisition device 712 determines that a face of the user U is detected (step S602, YES), the biometrics information acquisition device 712 captures the face of the user U and acquires the face image of the user U as a target face image (step S604).

[0295] Next, the CPU 702 of the boarding gate apparatus 70 transmits the target face image of the user U captured by the biometrics information acquisition device 712 to the management server 10 via the network NW together with a comparison request (step S606). Thereby, the CPU 702 requests the management server 10 to compare, at 1:N, the target face image of the user U captured by the biometrics information acquisition device 712 with a plurality of registered face images registered in the user information DB 106a of the management server 10.

[0296] In response to receiving a target face image and a comparison request from the boarding gate apparatus 70, the CPU 102 of the management server 10 performs comparison of the face image of the user U (step S608). The CPU 102 compares, at 1:N, a target face image received from the boarding gate apparatus 70 with a plurality of registered face images registered in the user information DB 106a. Furthermore, when a registered face image matching the target face image is found, the CPU 102 confirms that boarding information associated with the registered face image associates with an airplane boardable from the boarding gate P6.

[0297] Next, the CPU 102 transmits comparison result information indicating a result of the comparison to the boarding gate apparatus 70 via the network NW (step S610).

[0298] The CPU 702 of the boarding gate apparatus 70 receives comparison result information from the management server 10. If the comparison result information indicates that a registered face image matching the target face image is found and identity verification of the boarding user U is successful and that confirmation of boarding information is successful (step S612, YES), the CPU 702 performs an opening process of the gate 714 (step S614). Thereby, the CPU 702 transfers the gate 714 from a closed state for standby that blocks passage of the user U to an opened state that allows passage of the user U. The user U passes through the opened gate 714 and boards the airplane.

[0299] Next, the CPU 702 transmits, to the management server 10 via the network NW, status information indicating that the boarding on the airplane of the user U is completed after the comparison of the face image (step S616).

[0300] In response to receiving status information from the boarding gate apparatus 70, which is an external apparatus, the CPU 102 of the management server 10 updates the status of the user U of interest regarding the boarding on the airplane in the user information DB 106a from an unfinished state to a completed state (step S618). In such a way, the CPU 102 manages the status of the user U regarding boarding on an airplane based on a result of the comparison between a target face image acquired in the boarding on the airplane and a registered face image. That is, when there is a matching in the comparison between a target face image acquired in boarding on an airplane with registered face images, the CPU 102 updates the status of the user U of interest regarding the boarding on the airplane from an unfinished state to a completed state.

[0301] On the other hand, if the comparison result information indicates that no matching registered face image is found and identity verification of the user U fails or that confirmation of boarding information fails (step S612, NO), the CPU 702 of the boarding gate apparatus 70 notifies the user U (step S620). That is, the CPU 702 performs notification indicating that the identity verification failed or that the confirmation of boarding information failed. At this time, the CPU 702 causes the display 710 to display a notification window notifying that the identity verification failed or that the confirmation of boarding information failed. Note that the CPU 702 may notify the user U that the identity verification failed or that the confirmation of boarding information failed by using a voice, for example, in addition to the display of the notification window or instead of the display of the notification window.

[0302] In such a way, the user U who has passed through the boarding gate apparatus 70 boards an airplane.

[0303] As described above, the management server 10 recognizes and manages the status up to boarding on an airplane in the user information DB 106a for each of the plurality of users U who have performed a check-in procedure with the check-in terminal 20.

[0304] FIG. 16 is a schematic diagram illustrating an example of the user information DB 106a of the management server 10. As illustrated, the user information and the status on each user U are registered for each user identification (ID), which is an identifier that identifies each of the plurality of users U in the user information DB 106a.

[0305] The user information registered in the user information DB 106a includes identity information such as the name, the nationality, or the like, face information, namely, a face image of the user U, and boarding information such as a flight number, an origin, a destination, departure time, a boarding gate, or the like of an airplane which the user U boards in association with each other. Note that the identity information and the boarding information may include information other than illustrated in FIG. 16, respectively.

[0306] Further, the status of the user U registered in the user information DB 106a records that each procedure of a check-in procedure, a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane is completed or unfinished. Further, the status of the user U registered in the user information DB 106a records that presence or passage in front of the signage terminal 60 installed on the passage P5, for example, is completed or unfinished. Note that the signage terminal 60 in which the status regarding presence or passage in front thereof is recorded is not limited to that installed on the passage P5 and may be those installed in other places within the airport A. The status of the user U is updated at any time in accordance with the process illustrated in FIG. 10 to FIG. 15 described above.

[0307] Furthermore, for each procedure, the CPU 102 of the management server 10 calculates and sets deadline time by which the user U has to complete the procedure and generates an alert when the deadline time has expired with the procedure still unfinished. FIG. 17 is a schematic diagram illustrating an example of the deadline time of each procedure set by the CPU 102 of the management server 10. As illustrated, the deadline time is set for each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal 60, and boarding on an airplane with respect to the user U on a user ID basis. The CPU 102 of the management server 10 can record and manage the deadline time illustrated in FIG. 17 in the user information DB 106a in association with the user information and the status on the user U in the user information DB 106a illustrated in FIG. 16.

[0308] The staff member S is able to use the operation terminal 80 to access the management server 10 and confirm the details of the user information and the status on the user U and the deadline time for each procedure recorded in the user information DB 106a. In such a case, the CPU 802 of the operation terminal 80 can acquire, from the user information DB 106a, the user information and the status on the user U and the deadline time for each procedure recorded in the user information DB 106a and cause the display 810 to display the acquired the user information and the status on the user U and the deadline time for each procedure. The CPU 802 can display a part or whole of identity information

included in the user information and can display a part or whole of boarding information included in the user information. Further, for the status of the user U, the CPU 802 can use and display indication such as a bar-like or circular progress bar whose length changes in accordance with the progress status indicating which procedure or procedures of respective procedures have been completed so far, for example.

[0309] The operation including a status management operation and an alert operation performed by the management server 10 will be described below by using FIG. 18 to FIG. 20C. FIG. 18 and FIG. 19 are flowcharts illustrating the operation including a status management operation and an alert operation performed by the management server 10. FIG. 20A is a schematic diagram illustrating one example of an alert displayed on the operation terminal 80. FIG. 20B and FIG. 20C are schematic diagrams illustrating an example of a status review window displayed on the operation terminal 80. The management server 10 performs the process illustrated in FIG. 18 and FIG. 19 for each of the plurality of users U.

[0310] As illustrated in FIG. 18, the CPU 102 of the management server 10 determines whether or not user information is received from the check-in terminal 20 (step S702) and waits until user information is received from the check-in terminal 20 (step S702, NO).

[0311] In response to determining that user information is received from the check-in terminal 20 (step S702, YES), the CPU 102 registers the received user information to the user information DB 106a (step S704). Step S704 corresponds to step S118 illustrated in FIG. 10.

[0312] Next, the CPU 102 calculates and sets, from the boarding time based on the boarding information included in the user information, deadline time for each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal 60, and boarding on an airplane (step S706). Note that CPU 102 can appropriately change and set the deadline time for each procedure in accordance with the degree of congestion of the airport A, which is a facility where the user U boards an airplane. Further, the CPU 102 can record and manage the set deadline time for each procedure in the user information DB 106a in association with the user information and the status on the user U in the user information DB 106a, for example.

[0313] Next, the CPU 102 determines whether or not status information is received from the check-in terminal 20 (step S708) and waits until status information is received from the check-in terminal 20 (step S708, NO).

[0314] If the CPU 102 determines that status information is received from the check-in terminal 20 (step S708, YES), the CPU 102 updates the status of the user U of interest regarding the check-in procedure in the user information DB 106a from an unfinished state to a completed state (step S710). Step S710 corresponds to step S122 illustrated in FIG. 10.

[0315] Next, the CPU 102 determines whether or not status information is received from the automatic baggage deposit machine 30 (step S712).

[0316] If the CPU 102 determines that status information is not received from the automatic baggage deposit machine 30 (step S712, NO), the CPU 102 determines whether or not the deadline time for the baggage deposit procedure has expired (step S714).

[0317] If the CPU 102 determines that the deadline time for the baggage deposit procedure has expired (step S714, YES), the CPU 102 generates an alert indicating that the deadline time for the baggage deposit procedure has expired for the user U of interest and transmits the alert via the network NW (step S716). The destination of the alert is the operation terminal 80 of one or a plurality of staff members S.

[0318] If the CPU 102 determines that the deadline time for the baggage deposit procedure has not expired (step S714, NO), the CPU 102 determines whether or not status information is received from the security inspection apparatus 40 (step S718). Note that this step S718 is provided assuming that some users U have less baggage and do not deposit their baggage. If the CPU 102 determines that status information is received from the security inspection apparatus 40 (step S718, YES), the process proceeds to step S728 described later. Further, if the CPU 102 determines that status information is not received from the security inspection apparatus 40 (step S718, NO), the CPU 102 waits for reception of status information from the automatic baggage deposit machine 30 (step S712).

[0319] On the other hand, if the CPU 102 determines that status information is received from the automatic baggage deposit machine 30 (step S712, YES), the CPU 102 updates the status of the user U of interest regarding the baggage deposit procedure in the user information DB 106a from an unfinished state to a completed state (step S720). Step S720 corresponds to step S218 illustrated in FIG. 11.

[0320] Next, the CPU 102 determines whether or not status information is received from the security inspection apparatus 40 (step S722).

[0321] If the CPU 102 determines that status information is not received from the security inspection apparatus 40 (step S722, NO), the CPU 102 determines whether or not the deadline time for the security inspection procedure has expired (step S724).

[0322] If the CPU 102 determines that the deadline time for the security inspection procedure has expired (step S724, YES), the CPU 102 generates an alert indicating that the deadline time for the security inspection procedure has expired for the user U of interest and transmits the alert via the network NW (step S726). The destination of the alert is the operation terminal 80 of one or a plurality of staff members S.

[0323] If the CPU 102 determines that the deadline time for the security inspection procedure has not expired (step S724, NO), the CPU 102 waits for reception of status information from the security inspection apparatus 40 (step S722).

[0324] On the other hand, if the CPU 102 determines that status information is received from the security inspection apparatus 40 (step S722, YES), the CPU 102 updates the status of the user U of interest regarding the security inspection procedure in the user information DB 106a from an unfinished state to a completed state (step S728). Step S728 corresponds to step S318 illustrated in FIG. 12.

[0325] Next, as illustrated in FIG. 19, the CPU 102 determines whether or not status information is received from the automated gate apparatus 50 (step S730).

[0326] If the CPU 102 determines that status information is not received from the automated gate apparatus 50 (step



S730, NO), the CPU 102 determines whether or not the deadline time for the immigration procedure has expired (step S732).

[0327] If the CPU 102 determines that the deadline time for the immigration procedure has expired (step S732, YES), the CPU 102 generates an alert indicating that the deadline time for the immigration procedure has expired for the user U of interest and transmits the alert via the network NW (step S734). The destination of the alert is the operation terminal 80 of one or a plurality of staff members S.

[0328] If the CPU 102 determines that the deadline time for the immigration procedure has not expired (step S732, NO), the CPU 102 waits for reception of status information from the automated gate apparatus 50 (step S730).

[0329] On the other hand, if the CPU 102 determines that status information is received from the automated gate apparatus 50 (step S730, YES), the CPU 102 updates the status of the user U of interest regarding the immigration procedure in the user information DB 106a from an unfinished state to a completed state (step S736). Step S736 corresponds to step S418 illustrated in FIG. 13.

[0330] Next, the CPU 102 compares, at 1:N, a target face image received from the signage terminal 60 with a plurality of registered face images in the user information DB 106a and determines for the user U of interest whether or not there is presence or passage in front of the signage terminal 60 (step S738). As a result of the comparison, when a registered face image matching the target face image is found, which means that there is presence or passage in front of the signage terminal 60 for the user U of interest.

[0331] If the CPU 102 determines that there is no presence or passage in front of the signage terminal 60 (step S738, NO), the CPU 102 determines whether or not the deadline time for presence or passage in front of the signage terminal 60 has expired (step S740).

[0332] If the CPU 102 determines that the deadline time for presence or passage in front of the signage terminal 60 has expired (step S740, YES), the CPU 102 generates an alert and transmits the alert via the network NW (step S742). That is, the CPU 102 generates an alert indicating that the deadline time for presence or passage in front of the signage terminal 60 has expired for the user of interest and transmits the alert via the network NW. The destination of the alert is the operation terminal 80 of one or a plurality of staff members S.

[0333] If the CPU 102 determines that the deadline time for presence or passage in front of the signage terminal 60 has not expired (step S740, NO), the process proceeds to step S738.

[0334] On the other hand, if the CPU 102 determines that there is presence or passage in front of the signage terminal 60 (step S738, YES), the CPU 102 updates the status of the user U of interest (step S744). That is, the CPU 102 updates the status of the user U of interest regarding presence or passage in front of the signage terminal 60 in the user information DB 106a from an unfinished state to a completed state. Step S744 corresponds to step S512 illustrated in FIG. 14.

[0335] Further, the CPU 102 generates display information in accordance with the length of the remaining time to the boarding time of the identified user U in front of the signage terminal 60 and transmits the generated display

information to the signage terminal 60 via the network NW (step S745). Step S745 corresponds to steps S514 and S516 illustrated in FIG. 14.

[0336] Next, the CPU 102 determines whether or not status information is received from the boarding gate apparatus 70 (step S746).

[0337] If the CPU 102 determines that status information is not received from the boarding gate apparatus 70 (step S746, NO), the CPU 102 determines whether or not the deadline time for the boarding on an airplane has expired (step S748).

[0338] If the CPU 102 determines that the deadline time for the boarding on the airplane has expired (step S748, YES), the CPU 102 generates an alert indicating that the deadline time for the boarding on the airplane has expired for the user U of interest and transmits the alert via the network NW (step S750). The destination of the alert is the operation terminal 80 of one or a plurality of staff members S.

[0339] If the CPU 102 determines that the deadline time for the boarding on the airplane has not expired (step S748, NO), the CPU 102 waits for reception of status information from the boarding gate apparatus 70 (step S746).

[0340] On the other hand, if the CPU 102 determines that status information is received from the boarding gate apparatus 70 (step S746, YES), the CPU 102 updates the status of the user U of interest regarding the boarding on the airplane in the user information DB 106a from an unfinished state to a completed state (step S752). Step S752 corresponds to step S618 illustrated in FIG. 15.

[0341] Note that, while the CPU 102 determines whether or not the deadline time has expired for respective procedures sequentially in steps S714, S724, S732, S740, and S748 in the case illustrated in FIG. 18 and FIG. 19 described above, the determination is not necessarily required to be performed sequentially. The CPU 102 may determine whether or not the deadline time has expired for each procedure independently of each other.

[0342] In such a way, the management server 10 recognizes and manages the status of the user U regarding each procedure before the user U boards an airplane together with the deadline time and, if there is a procedure for which the deadline time has expired, generates and transmits an alert to the operation terminal 80.

[0343] When the staff member S selects alert display on the operation terminal 80, the CPU 802 of the operation terminal 80 acquires, from the management server 10 via the network NW, the user information and the status on the user U who missed the deadline time for the procedure identified by the alert. The CPU 802 acquires the user information and the status on the user U of interest from the user information DB 106a of the management server 10.

[0344] The CPU 802 displays the alert on the display 810 based on the acquired user information and status. FIG. 20A is a schematic diagram illustrating one example of an alert displayed on the operation terminal 80. Note that, unlike the case illustrated in FIG. 16, FIG. 20A illustrates a case where the user U whose user ID is 10105 has not yet completed a security inspection procedure while the deadline time for the security inspection procedure has expired.

[0345] As illustrated in FIG. 20A, an alert AL includes a user information display field AL10 that displays user information on the user U, a status display field AL12 that

displays the status of the user U, and a face image display field AL14 that displays a registered face image of the user U.

[0346] The user information display field AL10 displays identity information such as the name, the nationality, or the like of the user U and boarding information such as a flight number of an airplane which the user U is scheduled to board. Note that the user information display field AL10 may display a part or whole of identity information and may display a part or whole of boarding information.

[0347] The status display field AL12 displays the status of “unfinished” or “completed” for each procedure of the user U. For a procedure for which the deadline time has expired, the status display field AL12 displays the expiration. Note that, when an alert indicating that the deadline time has expired for a certain procedure is transmitted, this means that the deadline for the subsequent procedure expires.

[0348] The face image display field AL14 displays a captured face image or a passport face image of the user U captured by the check-in terminal 20 and registered in the user information DB 106a. The staff member S is able to rely on the registered face image of the user U displayed in the alert AL and easily search for the user U who has not yet completed procedures including the procedure in the airport A.

[0349] Further, the CPU 802 of the operation terminal 80 can display a status review window used for reviewing the status of the user U on the display 810 regardless of whether or not an alert from the management server 10 is received. In such a case, the CPU 802 acquires, from the user information DB 106a of the management server 10 via the network NW, user information and the status on the user U whose status has to be reviewed. FIG. 20B and FIG. 20C are schematic diagrams illustrating examples of a status review window displayed on the operation terminal 80. FIG. 20B and FIG. 20C are different in the form of a progress bar that displays the status of the user U. Note that, unlike the case illustrated in FIG. 16, FIG. 20B and FIG. 20C illustrate a case where the user U whose user ID is 10105 has not yet completed a security inspection procedure while the deadline time for the security inspection procedure has expired.

[0350] As illustrated in FIG. 20B and FIG. 20C, a status review window ST includes a user information display field ST10 that displays user information on the user U, a status display field ST12 that displays the status of the user U, and a face image display field ST14 that displays a registered face image of the user U.

[0351] The user information display field ST10 displays the same information as the user information display field AL10 of the alert AL. Further, the face image display field ST14 displays the same registered face image as the face image display field AL14 of the alert AL.

[0352] The status display field ST12 displays which procedure or procedures of respective procedures of the user U have been completed so far by using a progress bar whose length changes in accordance with a progress status. FIG. 20B illustrates a case where the status of the user U is displayed by using a bar-like progress bar. FIG. 20C illustrates a case where the status of the user U is displayed by using a circular progress bar.

[0353] Note that, also in the alert AL illustrated in FIG. 20A, the status of the user U may be displayed by a progress bar in the same manner as in FIG. 20B and FIG. 20C.

[0354] The CPU 802 of the operation terminal 80 can store the alert AL illustrated in FIG. 20A and user information to be displayed in the status review window ST illustrated in FIG. 20B and FIG. 20C in the local storage device 806 and display each window by using a local application. Further, the CPU 802 can also communicate with the management server 10 on a web browser to acquire the alert AL illustrated in FIG. 20A and user information to be displayed in the status review window ST illustrated in FIG. 20B and FIG. 20C and display each window on the web browser.

[0355] Further, the management server 10 generates different display information in accordance with the remaining time to the boarding time to the user U in front of the signage terminal 60 who is identified by the comparison with a target face image transmitted from the signage terminal 60 and displays the generated display information on the signage terminal 60. Thereby, the management server 10 manages display on the signage terminal 60.

[0356] A display management operation performed by the management server 10 that manages display on the signage terminal 60 will be described below by using FIG. 21. FIG. 21 is a flowchart illustrating the display management operation performed by the management server 10. The management server 10 performs the process illustrated in FIG. 21 for each of the signage terminals 60 in front of which the user is identified by the comparison with a target face image.

[0357] The CPU 102 of the management server 10 identifies the user U in front of the signage terminal 60 by the comparison with a face image in step S508 illustrated in FIG. 14 described above. The CPU 102 references the user information DB 106a and acquires boarding information on the identified user U.

[0358] Next, as illustrated in FIG. 21, the CPU 102 calculates the boarding time from the departure time included in the acquired boarding information or acquires the boarding time included in the boarding information and calculates the remaining time to the boarding time based on the boarding information calculated or acquired from the boarding information (step S802).

[0359] Next, the CPU 102 compares the calculated remaining time to the boarding time with a predetermined threshold (step S804). Thereby, the CPU 102 determines the length of the remaining time to the boarding time, that is, whether little time remains before the boarding time or enough time remains before the boarding time. Note that the CPU 102 may change and set the threshold that is a criterion of determination as appropriate in accordance with the degree of congestion of the airport A.

[0360] If the remaining time is less than or equal to the predetermined threshold (step S804, YES), the CPU 102 determines that the remaining time is short, that is, little time remains before the boarding time. Furthermore, the CPU 102 generates the first display information as display information to be displayed on the signage terminal 60 to the identified user U in front of the signage terminal 60 (step S806). As described above, the first display information includes the boarding information on the user U, the remaining time to the boarding time, the deadline time for each procedure, the progress status of each procedure, or the like, for example. Further, as described above, the first display information may include congestion information indicating the degree of congestion of the security inspection site P3, the immigration site P4, or the like that is a procedure site where the procedure is performed.

[0361] On the other hand, if the remaining time exceeds the predetermined threshold (step S804, NO), the CPU 102 determines that the remaining time is long, that is, enough time remains before the boarding time. Furthermore, the CPU 102 generates the second display information as display information to be displayed on the signage terminal 60 to the identified user U in front of the signage terminal 60 (step S808). As described above, the second display information includes advertisement information such as a general advertisement, a targeting advertisement, or the like, for example.

[0362] In such a way, the CPU 102 generates the first display information or the second display information different from each other as display information to be displayed on the signage terminal 60 in accordance with the remaining time to the boarding time of the identified user U. The CPU 102 transmits the generated first display information or second display information via the network NW to the signage terminal 60 where the user U in front thereof is identified (step S810).

[0363] The CPU 602 of the signage terminal 60 where the user U is identified in front thereof displays the first display information or the second display information received from the management server 10 on the display 608. FIG. 22 to FIG. 24 are a schematic diagram illustrating an example of a display window of the first display information on the signage terminal 60. FIG. 25 is a schematic diagram illustrating an example of a display window of the second display information on the signage terminal 60.

[0364] FIG. 22 and FIG. 23 illustrate a display window SNT on the display 608 of the signage terminal 60 that displays the boarding information included in the first display information, the remaining time to the boarding time, the deadline time for each procedure, and the status of each procedure. FIG. 22 illustrates a case where all the procedures of a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane are unfinished. FIG. 23 illustrates a case where, out of these procedures, the baggage deposit procedure and the security inspection procedure are completed, and the immigration procedure and the boarding on the airplane are unfinished.

[0365] As illustrated in FIG. 22 and FIG. 23, the display window SNT includes a boarding information display field SNT10 that displays boarding information on the user U and a remaining time display field SNT12 that displays the remaining time to the boarding time of the user U. Further, the display window SNT includes a deadline time display field SNT14 that displays a deadline time for each procedure and a status display field SNT16 that displays the status for each procedure.

[0366] The boarding information display field SNT10 displays boarding information such as the departure time, the flight number, the origin, the destination, or the like of the boarding flight which the user U boards. To identify the user U, the boarding information display field SNT10 may also display user identification information, which is information for identifying the user U, such as an identification number notified in advance to the user U, a nickname set in advance by the user U, or the like. In such a case, the CPU 102 of the management server may include the user identification information in the first display information. Further, in the CPU 102, while user information such as the name of the user U may be included in the first display information

as user identification information, the user information may be set so as not to be displayed in terms of privacy or the like.

[0367] The remaining time display field SNT12 displays the remaining time to the boarding time of the user U in comparison with the boarding time, for example. The display form to display the remaining time is not particularly limited and, for example, may be a display form of “hour:minute:second” as illustrated, may be a display form in which “second” is omitted from “hour:minute:second”, or may be a display form in a unit of minute. The remaining time display field SNT12 may display the remaining time to the deadline time for each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane additionally or alternatively the remaining time to the boarding time. In such a case, the remaining time display field SNT12 may regularly or irregularly switch and display the remaining time to be displayed out of the remaining time to the boarding time and the remaining time to the deadline time for each procedure or may list and display the remaining time to be displayed out of the remaining time to the boarding time and the remaining time to the deadline time for each procedure.

[0368] The deadline time display field SNT14 displays the deadline time for each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane with the procedure name being arranged. Note that, as illustrated in FIG. 23, in the deadline time display field SNT14, the procedure name and the deadline time for an already completed procedure is distinguished from the procedure name and the deadline time for an unfinished procedure and displayed in a gray-out display form or the like, for example. Further, when the deadline time for an already completed procedure is excluded from the first display information, the procedure name and the deadline time for the procedure are not displayed in the deadline time display field SNT14. That is, only the deadline time for the unfinished procedure is displayed in the deadline time display field SNT14.

[0369] The status field display SNT16 displays the progress status indicating the unfinished status or the completed status for each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, and boarding on an airplane with the procedure name being arranged, for example. The status display field SNT16 is not required to display all the deadline time for respective procedures and may display the deadline time for some of the procedures of the security inspection procedure, the immigration procedure, and the like. When the deadline time for the already completed procedure is excluded from the first display information, that is, only the deadline time for the unfinished procedure is displayed in the deadline time display field SNT14, the status display field SNT16 may be omitted.

[0370] In such a way, when little time remains before the boarding time, it is possible to display the remaining time to the boarding time, the deadline time for each procedure, or the remaining time to the deadline time in the display window SNT of the signage terminal 60 to urge the user U to take appropriate action for boarding. Further, the user U is able to suitably recognize his/her status by displaying the progress information in the display window SNT.

[0371] FIG. 24 illustrates a display window SNC on the display 608 of the signage terminal 60 that displays congestion information included in the first display information. The display window SNC illustrated in FIG. 24 illustrates the degree of congestion of the security inspection site as an example of congestion information. The CPU 602 of the signage terminal 60 may display the display window SNT and the display window SNC on the display 608 by switching the display window SNT and the display window SNC at a constant interval, for example, or may display the contents of the display window SNT and the display window SNC on the display 608 in a single window.

[0372] As illustrated in FIG. 24, the display window SNC includes an availability information display field SNC10 that identifies and displays a relatively less congested security inspection site out of the plurality of security inspection sites and a waiting time display field SNC12 that displays waiting time required for completion of a procedure at each security inspection site. Further, the display window SNC includes a place display field SNC14 that indicates a place of relatively less congested security inspection site.

[0373] The availability information display field SNC10 identifies and displays the security inspection site where the waiting time is the shortest out of the plurality of security inspection sites. In FIG. 24, the availability information display field SNC10 displays the security inspection site J where the waiting time is short as an available security inspection site out of the security inspection sites J and K.

[0374] The waiting time display field SNC12 displays the waiting time for each of the plurality of available security inspection sites. In FIG. 24, the waiting time display field SNC12 displays the waiting time for each of the security inspection sites J and K with a timespan from the upper limit to the lower limit of the expected waiting time. Note that the display form of the waiting time in the waiting time display field SNC12 may be a form that displays the expected waiting time with or without an expression such as “about”, “approximately”, “expected”, for example, other than the indication with a timespan.

[0375] The place display field SNC14 displays the place of a relatively less congested security inspection site out of the plurality of security inspection sites on a floor map with emphasis, for example. In FIG. 24, the place display field SNC14 displays the place of the relatively uncongested security inspection site J out of the security inspection sites J and K on a floor map of the departure lobby with emphasis.

[0376] Note that congestion information indicating the degree of congestion of a different procedure site other than the security inspection site, such as the immigration site, may be displayed on the signage terminal 60 in the same manner as the case of the security inspection site described above.

[0377] In such a way, when little time remains before the boarding time, it is possible to display congestion information of the procedure site on the display window SNC of the signage terminal 60 to urge the user U to take an appropriate action for boarding.

[0378] FIG. 25 illustrates a display window SNA on the display 608 of the signage terminal 60 that displays advertisement information included in the second display information. The display window SNA illustrated in FIG. 25 indicates an advertisement of a duty-free shop in the airport A as an example of advertisement information.

[0379] As illustrated in FIG. 25, the display window SNA includes an advertisement information display field SNA10 that displays advertisement information directed to the user U. The advertisement information display field SNA10 may display a general advertisement independent of the user U or may display a targeting advertisement in accordance with the user U based on attribute information, a purchase history, or the like on the user U.

[0380] In such a way, when enough time remains before the boarding time, advertisement information is displayed on the signage terminal 60 rather than information related to boarding for the user U, such as the remaining time to the boarding time, the deadline time for each procedure, or the like. Thus, it is possible to provide advertisement information to the user U effectively without excessively urging the user U to take an action for boarding.

[0381] As discussed above, according to the present example embodiment, for the user U who is scheduled to board an airplane, the deadline time for each procedure in the airport A is set based on boarding information acquired by the check-in terminal 20, and the status of the user U regarding each procedure is managed in the management server 10. Therefore, according to the present example embodiment, it is possible to recognize and manage in more detail the status of the user U who is scheduled to board an airplane.

[0382] Further, according to the present example embodiment, in the management server 10, different display information is displayed on the signage terminal 60 in accordance with the remaining time to the boarding time for the user U in front of the signage terminal 60 who is identified by the comparison of a target face image transmitted from the signage terminal 60. The management server 10 compares a target face image transmitted from the signage terminal 60 with a registered face image, which is a captured face image or a passport face image acquired by the check-in terminal 20 in a check-in procedure. Thus, the user U is not required to register in advance his/her face image to the server from the user terminal such as the mobile terminal thereof. Therefore, according to the present example embodiment, it is possible to realize display directed to the user U in front of the signage terminal 60 in accordance with a situation without involving a large burden on the user U scheduled for boarding.

[0383] Further, while the case where the first display information or the second display information different from each other is displayed on the signage terminal 60 in accordance with the remaining time to the boarding time for the identified user U has been described above, the invention is not limited thereto. Both of the first display information such as the remaining time to the boarding time and the second display information such as advertisement information may be displayed on the signage terminal 60. In such a case, the CPU 102 of the management server 10 may display the first display information and the second display information on the signage terminal 60 with a difference in the level of the size, the area, the font, or the like for information to be displayed, for example, in accordance with the remaining time to the boarding time. For example, when little time remains before boarding, the CPU 102 may cause the signage terminal 60 to display the first display information such as the remaining time to the boarding time to be larger than the second display information such as advertisement information in the display window SNTA of the signage

terminal 60, as shown in FIG. 26. On the other hand, when enough time remains before boarding, the CPU 102 may cause the signage terminal 60 to display the second display information such as advertisement information to be larger than the first display information such as the remaining time to the boarding time in the display window SNAT of the signage terminal 60, as shown in FIG. 27.

[0384] Further, while the case where different display information is displayed on the signage terminal 60 in accordance with the remaining time to the boarding time for the user U in front of the signage terminal 60 has been described above, the invention is not limited thereto. For example, the CPU 102 of the management server 10 may display different display information on a mobile information terminal such as a smartphone, a tablet terminal, or the like of the user U in accordance with the remaining time to the boarding time in the same manner as on the signage terminal 60 described above. In such a case, the CPU 102 transmits, to the mobile information terminal of a particular user U, the remaining time to the boarding time, the deadline time for each procedure, the remaining time to the deadline time, congestion information, or advertisement information as the display information in accordance with the remaining time to the boarding time. The particular user U to which the display information is to be transmitted is not particularly limited and may be, for example, the user U in front of the signage terminal 60 identified in the same manner as described above, the user located by using the video from the surveillance camera 90, or the like. The CPU 102 may identify a mobile information terminal of the user U to which information to be displayed is to be transmitted based on terminal identification information of the mobile information terminal registered in advance in association with user information or an ID of an application program installed in the mobile information terminal. Further, the CPU 102 may identify a mobile information terminal of the user U to which information to be displayed is to be transmitted based on a result of comparison between a face image captured by the mobile information terminal and a registered face image. The mobile information terminal of the user U may display information transmitted from the management server 10 by using an installed application program, for example. With the application program, it is possible to display information on a main window when executed, a lock window or the like of the mobile information terminal or display information with a popup notification or the like, for example. The user U is able to check the remaining time to the boarding time, the deadline time for each procedure, the remaining time to the deadline time, congestion information, or advertisement information displayed by the application program on a screen of the mobile information terminal.

#### Second Example Embodiment

[0385] An information processing system, an information processing apparatus, and an information processing method according to a second example embodiment of the present invention will be described by using FIG. 28. Note that the same components as those of the information processing system, the information processing apparatus, and the information processing method according to the first example embodiment described above will be labeled with the same references, and the description thereof will be omitted or simplified.

[0386] In the present example embodiment, an alert operation performed by the management server 10 that may transmit an alert in which the location of the user U in the airport A is determined will be described by using FIG. 28. FIG. 28 is a flowchart illustrating the alert operation performed by the management server 10 according to the present example embodiment.

[0387] The management server 10 according to the present example embodiment performs the process illustrated in FIG. 28 as the process of generating and transmitting the alert of steps S716 and S726 illustrated in FIG. 18 and steps S734, S742, and S750 illustrated in FIG. 19. Note that the process illustrated in FIG. 28 may be performed in all or some of steps S716, S726, S734, S742, and S750.

[0388] Further, the management server 10 utilizes a video captured by the plurality of surveillance cameras 90 installed in the airport A in order to determine the location of the user U. Each of the plurality of surveillance cameras 90 functions as a video acquisition unit that captures a predetermined place within the airport A and transmits the captured video to the management server 10 via the network NW.

[0389] The CPU 102 of the management server 10 functions as a locating unit that determines the location of the user U. The CPU 102 as the locating unit identifies the user U from a face image in a video captured by the plurality of surveillance cameras 90. The CPU 102 as the locating unit determines, as the location of the user U, the installation place of the surveillance camera 90 which captures a video in which the user U is determined out of the plurality of surveillance cameras 90.

[0390] As illustrated in FIG. 28, when determining that the deadline time for a particular procedure has expired, the CPU 102 of the management server 10 acquires a registered face image of the user U of interest from the user information DB 106a (step S902). The particular procedure is a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of a signage terminal 60, or boarding on an airplane.

[0391] Further, the CPU 102 acquires a video transmitted from the plurality of surveillance cameras 90 (step S904).

[0392] Next, the CPU 102 detects a face from a video captured by the plurality of surveillance cameras 90 and compares a detected face image, which is an image of the detected face, with a registered face image (step S906). In the comparison between a detected face image and a registered face image, the CPU 102 may compare a face feature amount extracted from the detected face image with a face feature amount extracted from the registered face image to compare both face images.

[0393] In particular, the CPU 102 can acquire a detected face image by detecting a face image from a video transmitted from the surveillance camera 90 installed between a place where a procedure for which the deadline time has expired is performed and a place where the latest completed procedure was performed. Thereby, face images to be detected can be narrowed, and comparison accuracy can be improved.

[0394] As a result of the comparison, if a detected face image matching a registered face image is found (step S908, YES), the CPU 102 determines the location of the user U, generates an alert indicating that the deadline time has expired, and transmits the alert via the network NW (step S910). In the alert, the CPU 102 determines, as the location of the user U, the installation place of the surveillance

camera **90** which captured the matching detected face image. Further, the destination of the alert is the operation terminal **80** of one or a plurality of staff members **S**. Because the location of the user **U** is determined in the alert, the staff member **S** receiving the alert on the operation terminal **80** is able to easily search for the user **U** who missed the deadline time for the particular procedure.

[0395] Further, the destination of an alert can be the operation terminal **80** of the staff member **S** closest to the location of the user **U** out of the plurality of staff members **S**. The operation terminal **80** carried by the staff member **80** transmits position information on the operation terminal **80** based on a position sensor that uses a Global Navigation Satellite System (GNSS), a magnetic sensor that detects geomagnetism, or the like, for example, to the management server **10**. The GNSS is not particularly limited and may be, for example, a Global Positioning System (GPS), a Global Navigation Satellite System (GLONASS), a Galileo compass, or the like. The CPU **102** of the management server **10** can identify the operation terminal **80** of the staff member **S** closest to the location of the user **U** based on the position information transmitted from the operation terminal **80**.

[0396] On the other hand, as a result of the comparison, if no detected face image matching a registered face image is found (step **S908**, NO), the CPU **102** generates an alert indicating that the deadline time has expired but the location of the user **U** is not determined and transmits the alert via the network **NW** (step **S912**).

[0397] As with the present example embodiment, the use of a video captured by the plurality of surveillance cameras **90** enables the management server **10** to generate and transmit an alert that determines the location of the user **U** who missed the deadline time for a particular procedure.

#### Other Example Embodiments

[0398] According to another example embodiment, the information processing apparatus described in the above example embodiments can be configured as illustrated in FIG. **29** and FIG. **30**. FIG. **29** and FIG. **30** are block diagrams each illustrating the configuration of an information processing apparatus according to another example embodiment.

[0399] As illustrated in FIG. **29**, an information processing apparatus **1000** according to another example embodiment has an information management unit **1002** and a display management unit **1004**. The information management unit **1002** registers boarding information on a user regarding boarding acquired by a check-in procedure performed by the user and biometrics information on the user acquired in the check-in procedure in association with each other. The display management unit **1004** causes a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user. The user is a user who is identified by the comparison between target biometrics information, which is biometrics information on the user acquired by the display terminal, and registered biometrics information, which is the biometrics information registered by the information management unit.

[0400] Further, as illustrated in FIG. **30**, the information processing apparatus **2000** according to another example embodiment has an information management unit **2002** and a display management unit **2004**. The information management unit **2002** is the same as the information management

unit **1002** as described above. The display management unit **2004** causes a display terminal to display a deadline for a procedure of the user up to the boarding in accordance with remaining time to boarding time based on the boarding information on the user. The user is the same as that in the case of information processing apparatus **1000** described above.

[0401] According to the above another example embodiment, a user is identified by the comparison between target biometrics information acquired by a display terminal and registered biometrics information acquired in a check-in procedure. Thus, the user is not required to register in advance his/her biometrics information from his/her user terminal such as a mobile terminal. Therefore, according to such other example embodiments, it is possible to realize display directed to a user in front of a signage terminal in accordance with a situation without involving a large burden on the user scheduled for boarding.

#### Modified Example Embodiments

[0402] The present invention can be changed in various manners without being limited to the example embodiments described above.

[0403] For example, while the case of managing the status of each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal **60**, and boarding on an airplane after a check-in procedure has been described in the above example embodiments, the invention is not limited thereto. For example, when the user boards an airplane of a domestic flight, management of the status regarding the immigration procedure is unnecessary. Further, the signage terminals **60** may be installed at a plurality of places within the airport **A**, and the status regarding presence or passage in front of each of the plurality of signage terminals **60** can be managed. In addition to the status regarding the procedures described above as examples, various status can be managed.

[0404] That is, while the case where the automatic baggage deposit machine **30**, the security inspection apparatus **40**, the automated gate apparatus **50**, the signage terminal **60**, and the boarding gate apparatus **70** are installed as the apparatus following the check-in terminal **20** has been described as an example in the above example embodiment, the invention is not limited thereto. It is not necessary that all of the automatic baggage deposit machine **30**, the security inspection apparatus **40**, the automated gate apparatus **50**, the signage terminal **60**, and the boarding gate apparatus **70** be installed, and at least some of these apparatuses may be installed. For example, only the signage terminal **60** may be installed as the apparatus subsequent to the check-in terminal **20**.

[0405] Further, while the case where identity verification is performed by comparison of a target face image in each procedure of a baggage deposit procedure, a security inspection procedure, an immigration procedure, presence or passage in front of the signage terminal **60**, and boarding airplane has been described as an example in the above example embodiment, the invention is not limited thereto. For example, identity verification using a code symbol such as a barcode, a QR code, or the like may be performed. In such a case, for example, when the user **U** performs a check-in procedure on the check-in terminal **20**, the check-in terminal **20** issues, to the user **U**, a ticket such as a boarding

pass in which a code symbol including information that identifies the user U is printed. The user U is able to perform identity verification by causing a code reader to read the code symbol in each procedure.

**[0406]** Further, while the case where the biometrics information acquisition devices **214**, **312**, **412**, **512**, **610**, and **712** capture and acquire a face image as biometrics information on the user U has been described as an example in the above example embodiment, the invention is not limited thereto. The biometrics information acquisition device **214** of the check-in terminal **20** may acquire a fingerprint image, an iris image, a finger vein image, a palm image, a palm vein image, or the like other than a face image as biometrics information registered as registered biometrics information in the user information DB **106a**. Further, other biometrics information acquisition devices **312**, **412**, **512**, **610**, and **712** may acquire a fingerprint image, an iris image, a finger vein image, a palm image, a palm vein image, or the like other than a face image as target biometrics information in accordance with registered biometrics information.

**[0407]** Further, the management server **10** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses. Further, the check-in terminal **20** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses. Further, the automatic baggage deposit machine **30** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses. Further, the security inspection apparatus **40** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses. Further, the automated gate apparatus **50** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses. Further, the signage terminal **60** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses. Further, the boarding gate apparatus **70** according to the example embodiments described above can be configured as a system formed of one or a plurality of apparatuses.

**[0408]** Further, the scope of each of the example embodiments includes a processing method that stores, in a storage medium, a program that causes the configuration of each of the example embodiments to operate so as to implement the function of each of the example embodiments described above, reads the program stored in the storage medium as a code, and executes the program in a computer. That is, the scope of each of the example embodiments also includes a computer readable storage medium. Further, each of the example embodiments includes not only the storage medium in which the computer program described above is stored but also the computer program itself.

**[0409]** As the storage medium, for example, a floppy (registered trademark) disk, a hard disk, an optical disk, a magneto-optical disk, a compact disc-read only memory (CD-ROM), a magnetic tape, a nonvolatile memory card, or a ROM can be used. Further, the scope of each of the example embodiments includes an example that operates on (operating system) OS to perform a process in cooperation with another software or a function of an add-in board without being limited to an example that performs a process by an individual program stored in the storage medium.

**[0410]** An example advantage according to the invention is that it is possible to realize display directed to a user in front of a display terminal in accordance with a situation without involving a large burden on the user scheduled for boarding.

**[0411]** While the invention has been particularly shown and described with reference to example embodiments thereof, the invention is not limited to these embodiments. It will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the present invention as defined by the claims.

**[0412]** The whole or part of the example embodiments disclosed above can be described as, but not limited to, the following supplementary notes.

(Supplementary Note 1)

**[0413]** An information processing apparatus comprising:

**[0414]** an information management unit that registers boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and

**[0415]** a display management unit that causes a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and registered biometrics information, which is the biometrics information registered by the information management unit.

(Supplementary Note 2)

**[0416]** The information processing apparatus according to supplementary note 1, wherein the display management unit causes the display terminal to display first display information when the remaining time is less than or equal to a predetermined threshold and causes the display terminal to display second display information different from the first display information when the remaining time exceeds the predetermined threshold.

(Supplementary Note 3)

**[0417]** The information processing apparatus according to supplementary note 1, wherein the display management unit causes the display terminal to display first display information to be larger than second display information different from the first display information when the remaining time is less than or equal to a predetermined threshold and causes the display terminal to display the second display information to be larger than the first display information when the remaining time exceeds the predetermined threshold.

(Supplementary Note 4)

**[0418]** The information processing apparatus according to supplementary note 2 or 3, wherein the display management unit causes the display terminal to display the first display information including the remaining time.

(Supplementary Note 5)

**[0419]** The information processing apparatus according to any one of supplementary notes 2 to 4 further comprising a deadline setting unit that sets a deadline for a procedure of the user up to the boarding based on the boarding information,

**[0420]** wherein the display management unit causes the display terminal to display the first display information including the deadline.

(Supplementary Note 6)

**[0421]** The information processing apparatus according to supplementary note 5 further comprising a status management unit that manages status regarding the procedure,

**[0422]** wherein the display management unit causes the display terminal to display the first display information including the deadline for the procedure having the status of an unfinished state.

(Supplementary Note 7)

**[0423]** The information processing apparatus according to any one of supplementary notes 2 to 6, wherein the display management unit causes the display terminal to display the first display information including congestion information indicating a degree of congestion in a place where a procedure of the user up to the boarding is performed.

(Supplementary Note 8)

**[0424]** The information processing apparatus according to any one of supplementary notes 2 to 7, wherein the display management unit causes the display terminal to display the second display information including advertisement information.

(Supplementary Note 9)

**[0425]** The information processing apparatus according to supplementary note 8, wherein the display management unit selects the advertisement information included in the second display information in accordance with the user.

(Supplementary Note 10)

**[0426]** An information processing apparatus comprising:

**[0427]** an information management unit that registers boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and

**[0428]** a display management unit that causes a display terminal to display a deadline for a procedure of the user up to the boarding in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and registered biometrics information, which is the biometrics information registered by the information management unit.

(Supplementary Note 11)

**[0429]** An information processing method comprising:

**[0430]** registering boarding information on a user regarding boarding acquired by a check-in procedure of

the user and biometrics information on the user acquired in the check-in procedure in association with each other; and

**[0431]** causing a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and the registered biometrics information.

(Supplementary Note 12)

**[0432]** A non-transitory storage medium storing a program that causes a computer to perform:

**[0433]** registering boarding information on a user regarding boarding acquired by a check-in procedure of the user and biometrics information on the user acquired in the check-in procedure in association with each other; and

**[0434]** causing a display terminal to display display information in accordance with remaining time to boarding time based on the boarding information on the user identified by a comparison between target biometrics information, which is biometrics information acquired by the display terminal for the user, and the registered biometrics information.

What is claimed is:

1. An information processing apparatus comprising:
  - a memory configured to store instructions; and
  - a processor configured to execute the instructions to:
    - receive a first facial image of a user who is present in the vicinity of a display terminal; and
    - display first display information related to boarding information associated with a second facial image of the user, the first display information being displayed larger than second display information that is different from the first display information,
 wherein the first display information is displayed based on a comparison between the first facial image and the second facial image.
2. The information processing apparatus according to claim 1, wherein the second display information is different from the boarding information associated with the user.
3. The information processing apparatus according to claim 1, wherein the processor is further configured to execute the instructions to:
  - display the first information including a remaining time for boarding indicated in the boarding information.
4. The information processing apparatus according to claim 3, wherein the processor is further configured to execute the instructions to:
  - set a deadline for at least one stage between a check-in stage and a boarding stage, among a plurality of stages, based on the boarding information; and
  - display the first information including the deadline when the remaining time is less than or equal to a predetermined threshold.
5. The information processing apparatus according to claim 4, wherein the processor is further configured to execute the instructions to:
  - receive a status regarding the plurality of stages before the boarding stage, and
  - display the first information including the deadline for the user for a certain stage, among the plurality of stages.



6. The information processing apparatus according to claim 5, wherein the processor is further configured to execute the instructions to:

receive the status corresponding to the certain stage based on status information indicating that a certain stage is completed received from an external apparatus related to each of the plurality of stages before the boarding stage, and

display the first information including the deadline for the stage having the status of an unfinished state.

7. The information processing apparatus according to claim 1, wherein the processor is further configured to execute the instructions to:

display the first information including congestion information based on a number of people determined based on sensors installed at each stage.

8. The information processing apparatus according to claim 7, wherein the congestion information indicates a degree of congestion in a place for a stage of the user before the boarding stage.

9. The information processing apparatus according to claim 7, wherein the congestion information indicates an availability information, waiting time or the number of waiting persons in a line.

10. The information processing apparatus according to claim 1, wherein the processor is further configured to execute the instructions to:

display the second display information including advertisement information.

11. The information processing apparatus according to claim 10, wherein the processor is further configured to execute the instructions to:

select the advertisement information included in the second display information in accordance with the user.

12. An information processing apparatus comprising:

a memory configured to store instructions; and

a processor configured to execute the instructions to:

receive a first facial image of a user who is present in front of a display terminal;

match the first facial image with a second facial image, among a plurality of second facial images; and

output information to display first display information related to boarding information associated with the second facial image of the user, the first display information output to be displayed larger than second display information that is different from the first display information.

13. An information processing apparatus comprising:

a memory configured to store instructions; and

a processor configured to execute the instructions to:

receive a first facial image of a user who is present in front of a display terminal;

match the first facial image with a second facial image, among a plurality of second facial images; and

display first display information related to boarding information associated with the second facial image of the user, the first display information being displayed larger than second display information that is different from the first display information.

14. An information processing method comprising:

receiving a first facial image of a user who is present in the vicinity of a display terminal; and

displaying first display information related to boarding information associated with a second facial image of

the user, the first display information being displayed larger than second display information that is different from the first display information,

wherein the first display is displayed based on a comparison between the first facial image and the second facial image.

15. A non-transitory storage medium storing a program that causes a computer to perform:

receiving a first facial image of a user who is present in the vicinity of a display terminal; and

displaying first display information related to boarding information associated with a second facial image of the user, the first display information being displayed larger than second display information that is different from the first display information,

wherein the first display information is displayed based on a comparison between the first facial image and the second facial image.

16. An information processing method comprising:

receiving a first facial image of a user who is present in front of a display terminal;

matching the first facial image with a second facial image, among a plurality of second facial images; and

outputting information to display first display information related to boarding information associated with the second facial image of the user, the first display information output to be displayed larger than second display information that is different from the first display information.

17. A non-transitory storage medium storing a program that causes a computer to perform:

receiving a first facial image of a user who is present in front of a display terminal;

matching the first facial image with a second facial image, among a plurality of second facial images; and

outputting information to display first display information related to boarding information associated with the second facial image of the user, the first display information output to be displayed larger than second display information that is different from the first display information.

18. An information processing apparatus comprising:

a memory configured to store instructions; and

a processor configured to execute the instructions to:

receive a first facial image of a user who is present in the vicinity of a display terminal;

display first information related to boarding information of the user; and

display second information different from the boarding information of the user,

wherein a visual feature of the first information related to boarding information of the user is configured for emphasis based on a match between the first facial image of the user and a second facial image, among a plurality of second facial images, and the first information being displayed larger than the second information.

19. An information processing method comprising:

receiving a first facial image of a user who is present in the vicinity of a display terminal;

displaying first information related to boarding information of the user; and

displaying second information different from the boarding information of the user,

wherein a visual feature of the first information related to boarding information of the user is configured for emphasis based on a match between the first facial image of the user and a second facial image, among a plurality of second facial images, and the first information being displayed larger than the second information.

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